AI BASED DIAGNOSIS OF ECCENTRICITY FAULTS IN AXIAL AND RADIAL FLUX PMSMS

A PROJECT REPORT

submitted by

CB.EN.U4EEE23110	**HARIKRISHNAN N**
CB.EN.U4EEE23125	**MANISH P**
CB.EN.U4EEE23151	**AMATI RITHWIK**
CB.EN.U4EEE23152	**SRIVARSHAN P**

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AMRITA SCHOOL OF ENGINEERING, COIMBATORE

AMRITA VISHWA VIDYAPEETHAM

COIMBATORE- 641112

September 2026

AMRITA VISHWA VIDYAPEETHAM
AMRITA SCHOOL OF ENGINEERING, COIMBATORE, 641112



BONAFIDE CERTIFICATE

This is to certify that the project report entitled, "**AI Based Diagnosis of Eccentricity Faults in Axial and Radial Flux PMSMs**" submitted by

CB.EN.U4EEE23110	**HARIKRISHNAN N**
CB.EN.U4EEE23125	**MANISH P**
CB.EN.U4EEE23151	**AMATI RITHWIK**
CB.EN.U4EEE23152	**SRIVARSHAN P**

in partial fulfillment of the requirements for the award of the Degree of **Bachelor of Technology in ELECTRICAL & ELECTRONICS ENGINEERING** is a bonafide record of the work carried out under my guidance and supervision at Amrita School of Engineering, Coimbatore.

Dr. N Praveen Kumar
Supervisor
Assistant Professor (Sl.Gr)
Department of Electrical and Electronics Engineering
Amrita School of Engineering
Coimbatore-641112

Dr. S. Balamurugan
Chairperson
Professor
Department of Electrical and Electronics Engineering
Amrita School of Engineering
Coimbatore-64112

This project report was evaluated by us on ……. (Date)

INTERNAL EXAMINER

EXTERNAL EXAMINER

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HARIKRISHNAN N
MANISH P
AMATI RITHWIK
SRIVARSHAN P

ABSTRACT

Permanent Magnet Synchronous Motors (PMSMs) are widely considered and have become the most preferred choice when it comes to electric vehicle systems mainly due to their high-power density, reliable efficiency and due to its geometric features like shape and size. However, these advantages tend to degrade due to eccentricity faults which are mainly caused due to misalignment and irregularities faced due to the air gap flux which tends to degrade the quality and performance of the motor due to increased losses. These faults may become a threat to the machine if left undetected. The various conventional fault-diagnosis methods are not reliable due to their inability to provide accurate and early detection under various operating conditions. This proposed work mainly focuses on providing an optimal solution by incorporating an Artificial Intelligence-based approach for the diagnosis of eccentricity faults in both Radial Flux and Axial Flux Permanent Magnet Synchronous Motors. With the help of ANSYS Maxwell, both the healthy and faulty motors are developed and simulated to generate proper datasets for varying faulty conditions. The Stator current signals are obtained from the simulation models, and they are further analyzed using Motor Current Signature Analysis (MCSA) and signal processing techniques to extract the various fault-sensitive electrical and magnetic features. These features are properly extracted and utilized to train models for classification and estimation of the severity of faults. Using these characteristics, a comparative analysis of both AFPMSM and RFPMSM are done to test the motor's behavior under various conditions of eccentricity. Apart from the software part, a hardware-setup for RFPMSM is developed to enable reliable, predetermined fault detection, severity classification, and AI-assisted predictive maintenance to improve the reliability, efficiency and safety of the motor.

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LIST OF SYMBOLS

S.No	SYMBOL	DEFINITION
1	β	Rotor Inclination Angle
2	g_o	Minimum Air-Gap length
3	g_{max}	Maximum Air-Gap length
4	R_{mid}	Mean-Radius of the rotor
5	B	Magnetic Flux Density
6	T_e	Electromagnetic Torque
7	I_s	Stator Current
8	E	Back-EMF
9	f	Frequency
10	O_s	Stator Centre
11	O_r	Rotor Centre

LIST OF ABBREVIATIONS

S.No	ABBREVIATION	EXPANSION
1	AI	Artificial Intelligence
2	ML	Machine Learning
3	DL	Deep Learning
4	PMSM	Permanent Magnet Synchronous Motor
5	AFPMSM	Axial Flux Permanent Magnet Synchronous Motor
6	RFPMSM	Radial Flux Permanent Magnet Synchronous Motor
7	FEM	Finite Element Method
8	FEA	Finite Element Analysis
9	MCSA	Motor Current Signature Analysis
10	FFT	Fast Fourier Transform
11	CNN	Convolutional Neural Network
12	LSTM	Long Short-Term Memory
13	HCNN	Hierarchical Convolutional Neural Network
14	CNN-LSTM	Convolutional Neural Network–Long Short-Term Memory
15	CNN-LSTM-Attention	Convolutional Neural Network–Long Short-Term Memory with Attention
16	CSA	Current Signature Analysis
17	MVSA	Magnetic Vector Spectrum Analysis
18	UMP	Unbalanced Magnetic Pull
19	Back-EMF	Back Electromotive Force
20	EV	Electric Vehicle
21	PM	Permanent Magnet
22	RF	Radial Flux
23	AF	Axial Flux
24	SVM	Support Vector Machine
25	XGBoost	Extreme Gradient Boosting
26	DWT	Discrete Wavelet Transform
27	STFT	Short-Time Fourier Transform
28	HHT	Hilbert–Huang Transform
29	KNN	K-Nearest Neighbours
30	ELM	Extreme Learning Machine
31	SVR	Support Vector Regression
32	GRU	Gated Recurrent Unit
33	ROC	Receiver Operating Characteristic
34	AUC	Area Under the ROC Curve
35	MCC	Matthews Correlation Coefficient
36	MAE	Mean Absolute Error
37	SE / DE	Static / Dynamic Eccentricity

Chapter 1

INTRODUCTION

1.1 PREFACE

PMSMs are widely adopted in high-performance drives because of high efficiency, power density, torque capability, and dynamic response. They are used in electric vehicles, industrial automation, aerospace, robotics, renewable energy, and precision motion control. RFPMSMs are well established because of their mature design and robust construction, while AFPMSMs offer compact axial dimensions, high torque density, and low weight. PMSMs are nevertheless susceptible to bearing defects, shaft misalignment, rotor imbalance, and eccentricity. Eccentricity produces a non-uniform air gap, unbalanced magnetic pull, vibration, torque ripple, noise, and additional losses. Early diagnosis is challenging because fault signatures may be weak and masked by inverter harmonics and changing load conditions. Therefore, this project combines ANSYS Maxwell FEA, signal processing, and AI/ML for intelligent eccentricity diagnosis.

RFPMSMs are widely used because of their established design and manufacturing methods, while AFPMSMs have gained interest for compact, high-torque-density applications such as electric vehicles. Both topologies offer high efficiency and power density, but their electromagnetic behaviour differs because of geometry, winding arrangement, and flux path. PMSMs can experience electrical and mechanical faults, including winding faults, bearing defects, shaft misalignment, rotor imbalance, and eccentricity. These faults reduce efficiency and reliability and can cause progressive damage.

Eccentricity is particularly critical because it creates a non-uniform air gap, producing unbalanced magnetic forces, vibration, torque ripple, acoustic noise, additional losses, and reduced machine performance. If eccentricity remains undetected, the resulting mechanical and electromagnetic stresses can cause insulation degradation, bearing or shaft damage, and eventual motor failure. The increasing use of PMSMs in safety-critical systems creates a need for intelligent condition monitoring and predictive maintenance capable of detecting weak fault signatures under varying operating conditions.AI and ML provide an alternative by learning relationships between motor signals and fault conditions, enabling automated fault identification, classification, and severity estimation. Accordingly, this project integrates ANSYS Maxwell FEA with MCSA, signal processing, and AI/ML to diagnose eccentricity in AFPMSMs and RFPMSMs and support predictive maintenance.

1.2 PERMANENT MAGNET SYNCHRONOUS MOTORS (PMSMs): ADVANTAGES, LIMITATIONS, AND APPLICATIONS

PMSMs are highly efficient machines because permanent magnets eliminate rotor copper losses. They provide high power density, torque-to-weight ratio, rapid response, precise speed control, and low maintenance because they have no brushes. RFPMSMs are mature and robust, whereas AFPMSMs provide higher torque density, shorter axial length, lower weight, and suitability for compact high-performance applications. The absence of brushes reduces mechanical wear and maintenance requirements.

RFPMSMs have long been an industrial standard because of their established design and cost-effective manufacturing. AFPMSMs are attractive for electric vehicles, aerospace, robotics, and other compact applications because of their torque density, reduced axial dimensions, and low weight. PMSMs also have limitations: permanent magnets can increase cost, inverter-fed operation requires

advanced control and position sensing, and high temperature can weaken magnets and reduce performance.

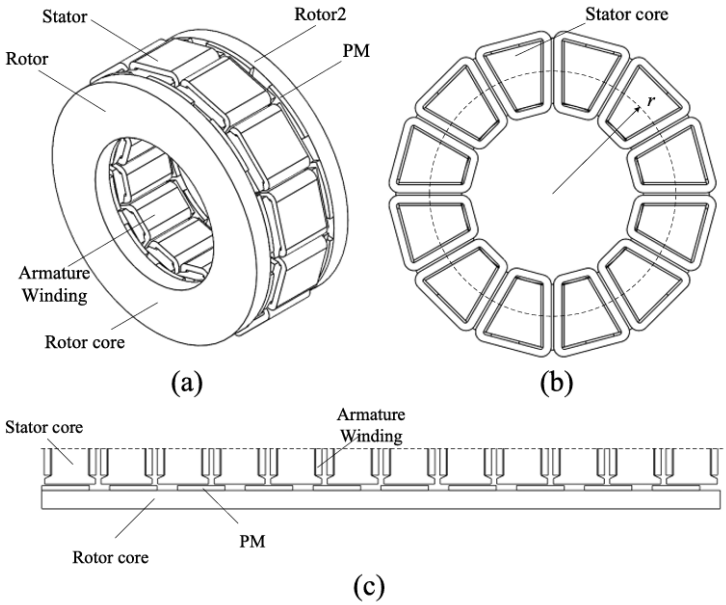


Fig.1.1 AFPMSM Dimensionality reduction schematic representation. (a) 3-D Modelling, (b) Cross-Sectional View, (c) Equivalent 2-D Modelling

Fig.1.1 shows a single-rotor, single-stator AFPMSM with permanent magnets on two rotor discs and a central stator. Its equivalent 2-D model simplifies the 3-D geometry for electromagnetic analysis while retaining the compact, high-torque-density configuration. PMSMs are widely used in EV traction, aerospace actuators, automation, robotics, renewable generation, railways, medical equipment, and precision manufacturing. These applications require reliable fault diagnosis to maintain safe and efficient operation.

1.3 OBJECTIVE

The main objective is to develop an AI/ML-based system for detecting and assessing eccentricity faults in AFPMSMs and RFPMSMs. Healthy operating characteristics are first established using FEA, including air-gap flux density, stator current, Back-EMF, torque, torque ripple, efficiency, and magnetic-field distribution. Static, dynamic, and mixed eccentricity conditions are then introduced to study their effects on both motor topologies. The effects on stator current, torque, flux linkage, vibration-related characteristics, harmonics, and magnetic-field distribution are evaluated, with emphasis on the relationship between fault severity and performance degradation. ANSYS Maxwell is used to generate healthy and faulty datasets through FEM simulations.

FFT and other signal-processing methods are used to analyse harmonics and extract fault-sensitive features for AI/ML models. Different AI/ML models are evaluated using suitable performance metrics for fault classification and severity estimation. AFPMSM and RFPMSM are compared in terms of eccentricity sensitivity, diagnostic accuracy, and suitability for predictive maintenance. The overall goal is early fault detection, severity assessment, reduced unexpected failures and maintenance cost, improved reliability, and safer, more efficient EV drive operation.

Chapter 2

LITERATURE REVIEW

2.1 INTRODUCTION

This chapter reviews research on PMSM design, eccentricity faults, FEM modelling, signal processing, AI/ML, deep learning, and predictive maintenance, with emphasis on AFPMSMs and RFPMSMs. The review establishes the basis for the proposed ANSYS Maxwell and AI-assisted diagnostic framework.

2.2 BASED ON PMSM DESIGN AND PERFORMANCE COMPARISON

PMSMs are studied extensively because of their efficiency, torque density, compactness, and dynamic performance. Research compares axial- and radial-flux topologies and optimizes their electromagnetic and structural characteristics for EV, aerospace, and industrial applications. Yang Lu et al. [1] proposed an axial-radial hybrid PMSM with optimized pole arc, skew, and phase-shift angles, achieving higher torque density and reduced cogging torque. Lin et al. [2] compared radial- and axial-flux PMSMs for aircraft propulsion under thermal and structural constraints. Eker et al. [4] experimentally compared radial- and axial-flux line-start PMSMs, while Khajuee-Zadeh and Tootoonchian [10] reported superior fault tolerance for axial-flux resolver structures. These studies show that both topologies have distinct performance advantages and motivate comparative fault-diagnosis studies.

2.3 BASED ON ECCENTRICITY FAULT ANALYSIS IN PMSMS

Rotor eccentricity produces non-uniform air gaps, unbalanced magnetic pull, vibration, torque ripple, acoustic noise, and performance degradation. Deng et al. [3] showed that static and dynamic eccentricity alter electromagnetic force harmonics and vibration. Du et al. [5] linked eccentricity-induced electromagnetic forces to vibration and acoustic behaviour. Khalesidoost et al. [6] studied displacement, tilt, and axial displacement in dual-rotor axial-flux PMSMs and observed changes in Back-EMF, force, and torque. Tiemblo et al. [7] showed that eccentricity also affects magnetization trajectories and flux manipulation.These studies establish eccentricity as a major mechanical fault in PMSMs.

2.4 BASED ON FEM-BASED ECCENTRICITY MODELLING

FEM accurately represents nonlinear magnetic materials, complex geometry, slotting, and air-gap irregularities, making it suitable for eccentricity analysis. Li et al. [8] combined electromagnetic analysis with diagnosis of demagnetization and eccentricity. Mortezaei et al. [9] combined analytical prediction, FEM, and experimental stray-flux sensing for non-invasive eccentricity detection. Wang et al. [11] used air-gap field modulation and FEM to show significant torque harmonics under static eccentricity. Jia and Lin [18] developed analytical and FEM models for asymmetric eccentricity in axial-flux PMSMs and validated changes in air-gap flux density and Back-EMF. Other studies investigated electromagnetic vibration, axial-force compensation, winding design, and vibration suppression [19], [20], [28], [31], [32]. Overall, FEM is a reliable method for modelling eccentricity-related electromagnetic behaviour.

2.5 BASED ON SIGNAL PROCESSING-BASED FAULT DIAGNOSIS

Signal-processing methods provide non-invasive fault information from electrical and magnetic signals. CSA, harmonic analysis, vibration monitoring, and stray-flux measurements are widely used for eccentricity diagnosis. Mortezaei et al. [9] demonstrated reliable eccentricity indicators from stray magnetic flux, while Du et al. [24] used Magnetic Vector Spectrum Analysis with Random Forest to distinguish rotor faults. Wu et al. [25], Sung et al. [29], and Kumar et al. [30] showed that harmonic features from stator current can identify eccentricity and other PMSM faults, providing useful inputs for AI diagnosis.

2.6 BASED ON AI-ML FAULT DIAGNOSIS

AI can learn complex nonlinear relationships between motor signals and fault conditions, enabling automated diagnosis. Shejwalkar et al. [12] compared ML algorithms for eccentricity classification and reported better performance from Random Forest than Decision Tree, SVM, and XGBoost. Krichen et al. [14] showed that combining signal processing with ML improves dynamic-eccentricity diagnosis while maintaining relatively low computational complexity

2.7 BASED ON DL FAULT DIAGNOSIS

Deep Learning can automatically learn representative features from motor signals, reducing dependence on manual feature extraction. Yusuf et al. [13] proposed an HCNN for fault classification and severity estimation with accuracy approaching 100%. Chu et al. [15] developed a CNN-LSTM model for static, dynamic, and mixed eccentricity with approximately 98.86% accuracy. Xu et al. [16] improved diagnosis using CNN-LSTM-Attention to capture spatial and temporal information. Other end-to-end and neural-network-based approaches [17], [22], [27] support classification, severity estimation, and intelligent monitoring. The reviewed studies indicate that Deep Learning can provide high classification accuracy and strong generalization.

2.8 RESEARCH GAP

Research has advanced in PMSM optimization, eccentricity analysis, FEM, signal processing, and AI-based diagnosis, but important gaps remain. Most studies examine AFPMSMs or RFPMSMs separately, with limited comparative diagnosis under identical operating conditions. Few studies integrate ANSYS Maxwell datasets with ML for eccentricity severity estimation, and many methods require additional sensors or hardware, increasing complexity and cost. These gaps motivate an integrated framework combining FEA, feature extraction, ML, and comparative evaluation for both PMSM topologies.

2.9 CONCLUSION

The literature confirms that eccentricity affects air-gap flux, electromagnetic force, Back-EMF, torque ripple, vibration, and noise. FEM provides accurate electromagnetic modelling, while signal processing and AI/ML support non-invasive fault detection, classification, and severity estimation. ML methods such as SVM, Decision Tree, Random Forest, and XGBoost provide efficient classification, while CNN, CNN-LSTM, HCNN, and CNN-LSTM-Attention offer strong deep-learning performance. Comparative studies of axial- and radial-flux PMSMs under identical conditions and integrated ANSYS Maxwell-to-AI severity estimation remain limited.

Chapter 3

FAULTS AND DIAGNOSIS

3.1 INTRODUCTION

PMSMs are widely used in EVs, aerospace, automation, robotics, and renewable applications because of their efficiency and controllability. RFPMSMs are common in conventional drives, while AFPMSMs offer high torque density and compactness. Both are susceptible to electrical, mechanical, and magnetic faults caused by manufacturing defects, assembly errors, overload, ageing, and wear.

Undetected faults reduce efficiency, increase losses, vibration, and noise, and may lead to failure. Rotor eccentricity is especially important because its non-uniform air gap produces unbalanced magnetic pull, torque ripple, vibration, noise, and additional losses. Eccentricity usually develops gradually, so early detection is important for reliability, safety, and maintenance cost reduction.

Conventional diagnosis can miss incipient eccentricity because signatures are weak and affected by inverter harmonics and operating conditions. FEM models healthy and faulty machines, MCSA and FFT extract fault-related harmonics, and AI/ML classifies fault conditions and estimates severity. This chapter covers PMSM fault types, rotor eccentricity, its effects and diagnostic challenges, and the FEM, MCSA, FFT, and AI methods used in the proposed work.

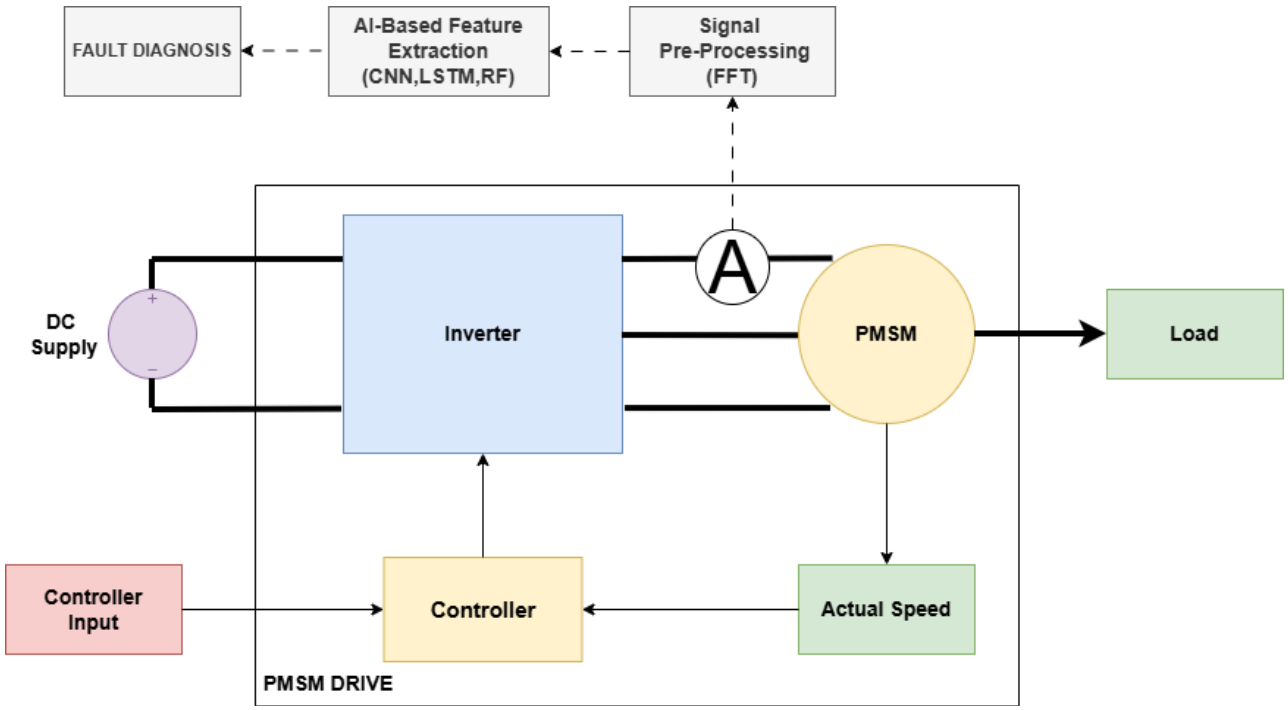


Fig 3.1 Block Diagram of the workflow

Fig. 3.1 shows the block diagram of the proposed AI-based fault diagnosis framework for a PMSM drive. The system has two parts: the PMSM drive, which runs the machine, and the diagnostic chain, which monitors the machine's health. The diagnostic chain works alongside normal operation and does not interfere with the drive control. This is why it is shown with dashed lines, while the power and control paths are shown with solid lines.

PMSM Drive

The drive is powered by a DC supply that feeds a three-phase inverter, which converts the DC voltage into a variable-voltage, variable-frequency supply for the PMSM. The motor shaft is connected to the mechanical load. The controller receives the reference input (the desired speed) and compares it with the actual speed measured from the motor. It uses the error to generate the control signals for the inverter. This closed-loop arrangement keeps the motor speed at the commanded value under different operating conditions.

Signal Acquisition and Pre-Processing

A current sensor, shown as the ammeter symbol, measures the stator current in one phase line between the inverter and the motor. Measuring current is non-intrusive and needs no extra sensors inside the machine. Eccentricity faults distort the air-gap field, so they leave characteristic harmonic components in the stator current. The measured current is passed to the signal pre-processing stage, where the Fast Fourier Transform (FFT) converts it from the time domain to the frequency domain. This makes the fault-related harmonics easier to identify.

AI-based Diagnosis

The frequency-domain data is passed to the AI-based feature extraction and classification stage, which uses methods such as Convolutional Neural Networks (CNN), Long Short-Term Memory (LSTM) networks and Random Forest (RF). These models learn the patterns that separate a healthy machine from one with static or dynamic eccentricity, and they also learn the fault severity. The final block, fault diagnosis, reports the detected condition. Because the current and torque signals show only subtle changes between eccentricity levels, this learning-based approach is well suited to detecting and classifying the faults automatically.

3.2 OCCURRENCE OF DIFFERENT FAULTS IN PMSMs

PMSM faults are broadly electrical, mechanical, or magnetic and depend on operating conditions, manufacturing quality, environment, loading, and maintenance. Bearing, stator-winding, rotor, and shaft-related faults are common; rotor eccentricity is important because it directly changes electromagnetic behaviour.

Bearing faults are among the most common rotating-machine failures, followed by stator and rotor-related faults. Eccentricity is a major research focus because of its direct effect on air-gap fields and machine performance. Bearing faults arise from poor lubrication, fatigue, contamination, excessive loading, misalignment, or improper installation. They increase vibration, noise, friction, and temperature and may contribute to secondary eccentricity and shaft faults. Progressive bearing degradation reduces efficiency and reliability, making early diagnosis important.

Stator Winding Faults

Stator winding faults include insulation degradation, inter-turn and phase faults, and open circuits caused by thermal ageing, overcurrent, voltage surges, manufacturing defects, or mechanical damage.

If undetected, winding faults spread and cause overheating, copper losses, reduced efficiency, and possible winding failure.

Rotor Faults

Rotor faults include demagnetization, magnet cracking, imbalance, deformation, and eccentricity. Eccentricity is particularly important because non-uniform air gaps alter flux distribution and cause unbalanced magnetic pull, torque pulsations, vibration, and noise; severity generally increases these effects.

Shaft and Mechanical Faults

Shaft and coupling faults include bending, cracking, misalignment, and deformation. These can displace the rotor, increase bearing loading, and accelerate mechanical wear. Mechanical deformation can therefore initiate eccentricity and reduce motor service life.

3.3 ROTOR ECCENTRICITY FAULTS IN PMSMs

Rotor eccentricity occurs when the rotor deviates from its concentric position, producing a non-uniform air gap and uneven magnetic-field distribution. The resulting unbalanced magnetic pull increases torque ripple, vibration, noise, losses, and reduces efficiency; severe faults can damage bearings and shafts and cause motor failure.

The distorted field creates unequal electromagnetic forces, or unbalanced magnetic pull, on the rotor. Increasing eccentricity raises torque ripple, vibration, noise, losses, and reduces efficiency; severe conditions may damage bearings, the shaft, and the motor. Rotor eccentricity is classified as static, dynamic, or mixed according to the rotor displacement.

3.3.1 STATIC ECCENTRICITY

Static eccentricity occurs when the rotor axis is displaced from the stator centre while rotating about its own axis; the minimum air gap remains fixed relative to the stator. It can result from manufacturing errors, improper assembly, or bearing installation. The uneven air gap produces asymmetric magnetic forces, torque ripple, and local saturation.

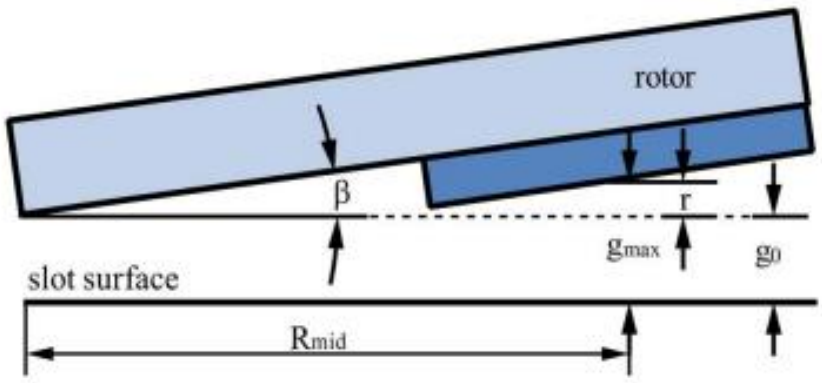


Fig.3.2 Schematic representation of SE in AFPMSMs

Fig.3.2 shows static eccentricity in an AFPMSM caused by rotor inclination. The rotor tilt β creates an axial air-gap variation from minimum g_0 to maximum g_{ax}, producing asymmetric fields and unbalanced forces.

3.3.2 DYNAMIC ECCENTRICITY

Dynamic eccentricity occurs when the rotor rotates about an axis different from its geometric centre, so the minimum air gap changes continuously with rotor position. Typical causes are shaft bending, bearing wear, rotor imbalance, and deformation. The changing air gap produces periodic electromagnetic forces, torque oscillations, and increased vibration.

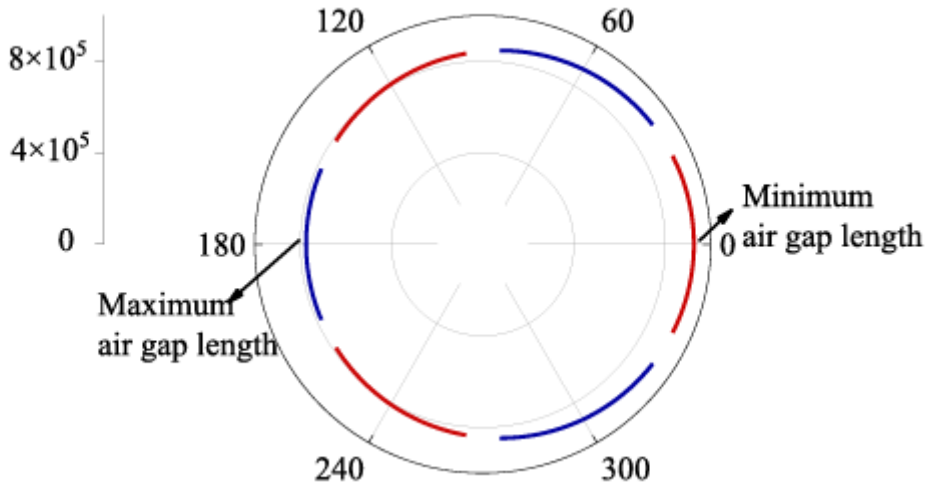


Fig.3.3 Distribution of redistributed magnetization

Fig.3.3 shows redistribution of magnetization caused by air-gap asymmetry. The field is stronger near the minimum gap and weaker near the maximum gap, producing unbalanced pull, torque ripples, and current harmonics.

3.3.3 MIXED ECCENTRICITY

Mixed eccentricity is the simultaneous occurrence of static and dynamic eccentricity and is common in practical machines because manufacturing errors and operating wear can coexist. It produces more complex field distortion and stronger current and torque harmonics, making diagnosis more difficult and increasing the need for advanced signal processing and AI.

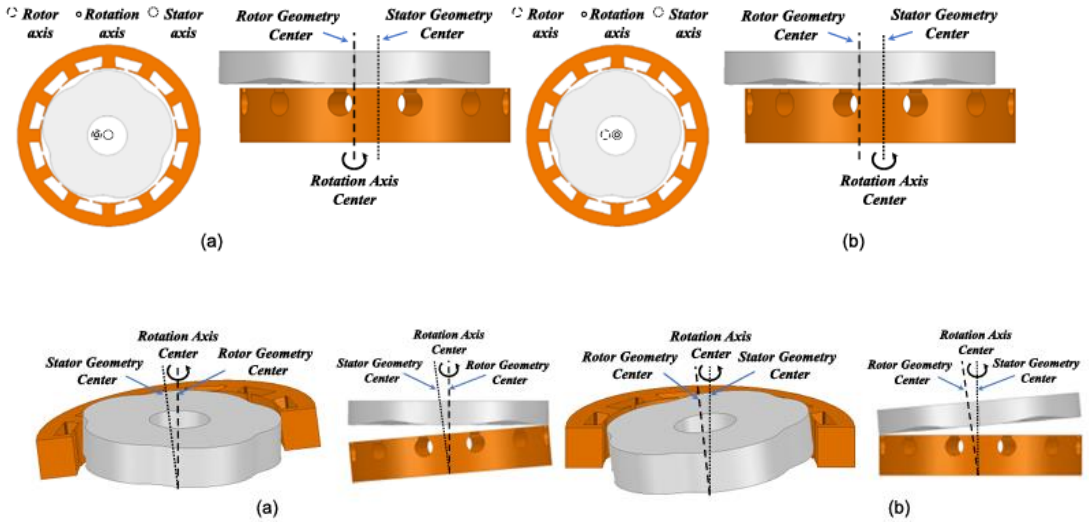


Fig.3.4 Axial and Radial flux resolvers under (a) static and (b) dynamic inclining

Fig.3.4 illustrates healthy, parallel, and inclined eccentricity conditions. Parallel eccentricity produces relatively uniform displacement, while inclined eccentricity causes a varying axial gap and strongly affects AFPMSM flux distribution and torque.

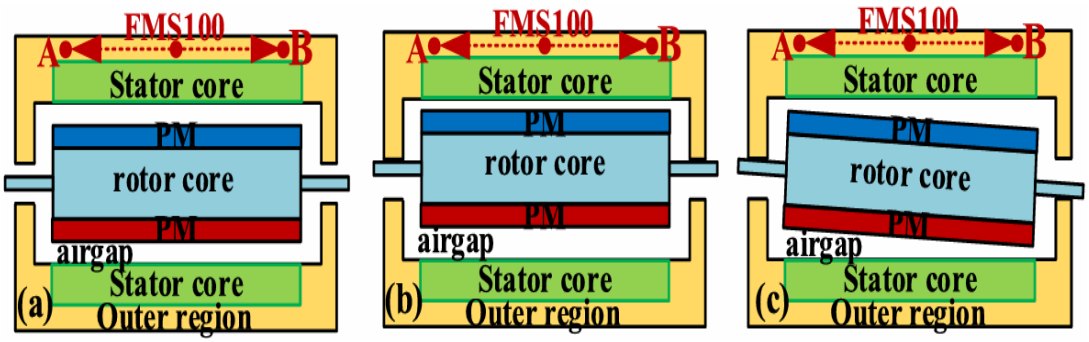


Fig.3.5 Possible Eccentricity cases (a) Healthy, (b) Parallel eccentricity, (c) Inclined eccentricity

Fig.3.5 compares healthy, parallel, and inclined eccentricity. The healthy air gap is uniform; parallel displacement gives a nearly uniform offset, while rotor tilt produces a varying axial gap and increased magnetic asymmetry and torque ripple.

3.4 SIGNIFICANCE AND CHALLENGES IN DIAGNOSING ROTOR ECCENTRICITY FAULTS

Eccentricity develops gradually and can significantly affect PMSM performance. Early diagnosis is important because the fault increases electromagnetic and mechanical stress and may eventually cause failure. However, if left undetected, the fault gradually becomes more severe, resulting in considerable electromagnetic and mechanical degradation. Therefore, early fault diagnosis is essential for improving the reliability, safety, and operational lifespan of PMSMs, especially in electric vehicle applications where uninterrupted operation is highly important.

3.4.1 SIGNIFICANCE OF EARLY ROTOR ECCENTRICITY DIAGNOSIS

- **Unbalanced magnetic pull:** uneven air gaps create unequal magnetic forces, increasing bearing and shaft stress.
- **Torque ripple:** air-gap distortion causes torque fluctuations, reducing speed regulation and dynamic performance.
- **Vibration and noise:** both increase with eccentricity severity and can loosen mechanical components.
- **Back-EMF distortion:** non-uniform fields alter the Back-EMF waveform and can affect inverter performance.
- **Efficiency reduction:** eccentricity increases iron, mechanical, and harmonic losses and therefore energy consumption.
- **Bearing and shaft damage:** continuous unbalanced forces shorten component life and can cause mechanical failure.
- **Unexpected failure and safety risk:** progressive eccentricity can cause costly downtime and, in EVs and other critical systems, compromise operational safety.

3.4.2 CHALLENGES IN DETECTING ROTOR ECCENTRICITY FAULTS

Early diagnosis is difficult because eccentricity signatures are weak, operating conditions vary, symptoms overlap with other faults, and measurement noise and inverter harmonics can mask the fault.

- **Weak initial signatures:** small eccentricity produces only slight changes in stator current, torque, and Back-EMF.
- **Load and speed variation:** eccentricity-related harmonics change with operating point, so the same fault can produce different signatures.
- **Similarity to other faults**: eccentricity can resemble bearing defects, shaft misalignment, and rotor imbalance.
- **PWM switching harmonics:** inverter switching components can mask eccentricity-related harmonics.
- **Measurement noise:** electrical noise, sensor errors, and environmental disturbances reduce signal quality. Advanced diagnosis is therefore required to separate weak fault signatures from normal variations.
- **Requirement for advanced diagnostic algorithms**: Reliable eccentricity diagnosis requires advanced signal processing methods together with intelligent classification algorithms capable of distinguishing between healthy and faulty operating conditions.

3.5 FINITE ELEMENT METHOD (FEM)

3.5.1 OVERVIEW OF FEM

FEM is a numerical method suited to electrical-machine field analysis because it can represent complex geometry, nonlinear magnetic materials, slotting, and air-gap irregularities.

In this work, ANSYS Maxwell models AFPMSMs and RFPMSMs under healthy and different eccentricity conditions to obtain stator current, torque, Back-EMF, flux density, and air-gap field data.

3.5.2 ADVANTAGES OF FEM

FEM provides accurate field computation, complex-geometry modelling, nonlinear material representation, air-gap variation modelling, torque and Back-EMF prediction, healthy/faulty simulation, and datasets for AI/ML.

- Accurate electromagnetic field computation.
- Capability to model complex motor geometries.
- Incorporation of nonlinear magnetic material characteristics.
- Precise representation of air-gap variations caused by eccentricity.
- Accurate prediction of torque ripple and Back-EMF.
- Simulation of healthy and faulty operating conditions.
- Generation of datasets for Artificial Intelligence and Machine Learning applications.

These capabilities make FEM effective for studying eccentricity effects on PMSM performance.

3.5.3 FEM ANALYSIS PROCEDURE

The FEM procedure consists of geometry modelling, material assignment, meshing, boundary conditions, healthy simulation, fault modelling, transient analysis, and post-processing.

- **Geometry modelling:** AFPMSM and RFPMSM geometries are created in ANSYS Maxwell according to the selected specifications.
- **Material assignment:** stator, rotor, permanent-magnet, and air-region properties are assigned, including nonlinear B-H characteristics where required.
- **Mesh generation:** the geometry is discretized into finite elements, with finer meshing in the air gap to improve accuracy.
- **Boundary conditions:** electrical excitation, rotation, and magnetic conditions are defined to represent operation.
- **Healthy simulation:** transient electromagnetic analysis establishes the baseline characteristics.
- **Fault modelling:** static, dynamic, and mixed eccentricity are introduced by changing rotor position and air-gap distribution.
- **Transient analysis:** the faulty motor is simulated under the selected operating conditions.
- **Post-processing**: stator current, torque, Back-EMF, flux density, and air-gap fields are extracted for signal processing and AI diagnosis.

3.6 MOTOR CURRENT SIGNATURE ANALYSIS (MCSA)

MCSA is a non-invasive fault-diagnosis method that uses stator current already available in the drive. Eccentricity changes the air-gap field and introduces characteristic current harmonics, enabling fault identification without additional mechanical sensors.

3.6.1 CONVENTIONAL MCSA DETECTION

Conventional MCSA uses the current spectrum, harmonic amplitudes, current imbalance, and sideband frequencies to identify abnormal components.

The principal techniques include:

- Frequency spectrum analysis.
- Harmonic amplitude monitoring.
- Current imbalance analysis.
- Sideband frequency identification.

Although simple and inexpensive, conventional MCSA is less effective for early eccentricity because fault harmonics are weak and may be masked by inverter harmonics and electrical noise.

3.6.2 SIGNAL ENHANCED MCSA

Signal-processing methods are applied before classification to improve the visibility of fault-related components.

- **Fast Fourier transform: FFT** converts the stator-current waveform from time to frequency domain and is used here to identify eccentricity-related harmonics.
- **Short-time Fourier transform:** STFT provides time-frequency information and is useful under variable-speed operation.
- **Wavelet transform:** Wavelet Transform provides multi-resolution analysis and is useful for transient fault signatures.
- **Feature extraction**: Features such as harmonic amplitudes, RMS current, THD, torque ripple, and Back-EMF harmonics are extracted for AI classification.

Combining MCSA with signal processing improves incipient eccentricity detection while remaining non-invasive and cost-effective.

3.6.3 SIGNAL ACQUISITION AND PRE-PROCESSING

The three-phase stator currents Ia, Ib and Ic obtained from the PMSM simulations in ANSYS Maxwell are used as the primary input signals for the signal-processing and machine-learning pipeline. The current waveforms contain information regarding the electromagnetic and magnetic changes introduced by eccentricity faults. For each operating condition, the current data are stored in CSV format. Each dataset contains **3001 samples** corresponding to Phase-A, Phase-B, and Phase-C currents. The original sampling frequency of the simulated current signal is **1 MHz.**

Before feature extraction, the raw current signals are subjected to preprocessing to obtain a consistent dataset. Invalid and duplicate samples are removed, and the sampling frequency is reduced from **1 MHz to 10 kHz** for feature extraction. Signal normalization is then performed to reduce the influence of amplitude and offset variations. The same preprocessing procedure is applied to all three phases to maintain consistency between the datasets.

3.6.4 SIGNAL WINDOWING AND TIME-DOMAIN EXTRACTION

After preprocessing, the three-phase current signals are divided into smaller time windows for feature extraction. A window duration of 5 ms is selected at the feature sampling frequency of 10 kHz.

The number of samples contained in each window is calculated as:

$$\mathbf{N = Fs*Tw}$$

where 'Fs' is the sampling frequency and 'Tw' is the window duration.

Therefore, each feature-extraction window contains 50 samples. Approximately 60 windows are generated from each CSV dataset. Since nine CSV files are used for training, approximately 540 training windows are obtained. Time-domain statistical features are extracted from each window to describe the amplitude, distribution, and statistical characteristics of the current waveform. These features provide information about variations in the current signal produced by different eccentricity conditions.

3.6.5 FREQUENCY-DOMAIN AND TIME-FREQUENCY SIGNAL PROCESSING

Signal-processing techniques are applied to the preprocessed current signals to identify characteristics associated with eccentricity. In addition to conventional frequency-domain analysis using Fast Fourier Transform (FFT), Hilbert-Huang Transform (HHT), Short-Time Fourier Transform (STFT), and Discrete Wavelet Transform (DWT) are incorporated to obtain complementary information from the current signals.

3.6.5.1 Fast Fourier Transform (FFT)

Fast Fourier Transform is used to transform the stator current signal from the time domain into the frequency domain. The frequency spectrum provides information about the harmonic components present in the current signal. Eccentricity modifies the air-gap magnetic field and consequently introduces changes in the harmonic components of the stator current. Therefore, the FFT spectrum can be analysed to identify fault-sensitive frequency components.

The discrete Fourier transform can be represented as:

$$X(k) = \sum_{n=0}^{N-1} x(n) e^{-\frac{j2\pi kn}{N}}$$

where x(n) represents the sampled current signal, N is the number of samples, and X(k) represents the corresponding frequency-domain component. FFT-based characteristics are subsequently used as features for machine-learning classification and severity estimation.

3.6.5.2 Hilbert-Huang Transform (HHT)

The Hilbert-Huang Transform is used to analyse non-stationary characteristics of the current signal. Unlike conventional Fourier-based analysis, HHT provides instantaneous frequency information for signal components that vary with time.

The current signal is first decomposed into Intrinsic Mode Functions (IMFs) using Empirical Mode Decomposition (EMD). The dominant IMFs are then analysed to extract fault-sensitive characteristics.

The features extracted from the first and second dominant IMFs include:

- IMF RMS
- IMF energy
- Maximum amplitude
- Mean instantaneous frequency
- Median instantaneous frequency
- Maximum instantaneous frequency
- Energy entropy
- Marginal weighted frequency

These features provide additional information regarding the temporal and frequency characteristics of the current signal and complement the features obtained using FFT.

3.6.5.3 Short-Time Fourier Transform (STFT)

Short-Time Fourier Transform is used to analyse the time-varying frequency characteristics of the current signal. The signal is divided into short overlapping segments, and frequency-domain analysis is performed for each segment.

The extracted STFT features include:

- Mean magnitude
- Maximum magnitude
- Magnitude variation
- Total energy
- Dominant frequency

STFT is useful for identifying frequency components whose characteristics change with time and therefore provides information that may not be fully represented by a conventional FFT spectrum.

3.6.5.4 Discrete Wavelet Transform (DWT)

Discrete Wavelet Transform provides localized time-frequency analysis of the current signal. It is particularly useful for identifying transient and localized variations that may not be clearly represented in a global frequency spectrum. In the proposed signal-processing procedure, the Daubechies-4 (db4) wavelet is used and two levels of decomposition are performed. The decomposition generates approximation and detail coefficients, from which statistical and energy-related characteristics are extracted.

The DWT therefore provides multi-resolution information regarding the current signal and complements the frequency-domain information obtained from FFT and STFT.

3.6.6 Multi-Phase Feature Construction

Feature extraction is performed independently for the three stator current phases Ia, Ib, and Ic. A total of 111 features are extracted from each phase.

Therefore, the number of phase-wise features is:

111*3 = 333 features

In addition to these individual phase features, five cross-phase features are included to capture relationships between the three phase currents. Hence, the total number of features obtained from each window is:

333+5=338 features

The use of all three phase currents enables the machine-learning model to capture both individual phase characteristics and three-phase asymmetry associated with eccentricity faults.

3.7 ARTIFICIAL INTELLIGENCE-BASED FAULT DIAGNOSIS

AI improves fault diagnosis by learning complex relationships between motor signals and fault conditions, reducing dependence on manual interpretation. In the proposed work, ANSYS Maxwell data are processed to extract fault-sensitive features, which are supplied to AI models for fault classification and severity estimation.

3.7.1 MACHINE LEARNING-BASED DIAGNOSIS

Machine Learning (ML) techniques are used to identify the operating condition of the PMSM based on features extracted from the stator current signals. In the proposed diagnosis system, the extracted features are used to classify the motor condition into **Healthy, Static Eccentricity (SE), and Dynamic Eccentricity (DE)** conditions. Multiple supervised ML classification algorithms are evaluated to determine their ability to distinguish between the different motor conditions.

The following Machine Learning algorithms are used in the proposed system:

Random Forest (RF)

Random Forest is an ensemble learning algorithm that combines multiple Decision Trees to perform classification. Each tree produces a classification decision, and the combined output of the trees provides the final prediction. RF can handle nonlinear relationships and interactions between extracted signal features and is suitable for classification of different PMSM operating conditions.

Extra Trees (ET)

Extra Trees, or Extremely Randomized Trees, is an ensemble tree-based classification algorithm that introduces additional randomization during tree construction. It combines the predictions of multiple randomized Decision Trees to improve classification robustness. The algorithm is capable of handling nonlinear feature relationships and high-dimensional feature sets.

Support Vector Machine (SVM)

Support Vector Machine is a supervised learning algorithm that separates different fault classes by constructing a decision boundary in the feature space. SVM is suitable for classification problems where the classes may have complex or nonlinear boundaries.

K-Nearest Neighbours (KNN)

K-Nearest Neighbours classifies an unknown sample based on the classes of its nearest neighbouring samples in the feature space. The method uses the similarity between feature vectors to identify the operating condition of the motor and can be applied to distinguish different eccentricity conditions.

Gradient Boosting (GB)

Gradient Boosting is an ensemble learning technique in which multiple weak decision models are sequentially combined to improve classification performance. Each successive model attempts to reduce the errors produced by the previous models. It can learn nonlinear relationships within the extracted motor-condition features.

Extreme Gradient Boosting (XGBoost)

XGBoost is an optimized gradient-boosting algorithm that builds a sequence of decision trees to improve prediction accuracy. It can effectively model complex relationships between the extracted signal features and the corresponding PMSM fault conditions.

Light Gradient Boosting Machine (LightGBM)

LightGBM is a gradient-boosting framework based on decision trees. It is designed for efficient training while maintaining good predictive performance. The algorithm is used to classify the PMSM operating conditions based on the selected signal features.

Categorical Boosting (CatBoost)

CatBoost is a gradient-boosting algorithm that combines multiple decision trees to perform classification. It provides a robust approach for learning nonlinear relationships within the feature dataset and is used for distinguishing between the different PMSM operating conditions.

3.7.2 Dataset Generation

The dataset used for machine-learning analysis is generated from the PMSM simulations performed in ANSYS Maxwell. The dataset consists of healthy, static eccentricity, and dynamic eccentricity operating conditions.

The healthy condition corresponds to 0% eccentricity, while static and dynamic eccentricity conditions are simulated at 10%, 20%, 40%, and 60% eccentricity levels. The training dataset therefore consists of nine CSV files corresponding to the selected healthy and faulty conditions. The windowing process generates approximately 60 windows from each CSV file, resulting in a total of 540 training windows. Each window initially contains 338 extracted features obtained from the three-phase current signals. The dataset is subsequently subjected to feature-selection procedures before being supplied to the machine-learning models.

3.7.3 Feature Selection

The initial feature set consists of 338 features for each current-signal window. Since some features may contain redundant information or exhibit very little variation, feature-selection techniques are applied before machine-learning classification.

Variance filtering is used to remove features exhibiting insufficient variation, while correlation filtering is used to reduce highly correlated or redundant features. Thus, 39 selected features are retained and supplied to the classification models. This reduction decreases feature redundancy while retaining the information required for identifying eccentricity conditions. The overall machine-learning procedure consists of signal acquisition, preprocessing, windowing, feature extraction, feature selection, model training, and validation. The trained models are subsequently evaluated using unseen 15% test data to verify their ability to classify eccentricity conditions and estimate the corresponding severity

3.8 FEM AND MCSA: COMPLEMENTARY ROLES

FEM and MCSA play complementary roles in the proposed eccentricity diagnosis framework. ANSYS Maxwell provides physics-based electromagnetic simulations for healthy and faulty PMSM operating conditions, while MCSA uses the stator current signal to identify fault-related characteristics. The simulated three-phase currents are preprocessed and divided into suitable time windows before applying FFT, HHT, STFT, and DWT-based signal-processing techniques.

The extracted time-domain, frequency-domain, and time-frequency features are combined to form a multi-phase feature set. Initially, 338 features are obtained from each signal window. Variance and correlation-based filtering is then used to reduce the feature set to 39 selected features. These features are supplied to machine-learning models for automatic eccentricity classification and severity estimation.

3.9 CONCLUSION

This chapter presented a comprehensive overview of fault diagnosis in Permanent Magnet Synchronous Motors, with particular emphasis on rotor eccentricity faults. The occurrence of various electrical, mechanical, and magnetic faults affecting PMSMs was first discussed, followed by a detailed explanation of static, dynamic, and mixed eccentricity faults. The significance of early fault diagnosis and the practical challenges associated with eccentricity detection were also presented.

The Finite Element Method was introduced as an effective numerical technique for accurately modelling healthy and faulty motor conditions. Motor Current Signature Analysis, together with Fast Fourier Transform, was discussed as a reliable non-invasive approach for extracting fault-sensitive harmonic information from stator current signals. Furthermore, the application of Artificial Intelligence, including Machine Learning and Deep Learning algorithms, was presented as an intelligent solution for automated fault classification and fault severity estimation.

Finally, an integrated diagnosis framework combining FEM, FFT, MCSA, and Artificial Intelligence was proposed for the diagnosis of eccentricity faults in Axial Flux and Radial Flux Permanent Magnet Synchronous Motors. The concepts presented in this chapter provide the theoretical foundation for

the ANSYS Maxwell modelling, simulation methodology, and AI-based fault diagnosis discussed in the subsequent chapters.

Chapter 4

SIMULATION RESULTS AND DISCUSSION

4.1 INTRODUCTION

This chapter presents the Finite Element Analysis (FEA) results obtained from ANSYS Maxwell for the healthy Axial Flux Permanent Magnet Synchronous Motor (AFPMSM) and Radial Flux Permanent Magnet Synchronous Motor (RFPMSM) models. The stator current, induced voltage, electromagnetic torque, flux linkage, and air-gap flux waveforms obtained under transient simulation are presented for both motor topologies. These baseline healthy-state results establish the reference performance characteristics that will subsequently be compared against faulty eccentric conditions in later stages of this work. The machine parameters of the motors are as follows:

Parameter	Value
Rated Power	0.55 kW
Rated Voltage	220 V
Rated Speed	1500 rpm
Rated Torque	3.51 Nm
Operating Frequency	100 Hz
No. of Poles	8

Table 4.1 Machine Parameters of RFPMSM

Parameter	Value
Rated Power	0.55 kW
Rated Voltage	220 V
Rated Speed	1500 rpm
Rated Torque	3.50167 Nm
Operating Frequency	100 Hz
No. of Poles	8

Table 4.2 Machine Parameters of AFPMSM

4.2 STATOR CURRENT

The three-phase stator current waveforms obtained from the transient simulation are shown below for both motor topologies.

RFPMSM

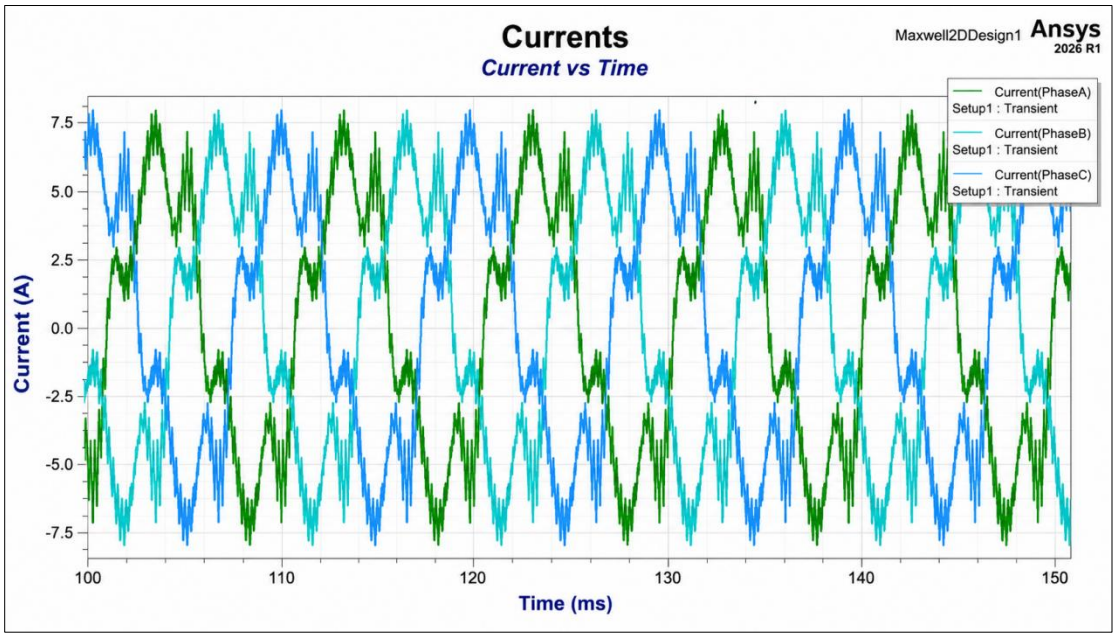


Fig. 4.1 Stator Current of RFPMSM

AFPMSM

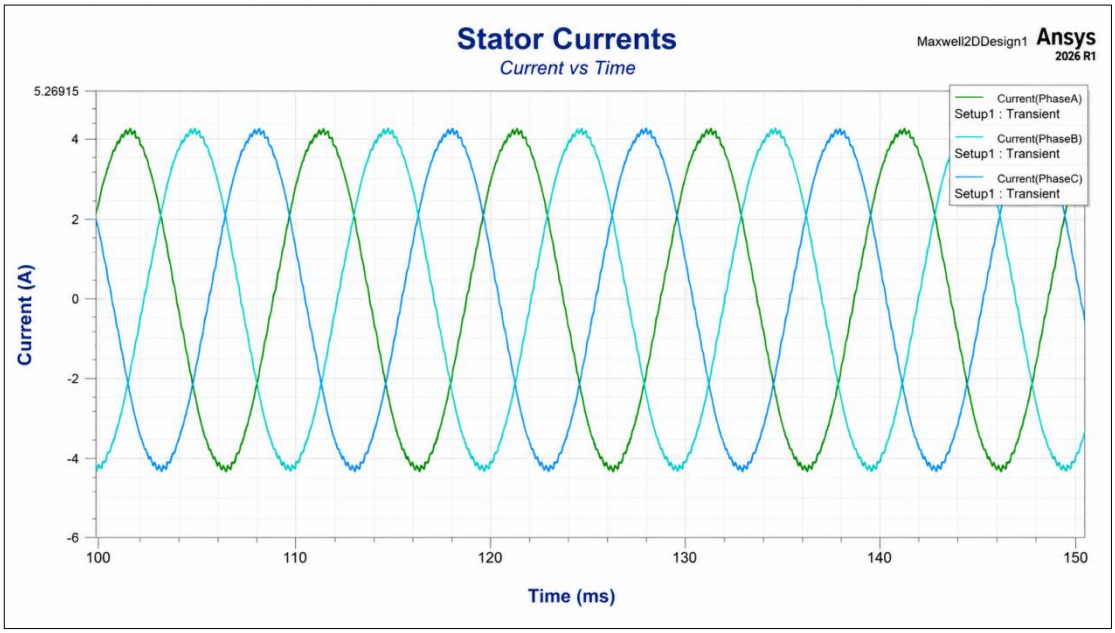


Fig. 4.2 Stator Current of AFPMSM

The AFPMSM exhibits balanced and nearly sinusoidal three-phase stator currents with a peak value of approximately 4 A and relatively low ripple. In comparison, the RFPMSM shows higher instantaneous current peaks of approximately 8 A along with noticeable waveform distortion and harmonic ripple. The difference in current characteristics can be attributed to differences in winding configuration, stator slot geometry, magnetic-field distribution, and machine electromagnetic parameters. Although both machines maintain balanced three-phase operation, the AFPMSM demonstrates a smoother current waveform under the simulated operating conditions.

4.3 INDUCED VOLTAGE (BACK-EMF)

The induced (Back-EMF) voltage waveforms for the three phases are presented below.

RFPMSM

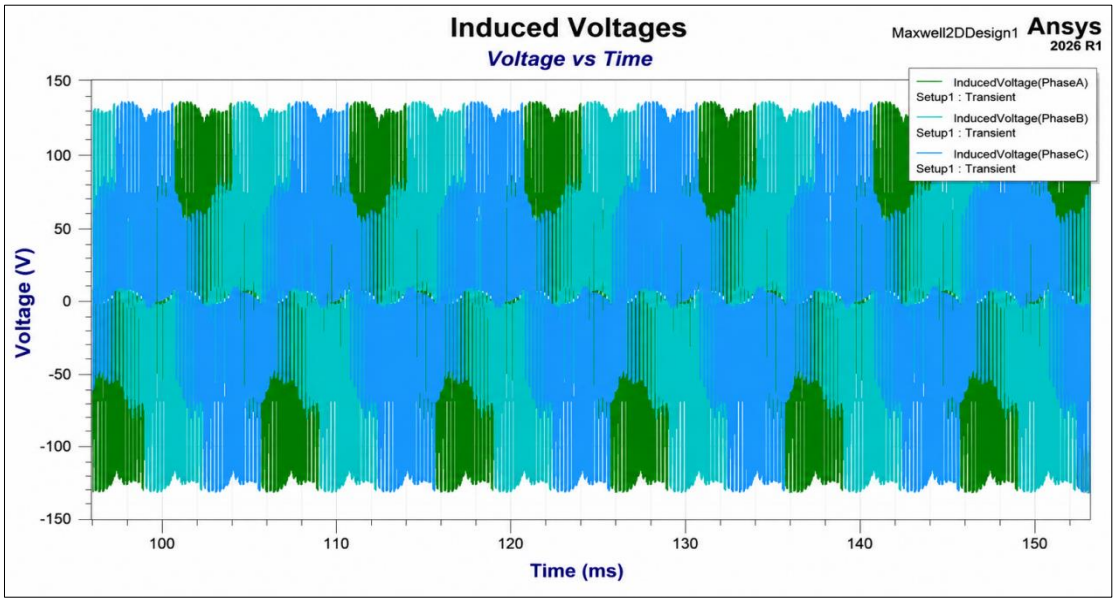


Fig. 4.3 Induced Voltage of RFPMSM

AFPMSM

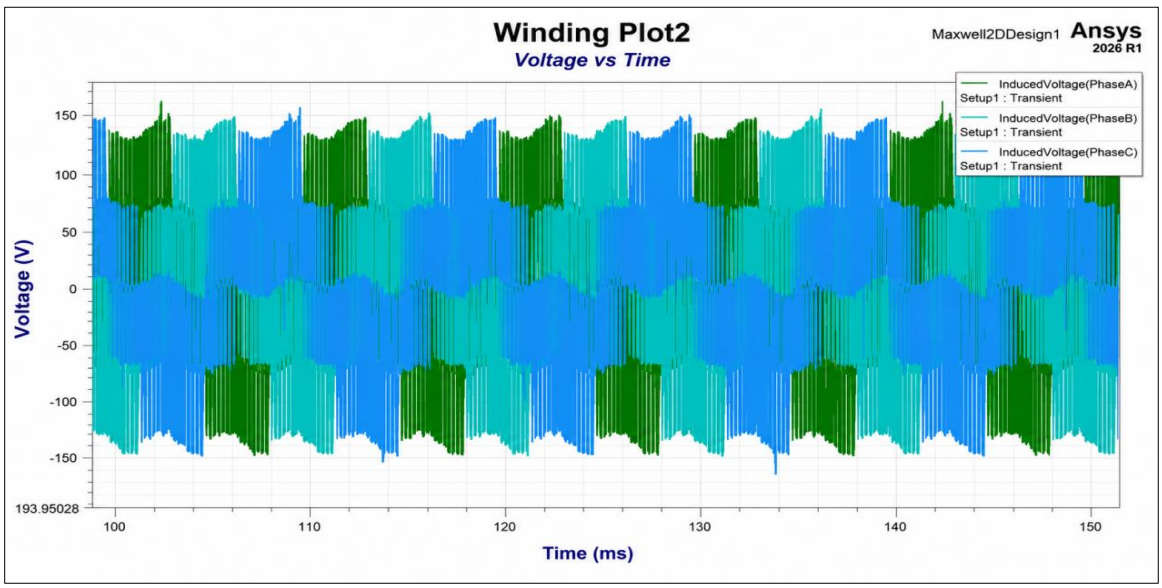


Fig. 4.4 Induced Voltage of AFPMSM

The induced voltage characteristics of the AFPMSM and RFPMSM were evaluated under the same simulated operating conditions to compare their electromagnetic response. Both machines exhibit balanced three-phase voltage waveforms with periodic switching-harmonic components. The AFPMSM produces a marginally higher peak induced voltage than the RFPMSM, while both maintain symmetrical and consistent voltage patterns throughout steady-state operation. The observed difference in voltage magnitude can be associated with variations in winding configuration, effective flux linkage, magnetic field distribution, and machine geometry. Overall, both topologies demonstrate stable three-phase induced voltage generation, with the AFPMSM exhibiting a slightly higher voltage amplitude under the considered operating condition.

4.4 ELECTROMAGNETIC TORQUE

The electromagnetic torque developed by each motor during the transient simulation is shown below.

RFPMSM

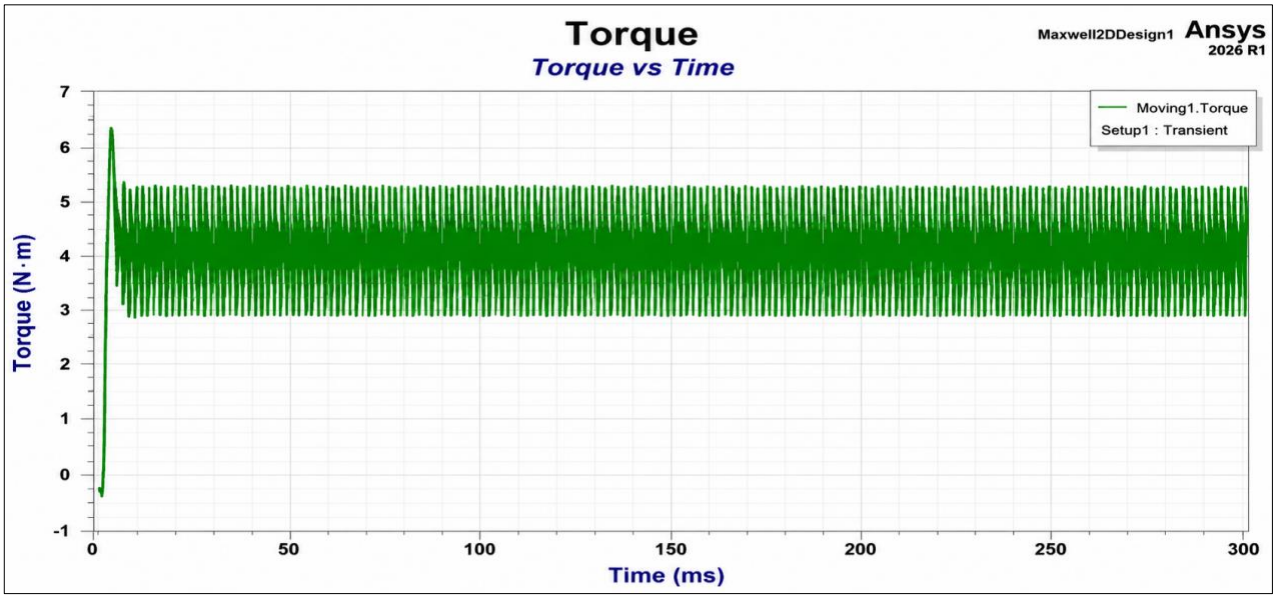


Fig. 4.5 Electromagnetic torque of RFPMSM

AFPMSM

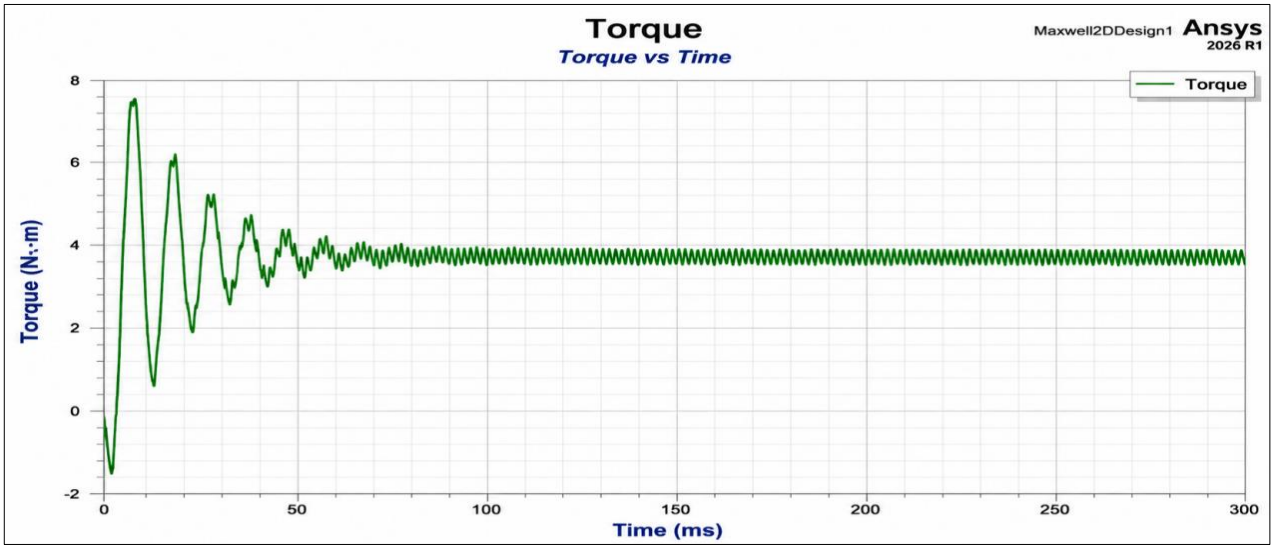


Fig-4.6 Electromagnetic torque of RFPMSM

The electromagnetic torque characteristics of the AFPMSM and RFPMSM show noticeable differences in both transient and steady-state operation. During startup, the AFPMSM exhibits a larger torque overshoot and takes a comparatively longer time to settle, whereas the RFPMSM reaches its steady-state torque more rapidly, indicating a faster transient response. However, after reaching steady state, the AFPMSM produces a significantly smoother torque with lower ripple, while the RFPMSM exhibits continuous torque oscillations around its average value. The higher torque ripple observed in the RFPMSM can be associated with stronger slotting effects, cogging torque, current harmonics, and variations in the air-gap magnetic field. Overall, the AFPMSM demonstrates better steady-state torque smoothness, whereas the RFPMSM provides a quicker torque response during the transient period.

4.5 AIR-GAP FLUX

The air-gap magnetic flux distribution along the circumferential/axial distance of the machine is shown below for both motors.

RFPMSM

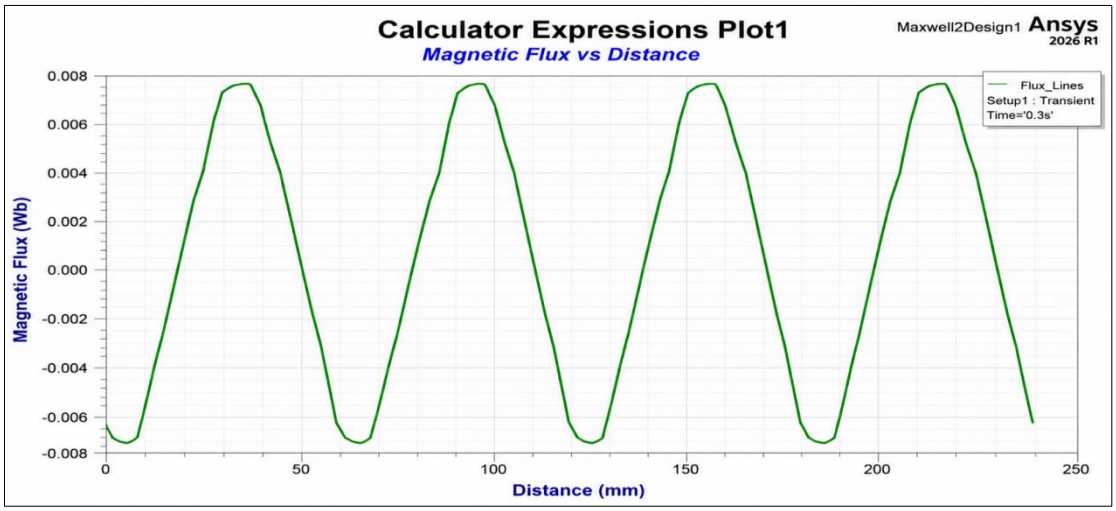


Fig. 4.7 Air-Gap Flux of RFPMSM

AFPMSM

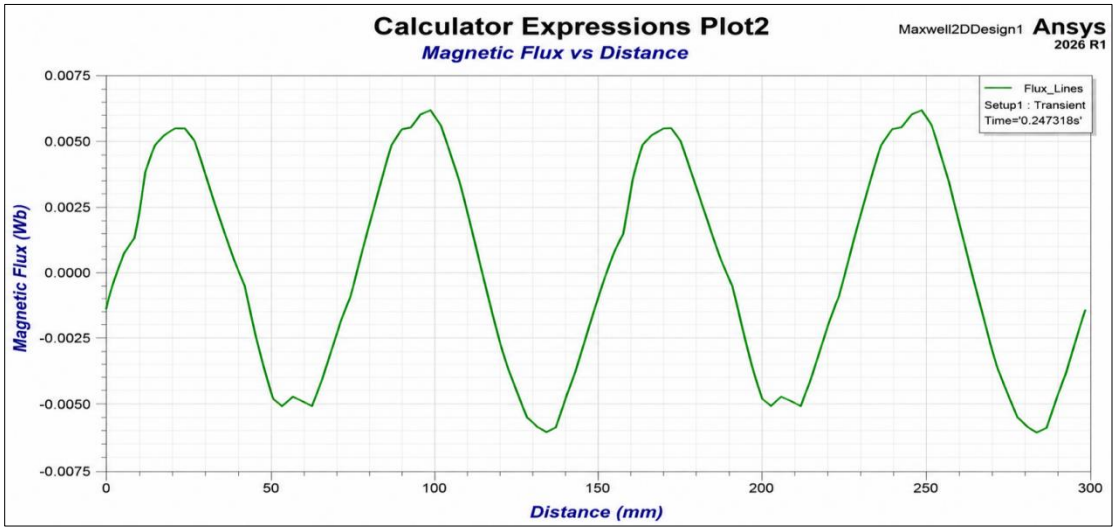


Fig.4.8 Air-Gap Flux of AFPMSM

The air-gap magnetic flux characteristics of the AFPMSM and RFPMSM show a clear difference in both magnitude and spatial distribution. The AFPMSM exhibits a comparatively smoother and more uniform flux variation along the air gap, whereas the RFPMSM shows a higher peak flux with a more pronounced variation in waveform shape. The higher flux magnitude in the RFPMSM indicates greater magnetic loading, while the differences in waveform shape can be associated with the machine geometry, magnet arrangement, and stator slotting effects. In both machines, the flux alternates between positive and negative values due to the reversal of magnetic-field direction between successive north and south rotor poles. Overall, the AFPMSM demonstrates a smoother air-gap flux distribution, while the RFPMSM exhibits a higher peak magnetic flux under the simulated operating conditions.

4.6 FAULT STATIC

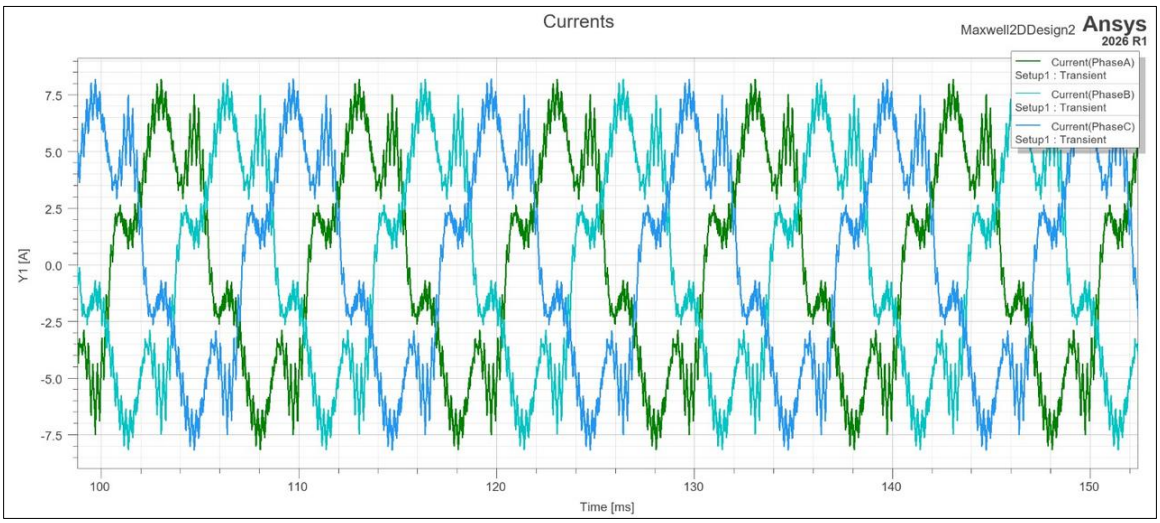


Fig. 4.9 Stator Current of RFPMSM with 10% Static Eccentricity

Fig. 4.9 shows the three-phase stator currents of the RFPMSM under 10% static eccentricity. The currents stay balanced and 120° apart, with peaks of about ±8 A and high-frequency ripple on the fundamental. This mildest static case is the starting point for comparison.

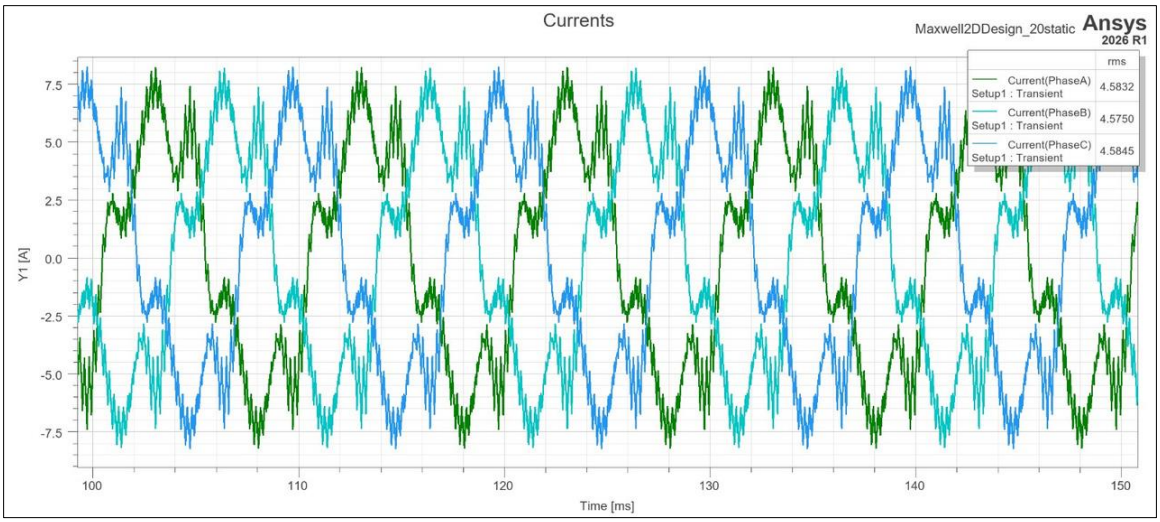


Fig. 4.10 Stator Current of RFPMSM with 20% Static Eccentricity

Fig. 4.10 shows the stator currents under 20% static eccentricity. The waveform shape and the ±8 A peak are essentially unchanged, and the phase RMS values of about 4.575–4.585 A differ only marginally, so the imbalance between phases is very small at this level.

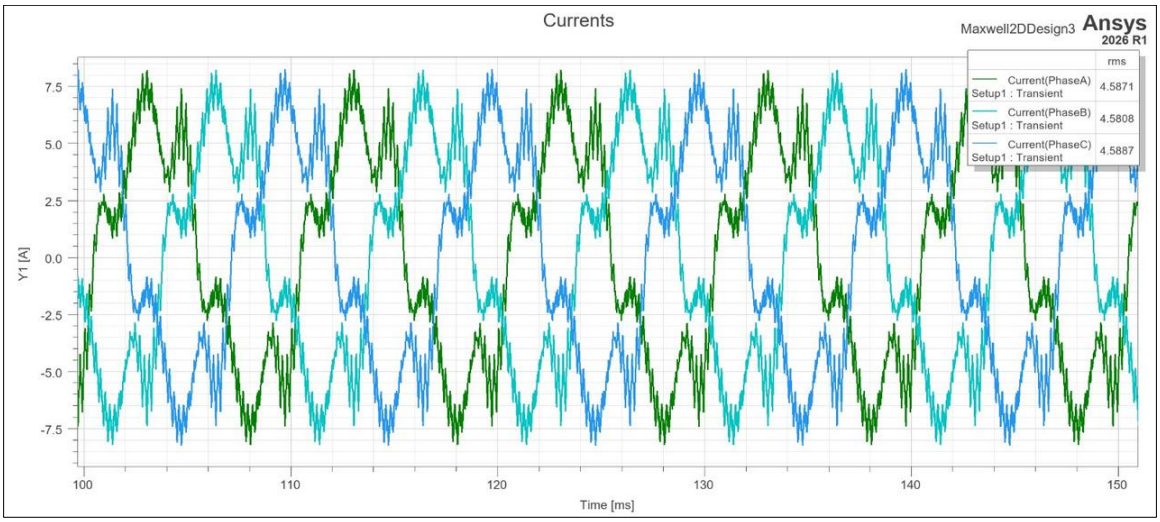


Fig. 4.11 Stator Current of RFPMSM with 40% Static Eccentricity

Fig. 4.11 shows the stator currents under 40% static eccentricity. The phase RMS values rise slightly to about 4.581–4.589 A, while the ±8 A peaks and ripple pattern stay similar, so the fault leaves only a subtle signature in the time-domain current.

Conclusion – Fault Static (10%, 20%, 40%)

The three phase currents stay balanced and 120° apart at about ±8 A for all three levels, with high-frequency ripple on the fundamental. The phase RMS values rise only marginally, from about 4.575–4.585 A at 20% to 4.581–4.589 A at 40%. Static eccentricity therefore leaves only a subtle signature in the time-domain current.

4.7 FAULT DYNAMIC

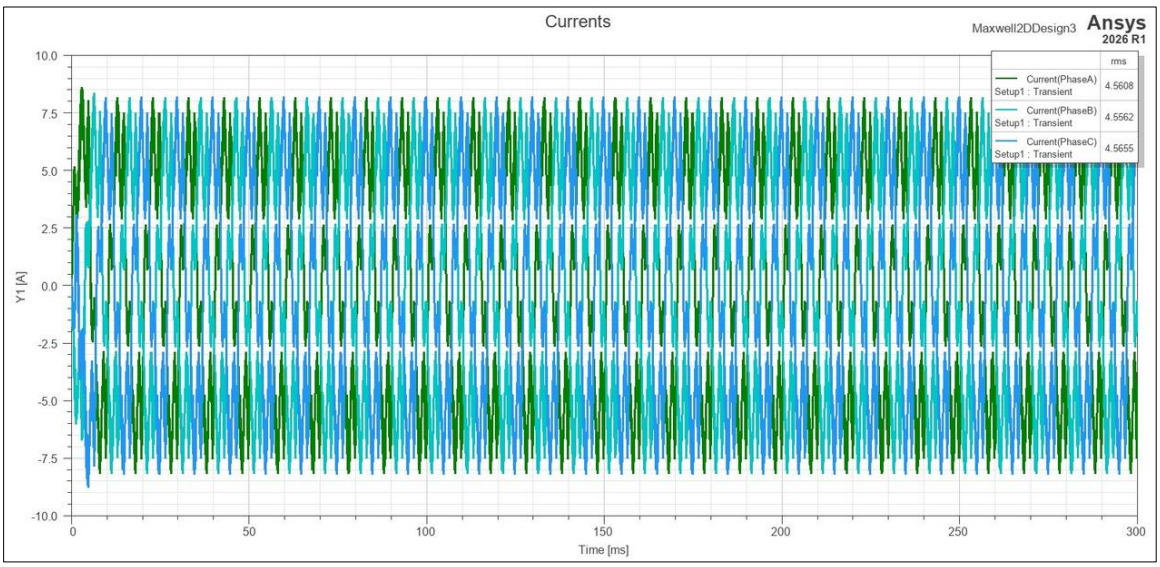


Fig. 4.12 Stator Current of RFPMSM with 10% Dynamic Eccentricity

Fig. 4.12 shows the stator currents under 10% dynamic eccentricity over the full 0–300 ms window. After a brief start-up transient the three phases settle into a balanced pattern of about ±8 A, with phase RMS values of roughly 4.556–4.566 A.

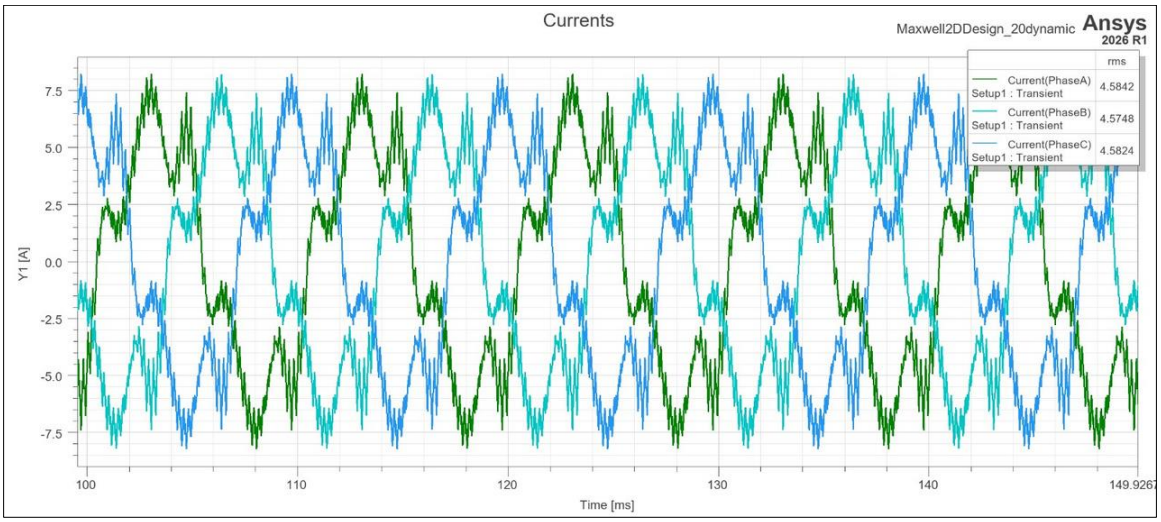


Fig. 4.13 Stator Current of RFPMSM with 20% Dynamic Eccentricity

Fig. 4.13 shows the stator currents under 20% dynamic eccentricity. The phase RMS values increase to about 4.575–4.584 A relative to the 10% case, while the waveforms remain balanced with peaks of about ±8 A.

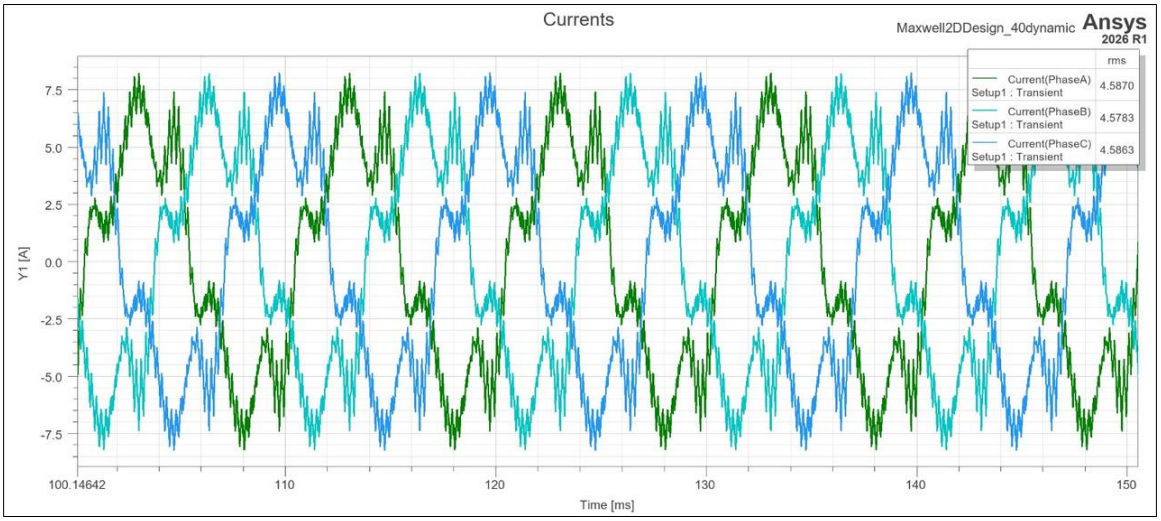


Fig. 4.14 Stator Current of RFPMSM with 40% Dynamic Eccentricity

Fig. 4.14 shows the stator currents under 40% dynamic eccentricity. The phase RMS values reach about 4.578–4.587 A, the highest of the dynamic cases, yet the waveforms remain balanced, so harmonic analysis is needed to bring out the fault signature.

Conclusion – Fault Dynamic (10%, 20%, 40%)

The currents remain balanced at about ±8 A, with phase RMS rising from about 4.556–4.566 A at 10% to 4.575–4.584 A at 20% and 4.578–4.587 A at 40%. The rise with severity is small, so harmonic analysis is needed to bring out the fault signature.

CURRENTS

	HEALTHY	10% STATIC	40% STATIC	10% DYNAMIC	15% STATIC	15% DYNAMIC	20% STATIC	20% DYNAMIC	40% DYNAMIC	60% STATIC	60% DYNAMIC	MIXED
PHASE-A	4.5536	4.5619	4.5871	4.5608	4.5628	4.5603	4.5832	4.5842	4.5870	4.5867	4.5769	4.5631
PHASE-B	4.5496	4.5610	4.5808	4.5562	4.5609	4.5567	4.5750	4.5748	4.5783	4.5878	4.5765	4.5604
PHASE-C	4.5544	4.5677	4.5887	4.5565	4.5669	4.5656	4.5845	4.5824	4.5863	4.5932	4.5791	4.5665

Table 4.3 3-Phase Currents

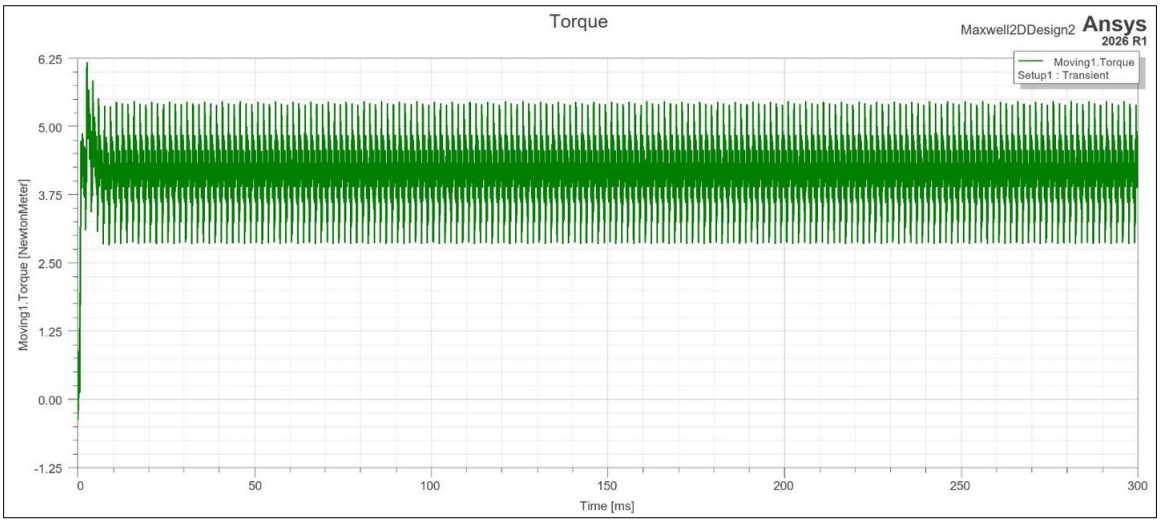


Fig. 4.15 Electromagnetic Torque of RFPMSM with 10% Static Eccentricity

Fig. 4.15 shows the electromagnetic torque under 10% static eccentricity. After a start-up overshoot of about 6.2 N·m, the torque settles into a steady oscillation between roughly 2.8 and 5.5 N·m, which indicates a considerable torque ripple about the mean value.

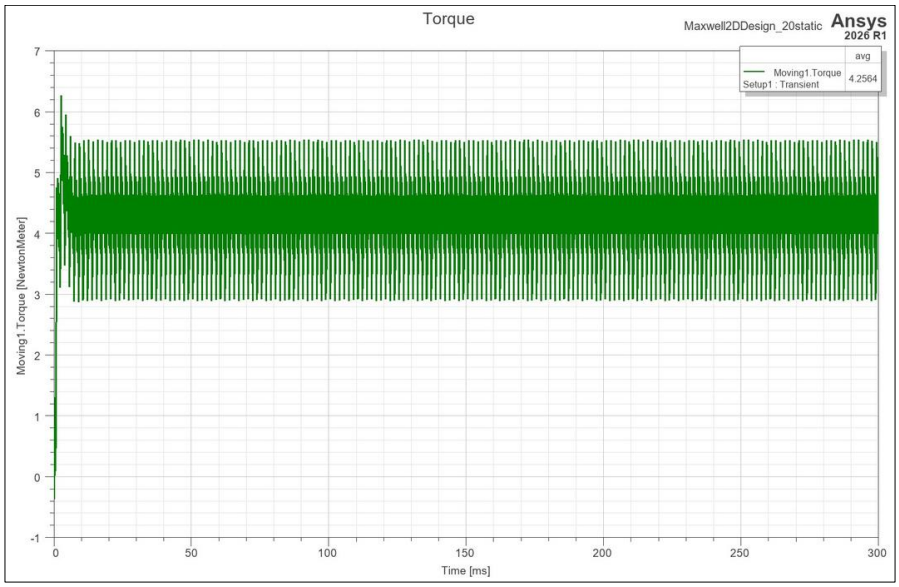


Fig. 4.16 Electromagnetic Torque of RFPMSM with 20% Static Eccentricity

Fig. 4.16 shows the electromagnetic torque under 20% static eccentricity. The average torque is 4.2564 N·m and the ripple band of roughly 2.9–5.5 N·m is similar to the 10% case, so the mean torque is not appreciably affected by the eccentricity level.

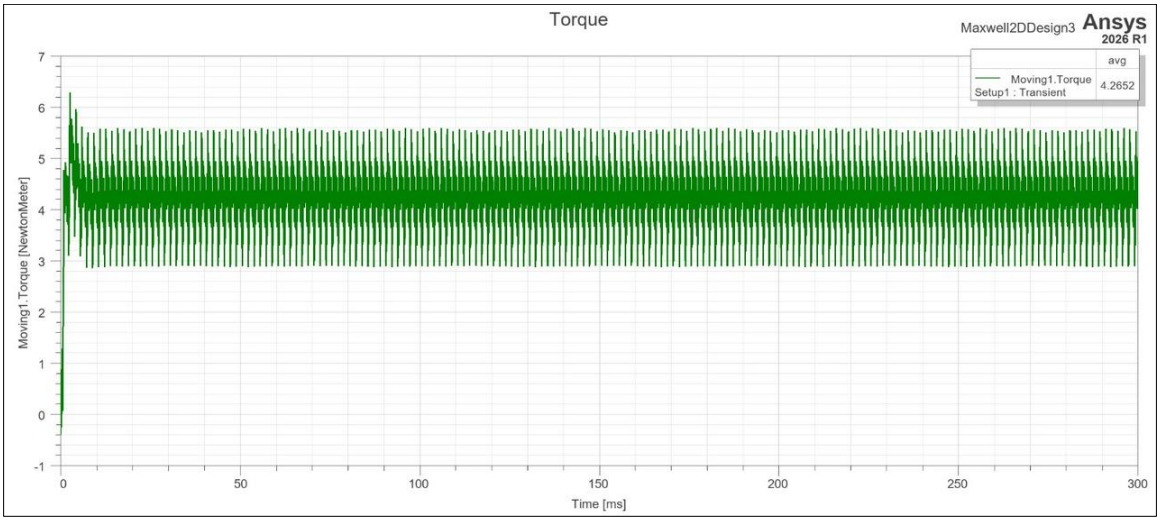


Fig. 4.17 Electromagnetic Torque of RFPMSM with 40% Static Eccentricity

Fig. 4.17 shows the electromagnetic torque under 40% static eccentricity. The average torque is 4.2652 N·m, almost the same as at 20%, and the ripple stays in a similar band, so the mean torque alone is a poor indicator of eccentricity severity.

Conclusion – Fault Static (10%, 20%, 40%)

After a start-up overshoot of about 6.2 N·m, the torque oscillates between roughly 2.8 and 5.5 N·m at every level. The average torque is nearly unchanged (4.2564 N·m at 20% and 4.2652 N·m at 40%). The mean torque is therefore a poor indicator of static eccentricity severity.

FAULT DYNAMIC

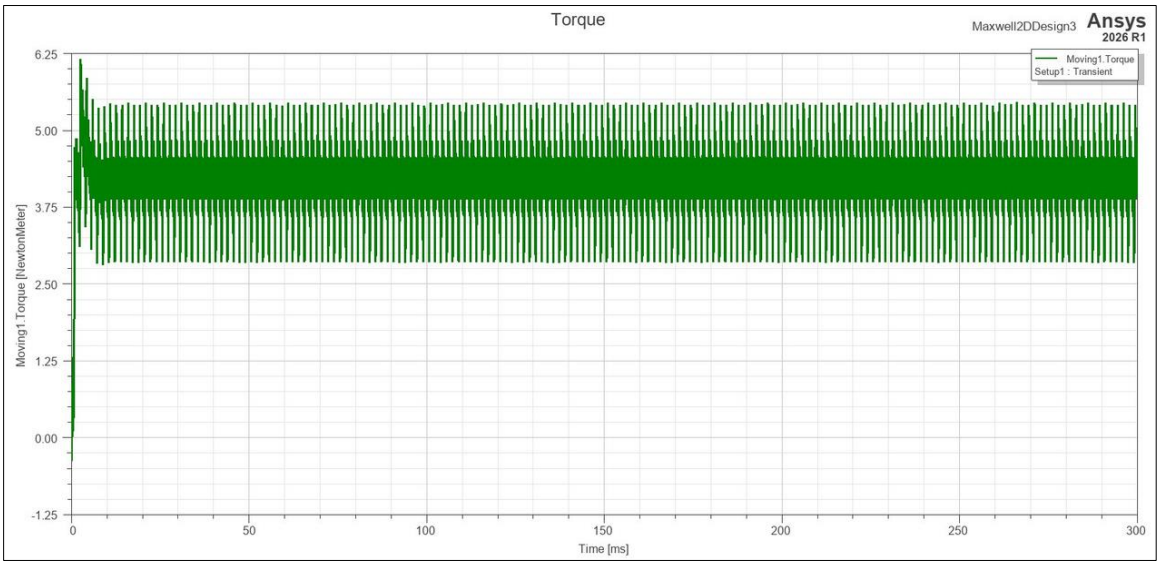


Fig. 4.18 Electromagnetic Torque of RFPMSM with 10% Dynamic Eccentricity

Fig. 4.18 shows the electromagnetic torque under 10% dynamic eccentricity. After the initial overshoot the torque oscillates between roughly 2.8 and 5.5 N·m about a mean near 4.2 N·m, with a uniform ripple envelope throughout the 300 ms simulation.

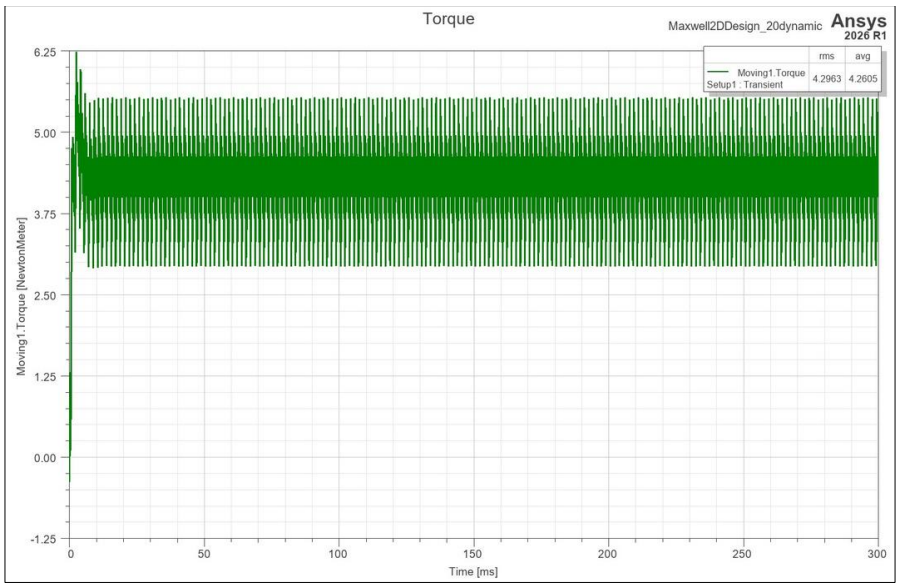


Fig. 4.19 Electromagnetic Torque of RFPMSM with 20% Dynamic Eccentricity

Fig. 4.19 shows the electromagnetic torque under 20% dynamic eccentricity. The average torque is 4.2605 N·m with an RMS value of 4.2963 N·m, and the ripple envelope is nearly identical to that of the other cases after the start-up overshoot.

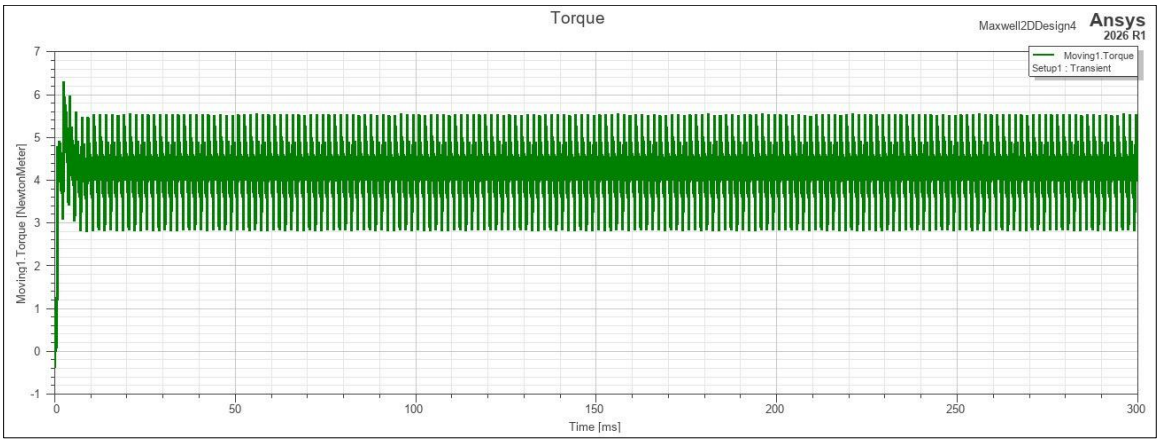


Fig. 4.20 Electromagnetic Torque of RFPMSM with 40% Dynamic Eccentricity

Fig. 4.20 shows the electromagnetic torque under 40% dynamic eccentricity. The torque again settles about a mean of roughly 4.2 N·m with a ripple of about 2.8–5.5 N·m, so the dynamic severity levels are barely separable from the torque waveform alone.

Conclusion – Fault Dynamic (10%, 20%, 40%)

The torque shows the same start-up overshoot and the same ripple band of about 2.8–5.5 N·m. The average is 4.2605 N·m at 20% (RMS 4.2963 N·m) and about 4.2 N·m at 10% and 40%. The three dynamic levels are barely separable from the torque waveform alone.

Torque

	HEALTHY	10% STATIC	10% DYNAMIC	15% STATIC	15% DYNAMIC	20% STATIC	20% DYNAMIC	40% STATIC	40% DYNAMIC	60% STATIC	60% DYNAMIC	MIXED
AVERAGE	4.2511	4.1753	4.1726	4.2090	4.2078	4.2564	4.2605	4.2652	4.2707	4.2052	4.2030	4.1711

Table 4.4 Tabulation of Torques

FAULT STATIC

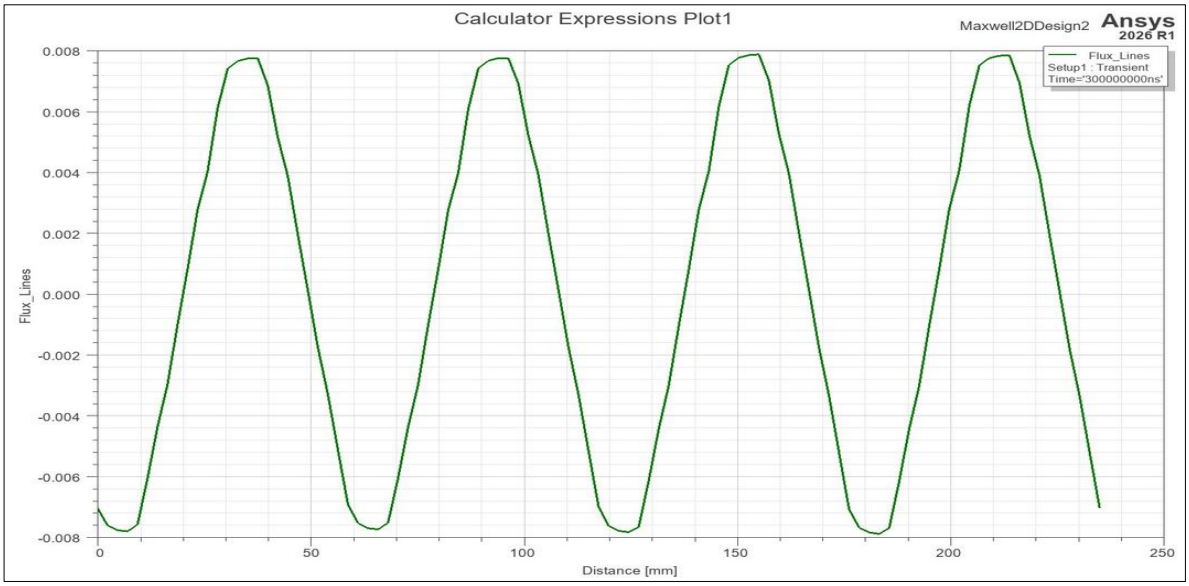


Fig. 4.21 Air-Gap Flux of RFPMSM with 10% Static Eccentricity

Fig. 4.21 shows the air-gap flux distribution (flux lines) around the air gap for 10% static eccentricity. Four equal periods appear, matching the eight-pole rotor, with a peak of about 0.0078 and nearly identical amplitudes under every pole pair.

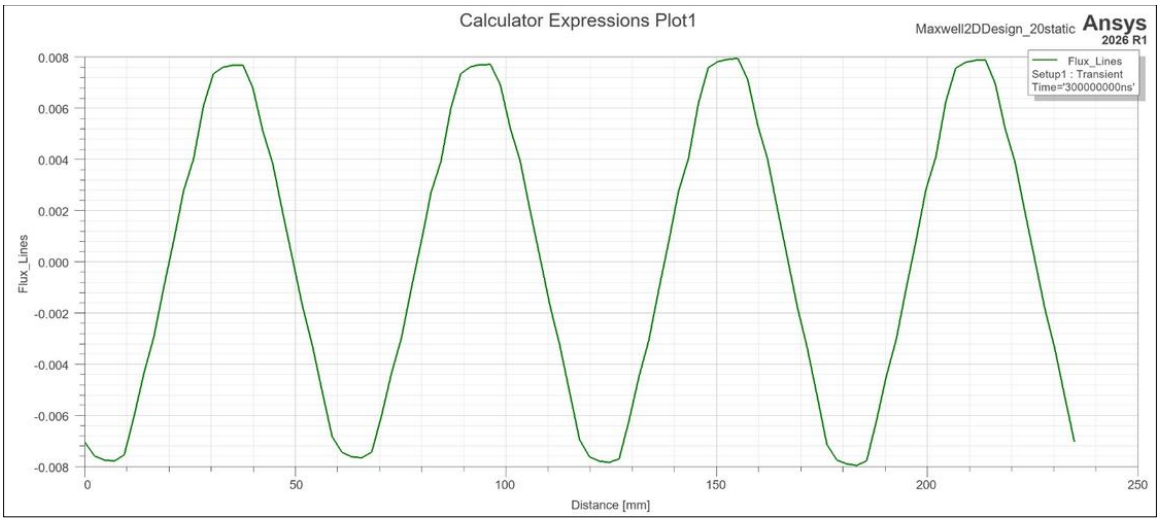


Fig. 4.22 Air-Gap Flux of RFPMSM with 20% Static Eccentricity

Fig. 4.22 shows the air-gap flux distribution for 20% static eccentricity. The four-period pattern is preserved, with peaks of about 0.0077–0.0079, so the field remains almost symmetric with only a marginal period-to-period variation.

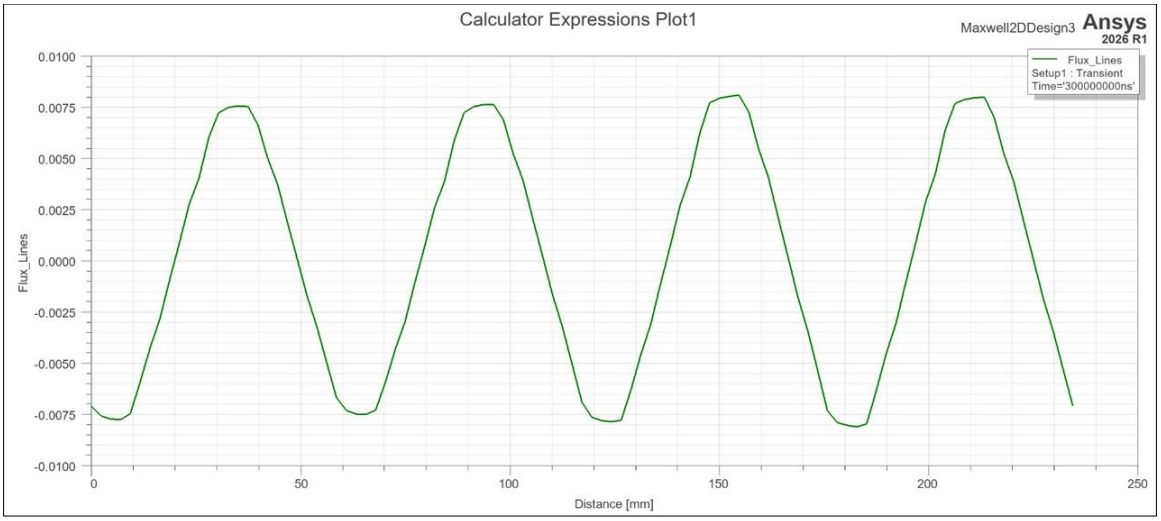


Fig. 4.23 Air-Gap Flux of RFPMSM with 40% Static Eccentricity

Fig. 4.23 shows the air-gap flux distribution for 40% static eccentricity. The peaks are now unequal, at about 0.0075 in the first two periods and 0.0080 in the last two, reflecting the non-uniform air gap around the rotor.

Conclusion – Fault Static (10%, 20%, 40%)

At 10% the four periods are nearly equal, with a peak of about 0.0078. At 20% the peaks are about 0.0077–0.0079, still almost symmetric. At 40% they become unequal, about 0.0075 in the first two periods and 0.0080 in the last two, reflecting the non-uniform air gap.

FAULT DYNAMIC

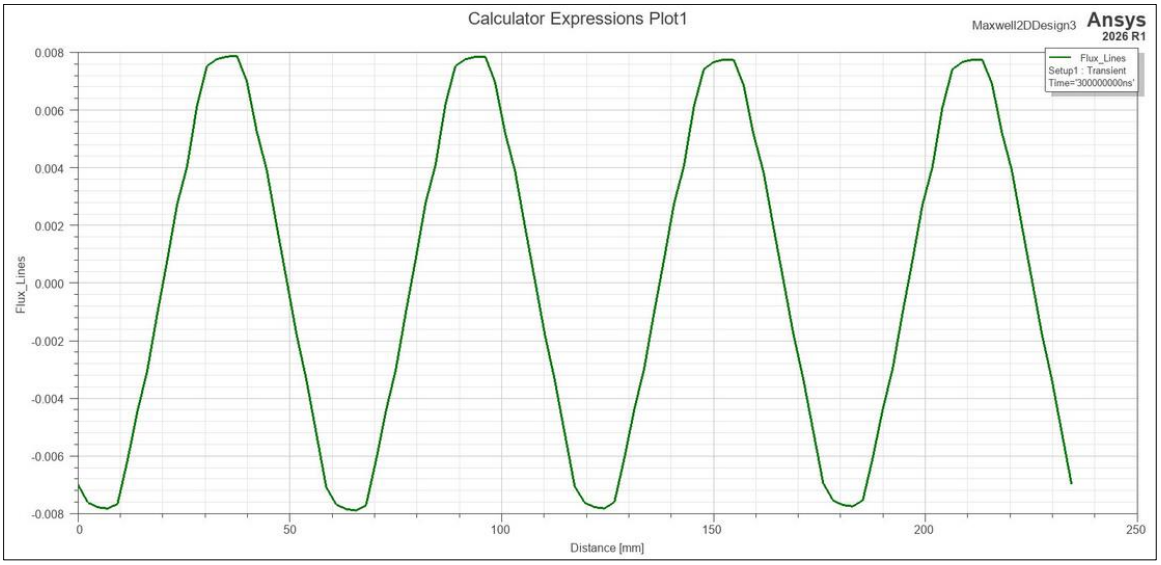


Fig. 4.24 Air-Gap Flux of RFPMSM with 10% Dynamic Eccentricity

Fig. 4.24 shows the air-gap flux distribution for 10% dynamic eccentricity. The four periods are almost identical in amplitude (about 0.0078), so the flux distribution is nearly symmetric at this small eccentricity level.

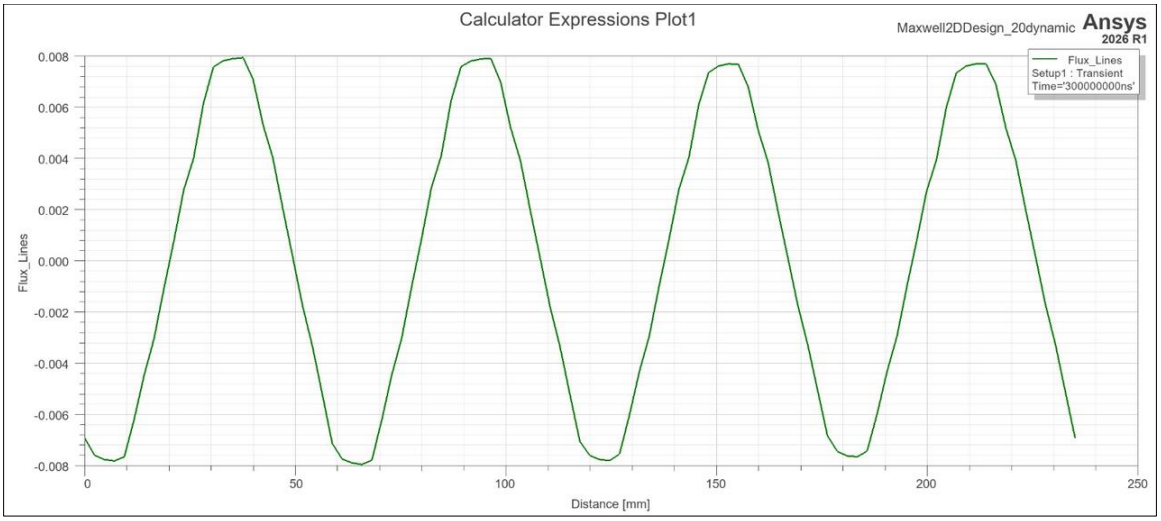


Fig. 4.25 Air-Gap Flux of RFPMSM with 20% Dynamic Eccentricity

Fig. 4.25 shows the air-gap flux distribution for 20% dynamic eccentricity. A small amplitude difference begins to appear, with the first two periods peaking near 0.0079 and the last two near 0.0077.

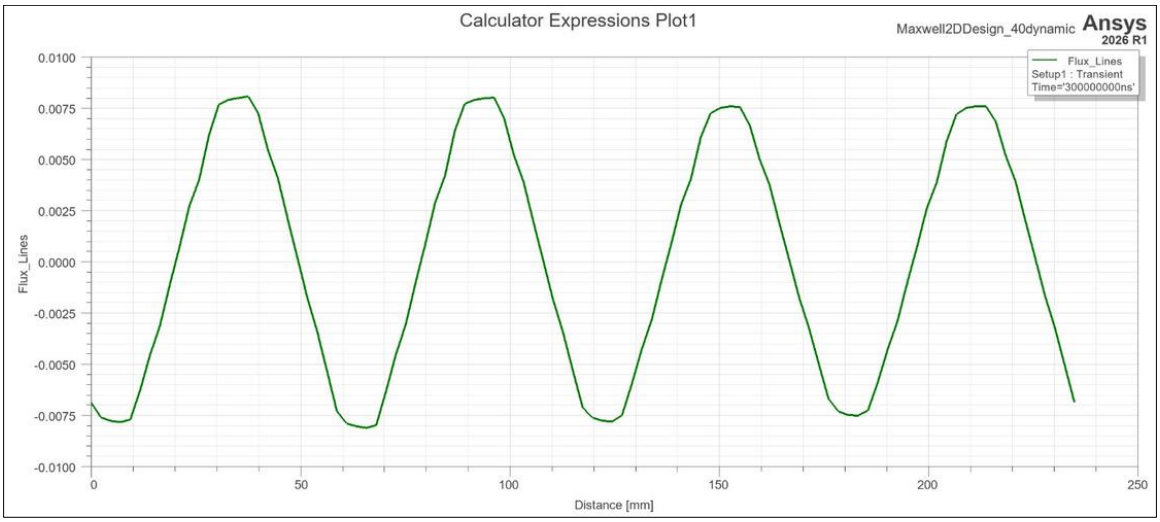


Fig. 4.26 Air-Gap Flux of RFPMSM with 40% Dynamic Eccentricity

Fig. 4.26 shows the air-gap flux distribution for 40% dynamic eccentricity. The difference between periods is now clear, about 0.0080 in the first two and 0.0075 in the last two, showing that the asymmetry grows with eccentricity severity.

Conclusion – Fault Dynamic (10%, 20%, 40%)

At 10% the four periods are almost identical (about 0.0078). At 20% a small difference appears (about 0.0079 vs 0.0077), and at 40% it is clear (about 0.0080 vs 0.0075). The asymmetry of the air-gap flux therefore grows gradually with severity.

FAULT STATIC

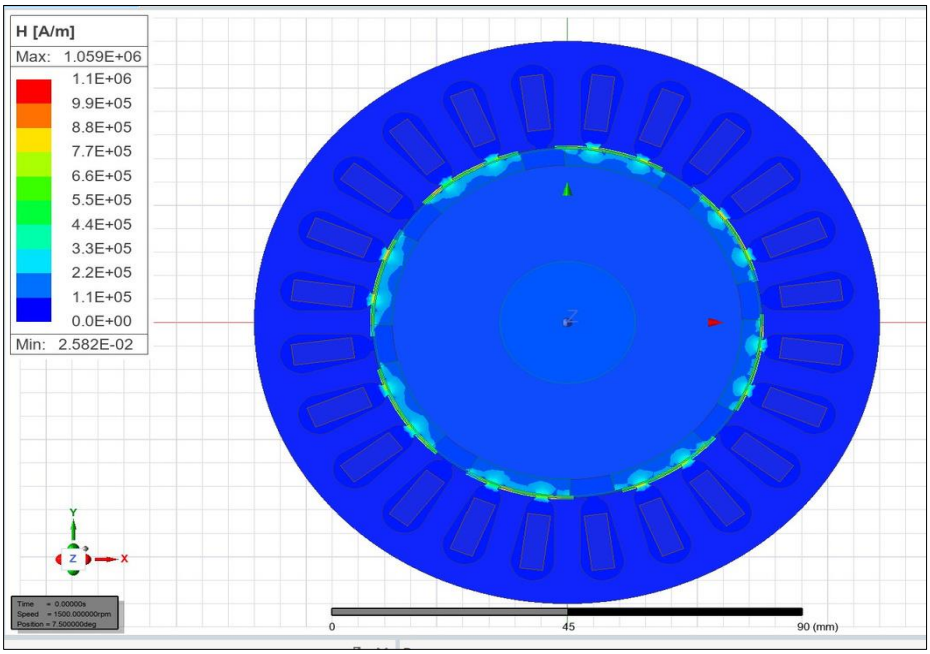


Fig. 4.27 Magnetic Field Intensity (H) Distribution of RFPMSM with 10% Static Eccentricity

Fig. 4.27 shows the magnetic field intensity (H) in the machine cross-section for 10% static eccentricity. The field is concentrated in the air-gap region and around the magnet edges, with a maximum of 1.059×10^6 A/m, while the stator core remains at low H.

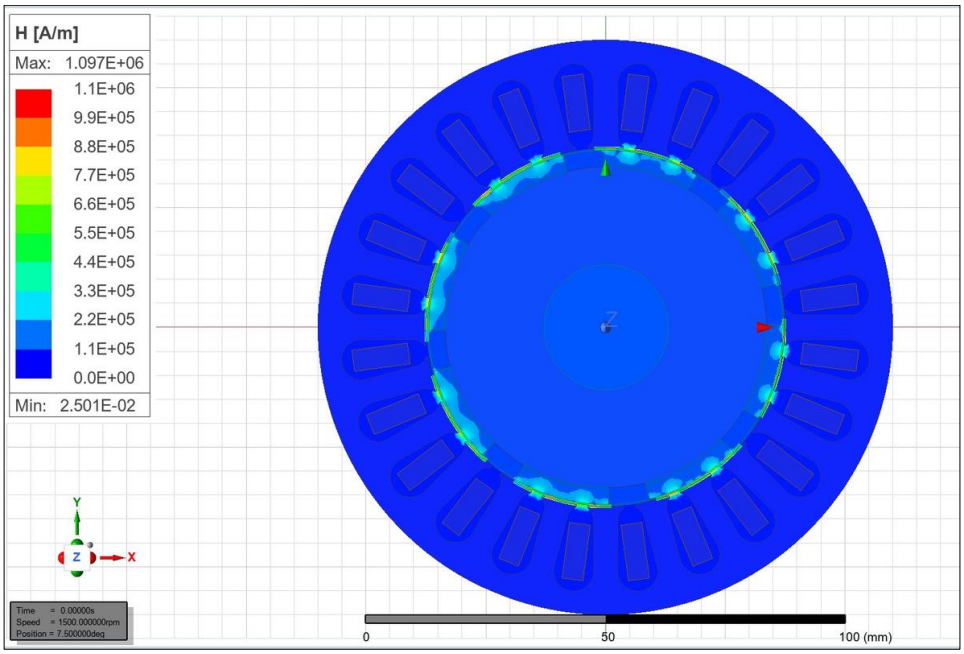


Fig. 4.28 Magnetic Field Intensity (H) Distribution of RFPMSM with 20% Static Eccentricity

Fig. 4.28 shows the magnetic field intensity for 20% static eccentricity. The maximum rises to 1.097×10^6 A/m, and the high-H spots remain localised along the air gap and the magnet edges.

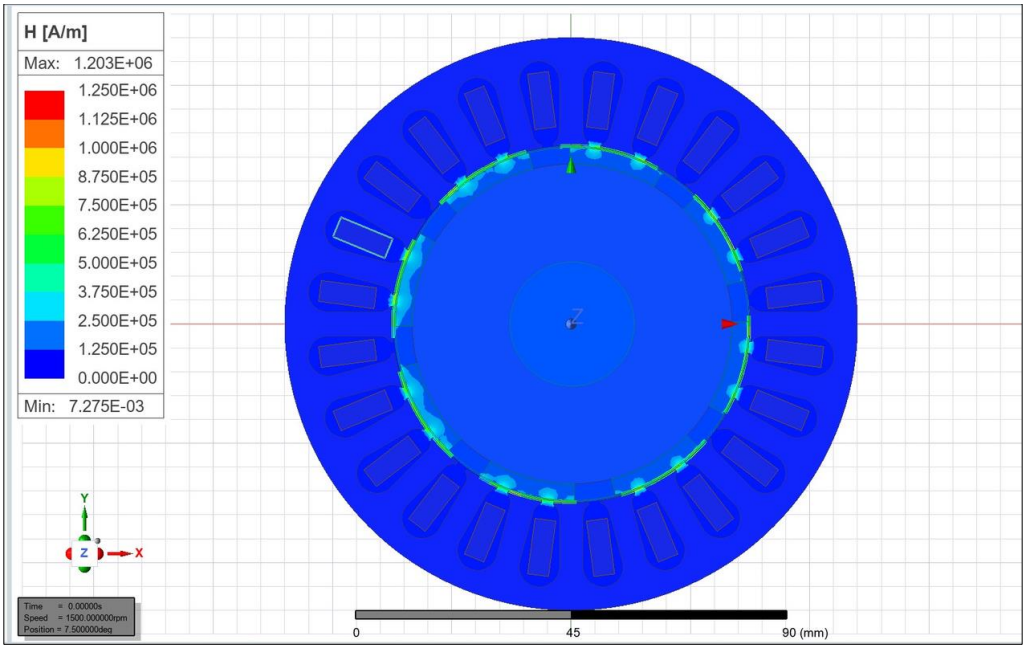


Fig. 4.29 Magnetic Field Intensity (H) Distribution of RFPMSM with 40% Static Eccentricity

Fig. 4.29 shows the magnetic field intensity for 40% static eccentricity. The maximum increases further to 1.203×10^6 A/m, the highest of all cases, indicating a stronger local field concentration where the air gap is reduced.

Conclusion – Fault Static (10%, 20%, 40%)

The field is concentrated in the air-gap region and around the magnet edges, while the stator core stays at low H. The maximum rises with severity: 1.059, 1.097 and 1.203×10^6 A/m for 10%, 20% and 40%. The 40% static case is the highest of all six cases.

FAULT DYNAMIC

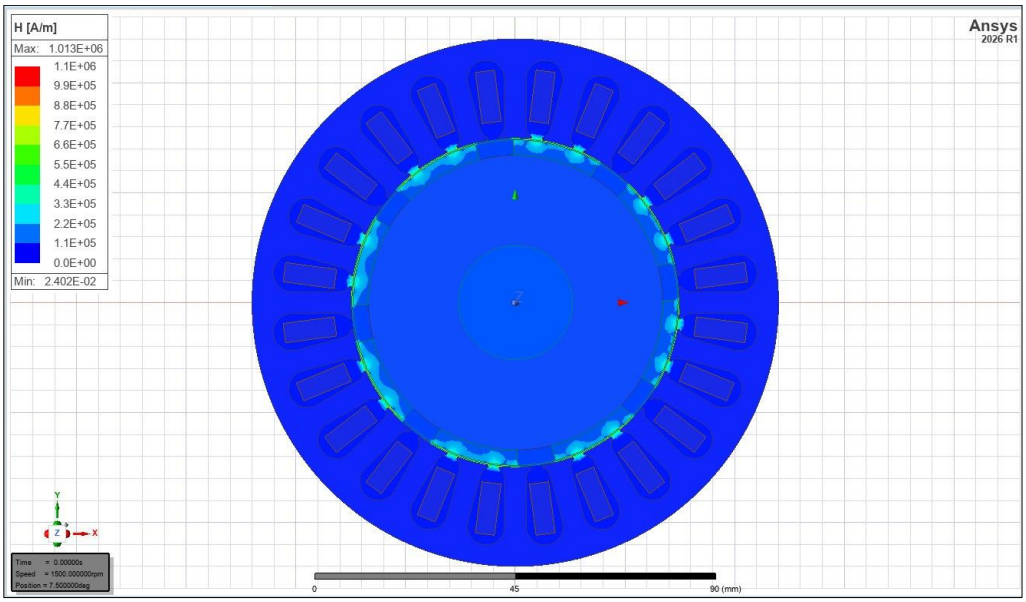


Fig. 4.30 Magnetic Field Intensity (H) Distribution of RFPMSM with 10% Dynamic Eccentricity

Fig. 4.30 shows the magnetic field intensity for 10% dynamic eccentricity. The maximum is 1.013×10^6 A/m, the lowest among the eccentric cases, with the field again confined to the air-gap and magnet region.

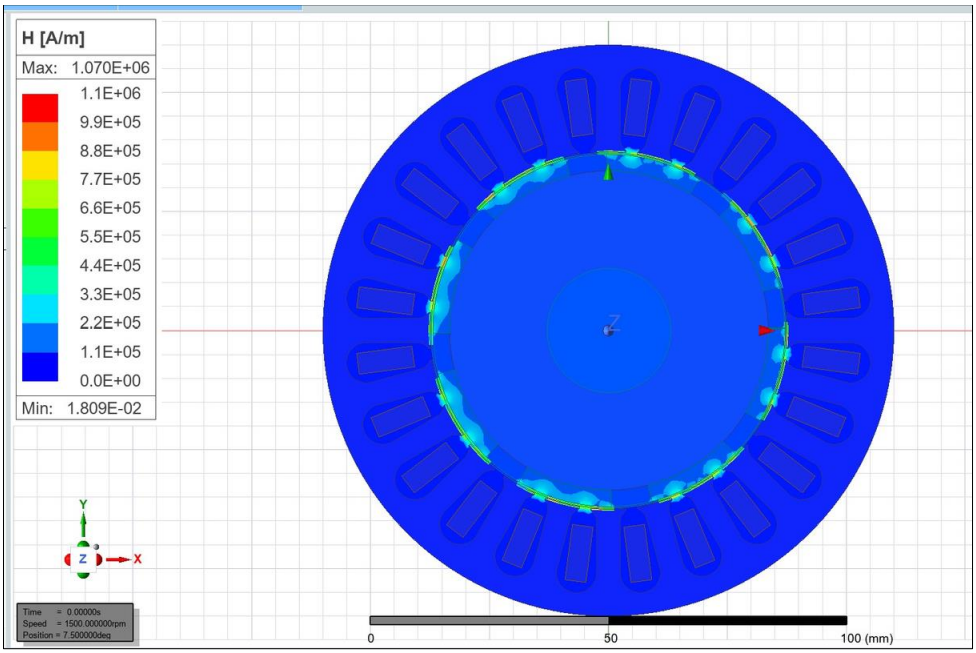


Fig. 4.31 Magnetic Field Intensity (H) Distribution of RFPMSM with 20% Dynamic Eccentricity

Fig. 4.31 shows the magnetic field intensity for 20% dynamic eccentricity. The maximum rises to 1.070×10^6 A/m compared with the 10% dynamic case, showing that the peak field grows with eccentricity severity.

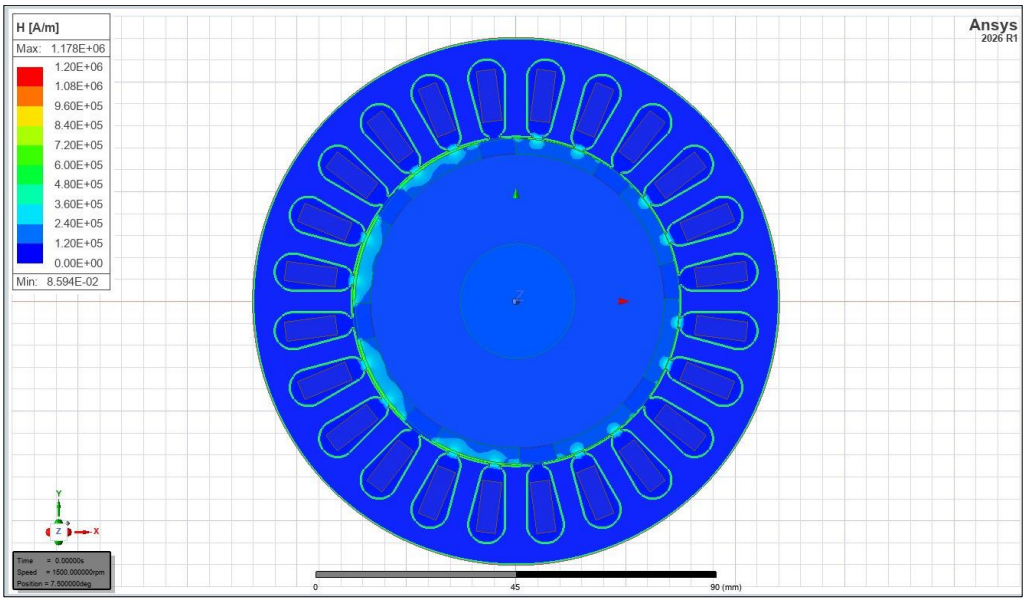


Fig. 4.32 Magnetic Field Intensity (H) Distribution of RFPMSM with 40% Dynamic Eccentricity

Fig. 4.32 shows the magnetic field intensity for 40% dynamic eccentricity. The maximum reaches 1.178×10^6 A/m, the highest dynamic value, so both static and dynamic eccentricity raise the peak field as severity increases.

Conclusion – Fault Dynamic (10%, 20%, 40%)

The field pattern is the same, confined to the air gap and magnet region. The maximum rises from 1.013 to 1.070 to 1.178×10^6 A/m. The peak field therefore grows with severity for both fault types.

FALUT STATIC

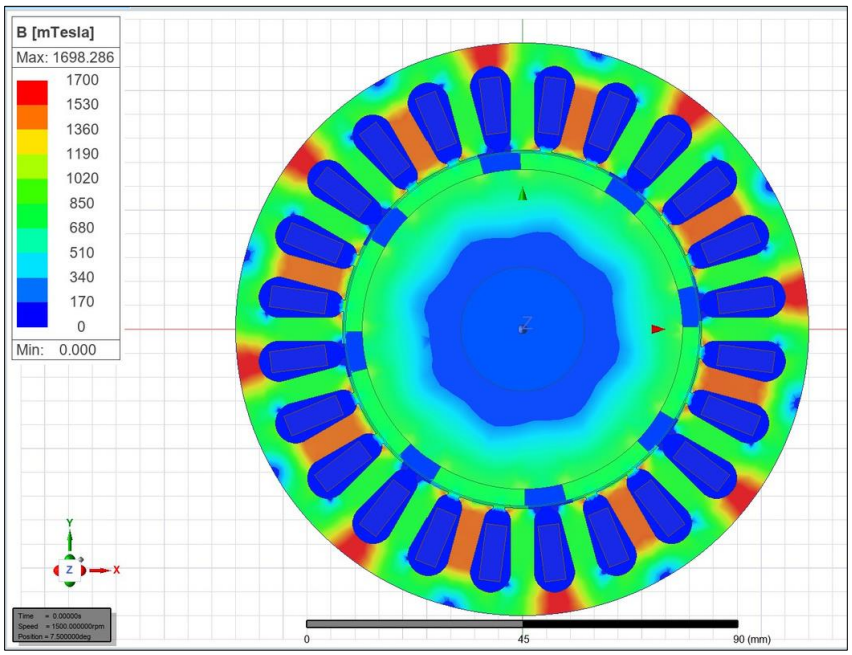


Fig. 4.33 Flux Density (B) Distribution of RFPMSM with 10% Static Eccentricity

Fig. 4.33 shows the flux density (B) distribution for 10% static eccentricity. The highest values, up to about 1.70 T, appear in some stator teeth and yoke regions, while the central rotor region stays at low flux density.

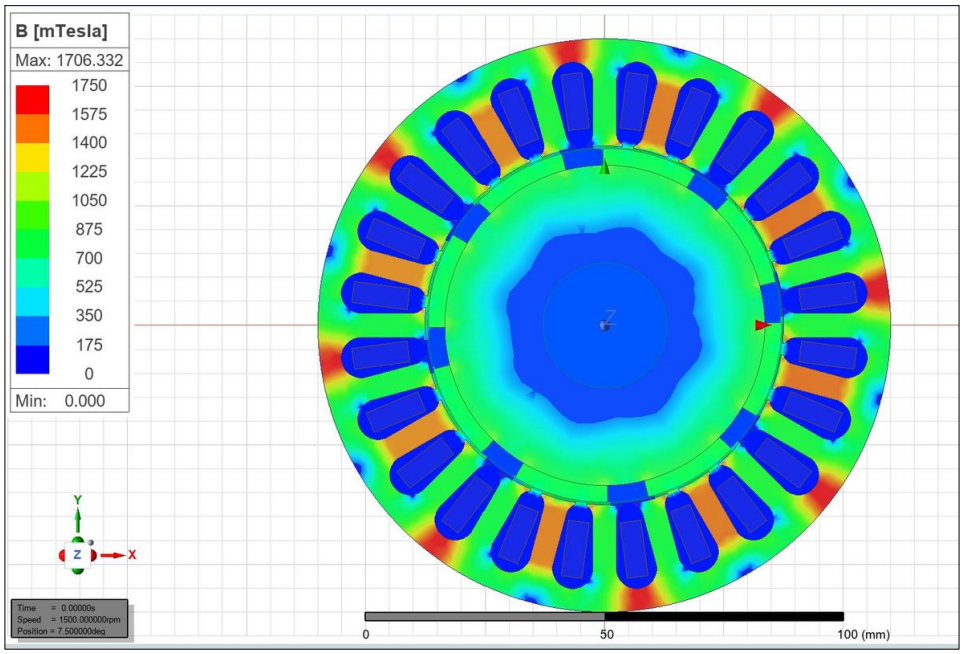


Fig. 4.34 Flux Density (B) Distribution of RFPMSM with 20% Static Eccentricity

Fig. 4.34 shows the flux density distribution for 20% static eccentricity. The maximum is about 1.71 T, only marginally above the 10% case, and the tooth and yoke flux pattern is very similar.

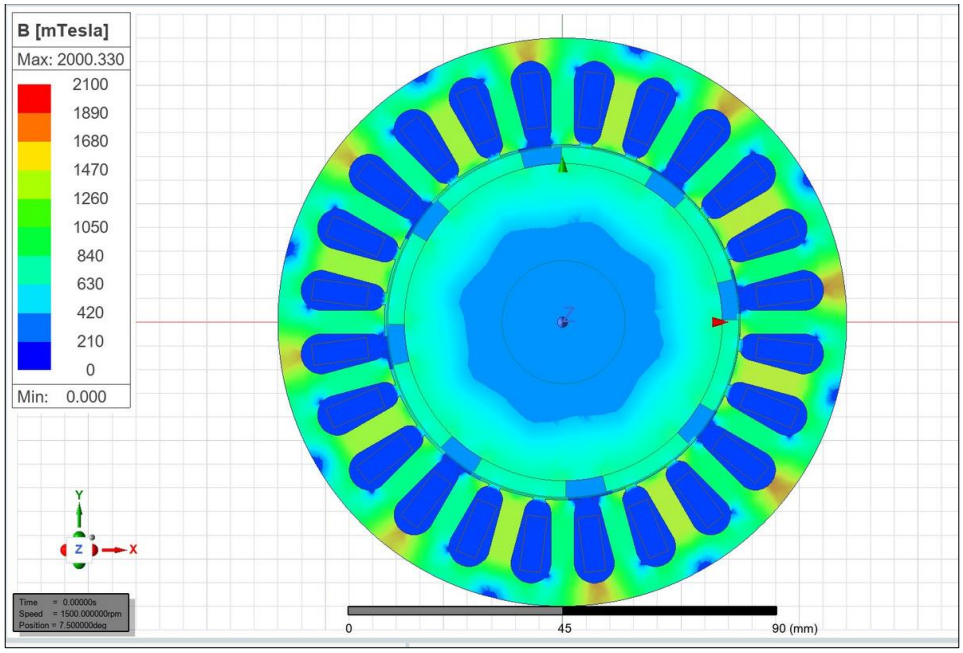


Fig. 4.35 Flux Density (B) Distribution of RFPMSM with 40% Static Eccentricity

Fig. 4.35 shows the flux density distribution for 40% static eccentricity. The maximum jumps to about 2.00 T, well above the other cases, which points to local saturation of the stator core caused by the strong air-gap asymmetry.

Conclusion – Fault Static (10%, 20%, 40%)

The highest flux density appears in some stator teeth and yoke regions. The maximum is about 1.70 T at 10% and 1.71 T at 20%, then jumps to about 2.00 T at 40%. This points to local saturation caused by the strong air-gap asymmetry.

FAULT DYNAMIC

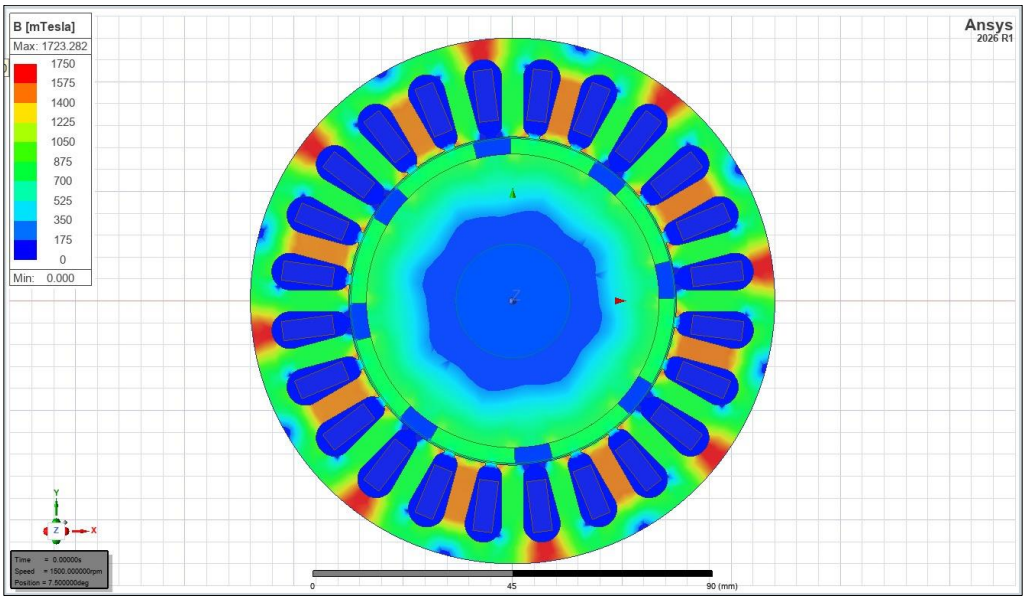


Fig. 4.36 Flux Density (B) Distribution of RFPMSM with 10% Dynamic Eccentricity

Fig. 4.36 shows the flux density distribution for 10% dynamic eccentricity. The maximum is about 1.72 T, slightly higher than in the 10% static case, with the highest flux density again located in the stator teeth and yoke.

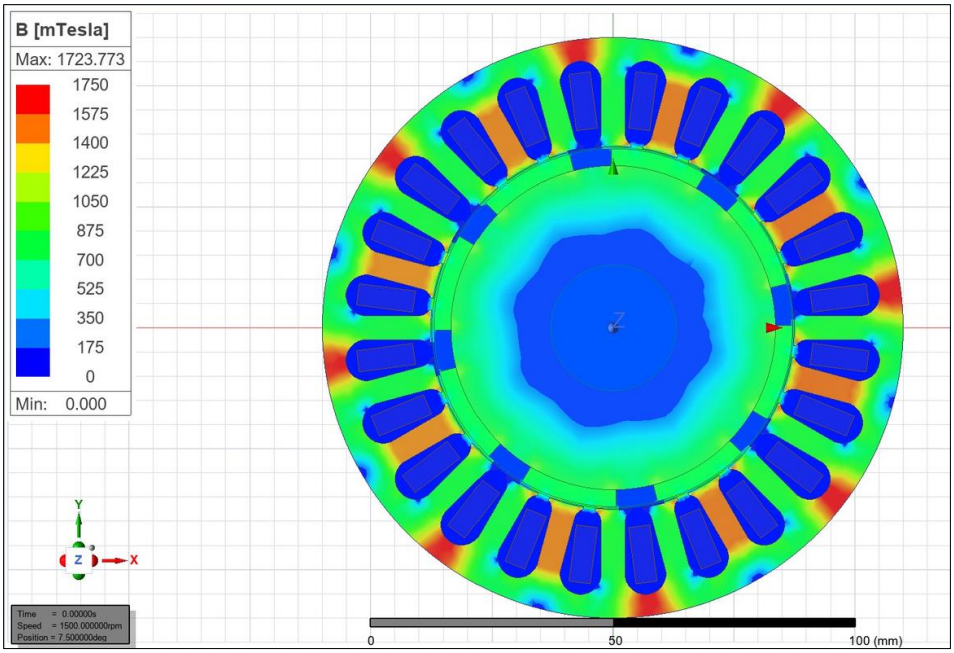


Fig. 4.37 Flux Density (B) Distribution of RFPMSM with 20% Dynamic Eccentricity

Fig. 4.37 shows the flux density distribution for 20% dynamic eccentricity. The maximum is about 1.72 T, practically the same as at 10%, and the flux pattern in the teeth and yoke changes very little.

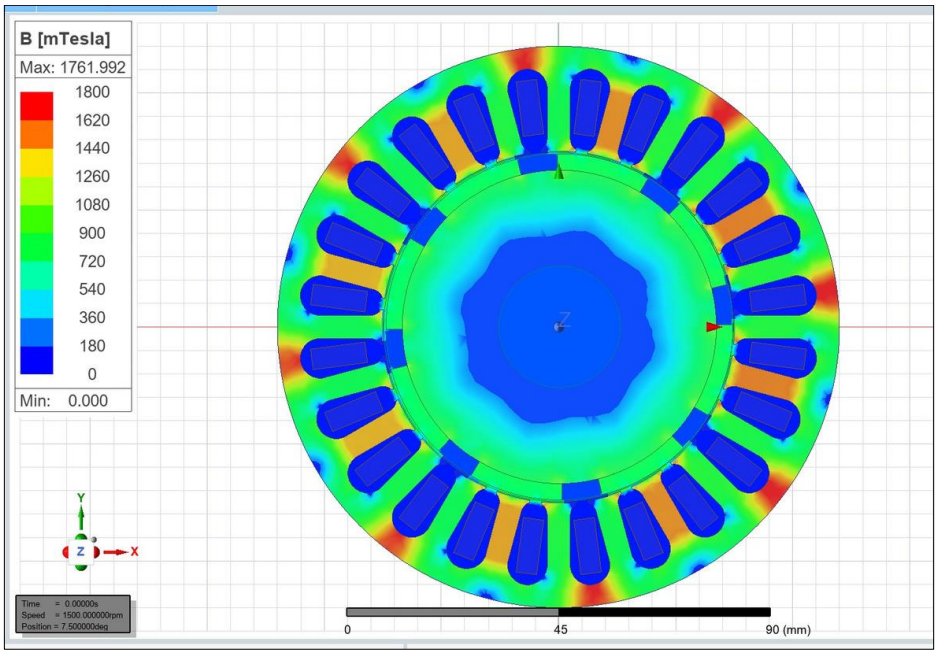


Fig. 4.38 Flux Density (B) Distribution of RFPMSM with 40% Dynamic Eccentricity

Fig. 4.38 shows the flux density distribution for 40% dynamic eccentricity. The maximum rises to about 1.76 T, the highest dynamic value, but still well below the 2.00 T seen for 40% static eccentricity.

Conclusion – Fault Dynamic (10%, 20%, 40%)

The maximum rises only slightly, from about 1.72 T at 10% and 20% to about 1.76 T at 40%. This stays well below the 2.00 T of the 40% static case, so dynamic eccentricity is less severe magnetically at the same percentage.

FAULT STATIC

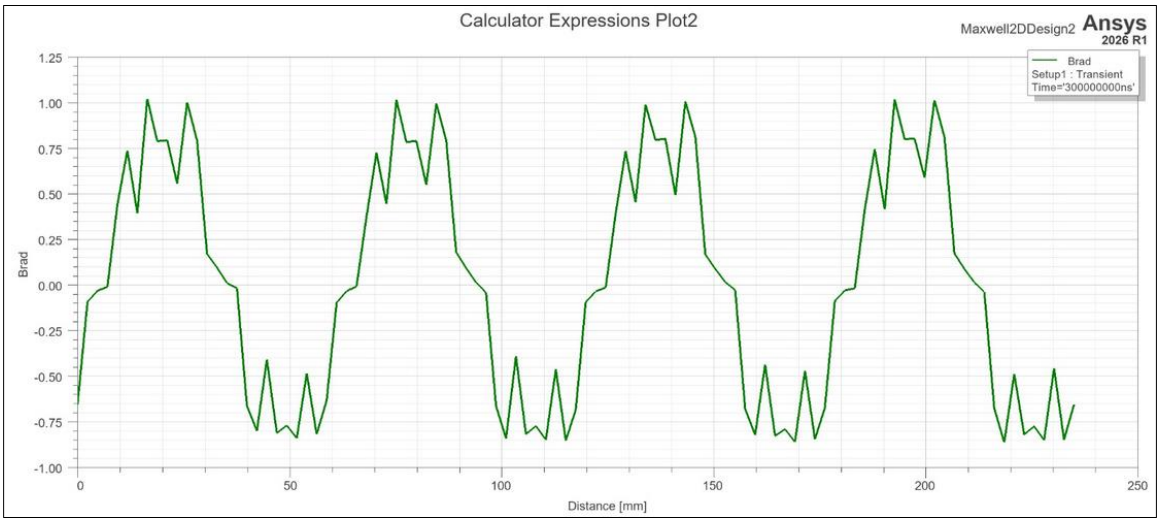


Fig. 4.39 Radial Air-Gap Flux Density of RFPMSM with 10% Static Eccentricity

Fig. 4.39 shows the radial component of the air-gap flux density (Brad) along the air-gap path for 10% static eccentricity. The four flat-topped periods carry slot-induced ripple and each peak at about 1.0 T, so the field is nearly uniform under all poles.

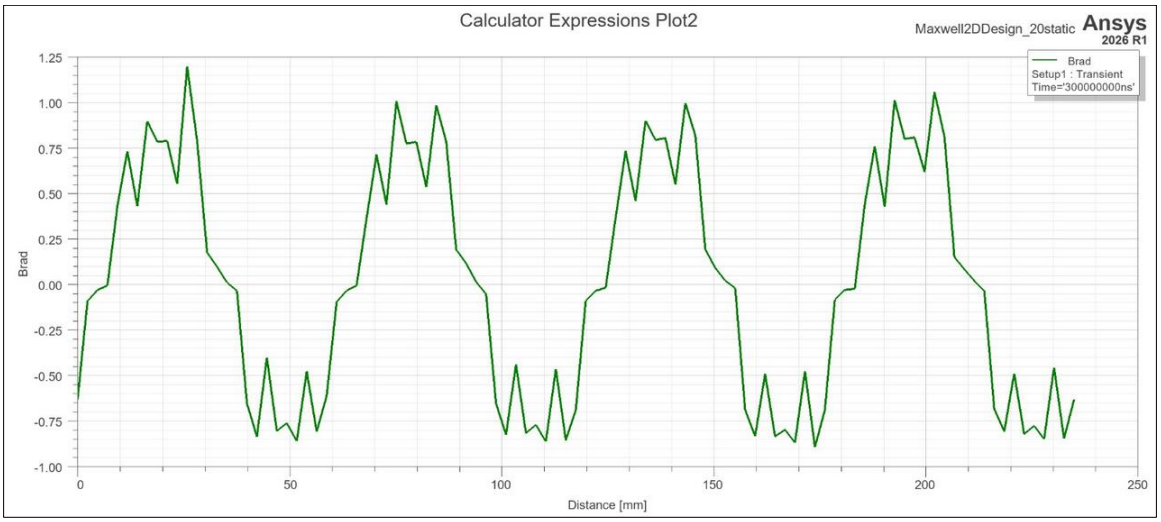


Fig. 4.40 Radial Air-Gap Flux Density of RFPMSM with 20% Static Eccentricity

Fig. 4.40 shows the radial air-gap flux density for 20% static eccentricity. The first period now peaks at about 1.2 T while the others remain near 1.0 T, a clear local distortion of the air-gap field caused by the reduced gap on one side.

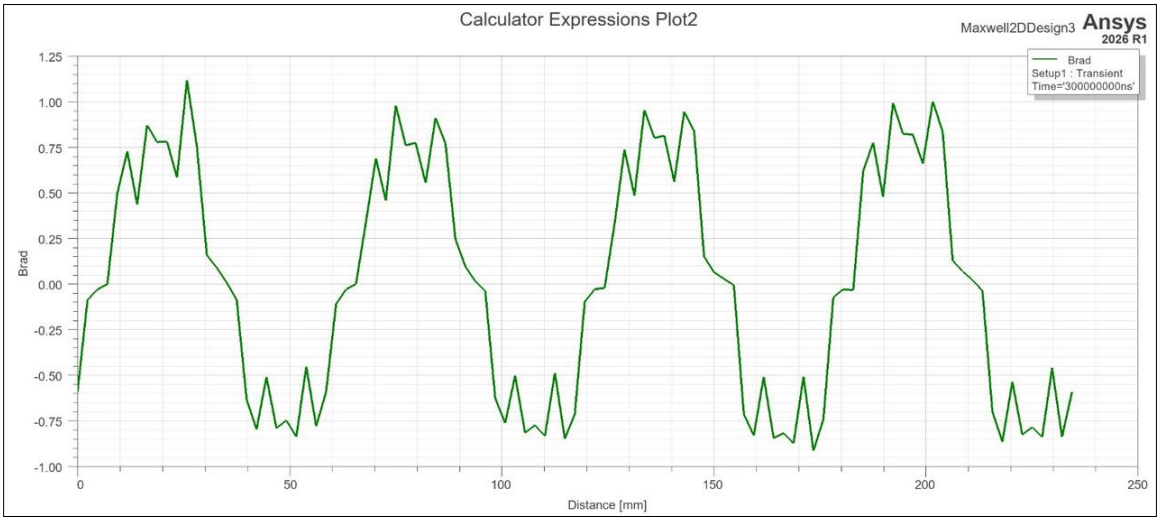


Fig. 4.41 Radial Air-Gap Flux Density of RFPMSM with 40% Static Eccentricity

Fig. 4.41 shows the radial air-gap flux density for 40% static eccentricity. The peaks vary between about 0.95 T and 1.1 T across the four periods, confirming the pole-to-pole asymmetry produced by the displaced rotor.

Conclusion – Fault Static (10%, 20%, 40%)

At 10% the four flat-topped periods each peak at about 1.0 T. At 20% the first period rises to about 1.2 T while the others stay near 1.0 T. At 40% the peaks vary between about 0.95 T and 1.1 T, confirming the pole-to-pole asymmetry produced by the displaced rotor.

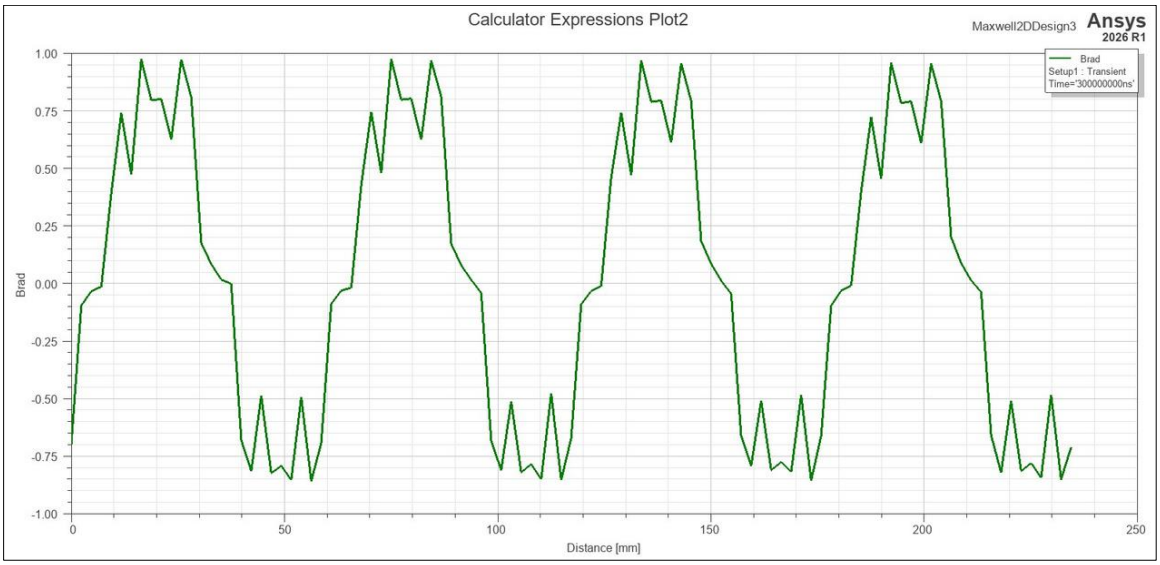


Fig. 4.42 Radial Air-Gap Flux Density of RFPMSM with 10% Dynamic Eccentricity

Fig. 4.42 shows the radial air-gap flux density for 10% dynamic eccentricity. All four periods peak at about 0.97 T with the same slot-ripple pattern, so the field remains nearly symmetric at this level.

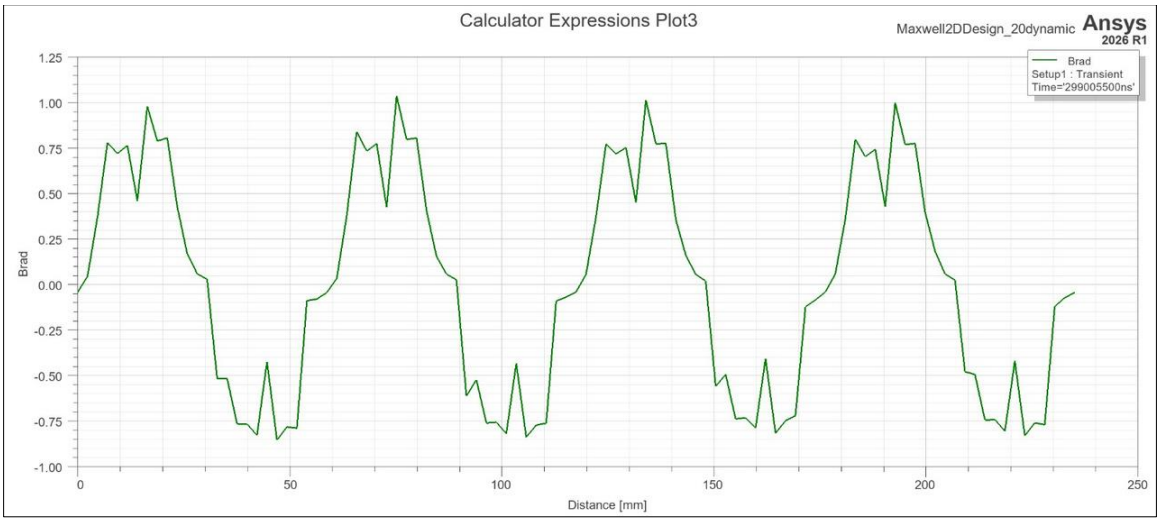


Fig. 4.43 Radial Air-Gap Flux Density of RFPMSM with 20% Dynamic Eccentricity

Fig. 4.43 shows the radial air-gap flux density for 20% dynamic eccentricity. The peaks stay close to 1.0 T with only slight differences between the periods, indicating a mild distortion of the air-gap field.

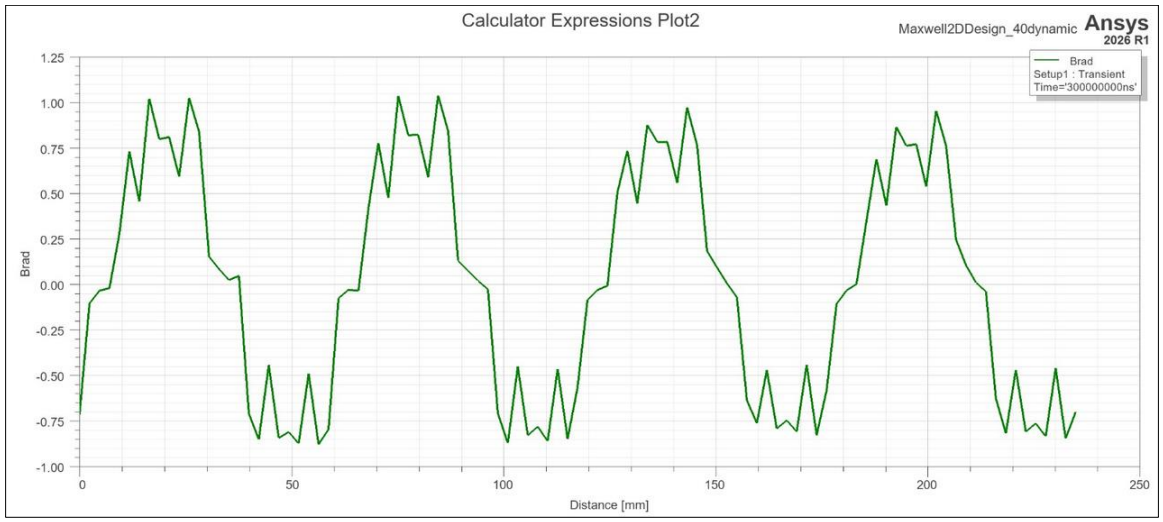


Fig. 4.44 Radial Air-Gap Flux Density of RFPMSM with 40% Dynamic Eccentricity

Fig. 4.44 shows the radial air-gap flux density for 40% dynamic eccentricity. The first two periods peak at about 1.03 T and the last two at about 0.95–0.97 T, showing that the field asymmetry grows with the eccentricity level.

Conclusion – Fault Dynamic (10%, 20%, 40%)

At 10% all periods peak at about 0.97 T. At 20% they stay close to 1.0 T with slight differences. At 40% the first two periods peak at about 1.03 T and the last two at about 0.95–0.97 T, so the asymmetry grows with the eccentricity level.

FAULT STATIC

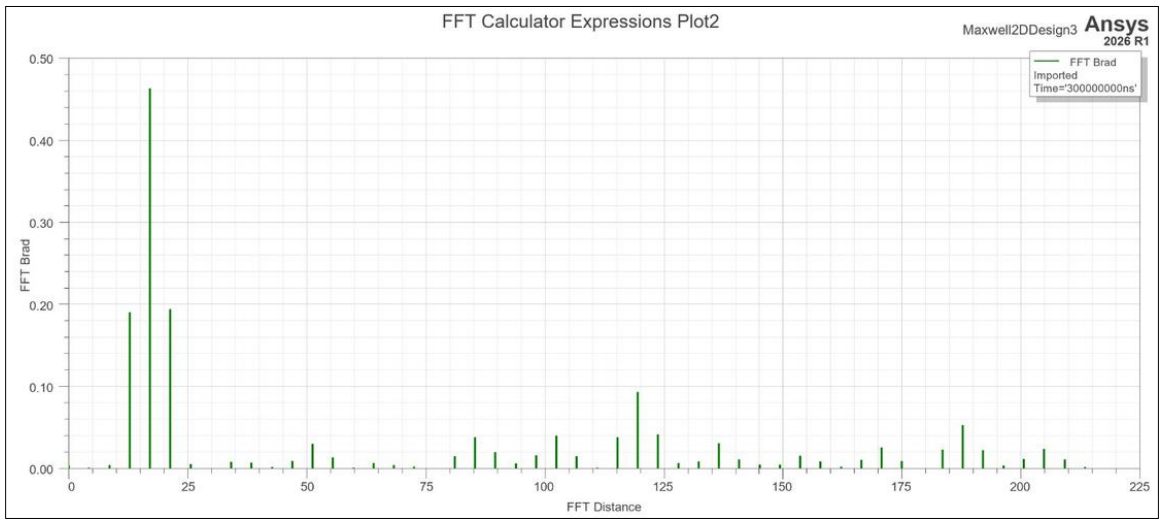


Fig. 4.45 FFT of Radial Air-Gap Flux Density of RFPMSM with 40% Static Eccentricity

Fig. 4.45 shows the spatial FFT of the radial air-gap flux density for 40% static eccentricity. The dominant component (about 0.46) is the four-pole-pair fundamental, flanked by sidebands of about 0.19 at the adjacent orders, which are typical of an uneven air gap, plus smaller slot harmonics.

Conclusion – Fault Static (10%, 20%, 40%)

The dominant component (about 0.46) is the four-pole-pair fundamental. It is flanked by sidebands of about 0.19 at the adjacent orders, which are typical of an uneven air gap, and by smaller slot harmonics.

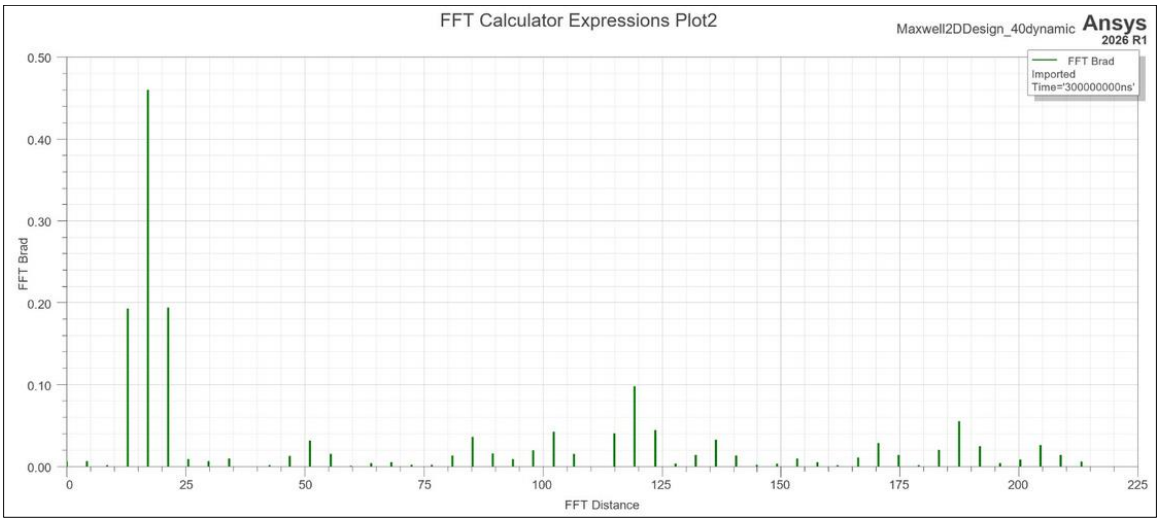


Fig. 4.46 FFT of Radial Air-Gap Flux Density of RFPMSM with 40% Dynamic Eccentricity

Fig. 4.46 shows the spatial FFT of the radial air-gap flux density for 40% dynamic eccentricity. The spectrum is almost identical to the static case, with the same 0.46 fundamental and 0.19 sidebands, so a single air-gap snapshot cannot by itself separate static from dynamic eccentricity.

Conclusion – Fault Dynamic (10%, 20%, 40%)

The spectrum is almost identical to the static one, with the same 0.46 fundamental and 0.19 sidebands. A single air-gap snapshot therefore cannot separate static from dynamic eccentricity, and additional features or AI-based classification are needed

Eccentricity Modeling in ANSYS – Axial Flux PMSM

ANSYS Implementation	Literature Terminology
Rotor shifted in X/Y, axis fixed	Static Axis Eccentricity
Rotor shifted in X/Y and rotated about original axis	Dynamic Axis Eccentricity
Rotor shifted along Z-axis	Axial Eccentricity
Rotor tilted with fixed axis	Static Angular/Inclined Eccentricity
Rotor tilted with the tilt rotating	Dynamic Angular Eccentricity
Rotor tilted + Z-axis shift	Mixed Angular–Axial Eccentricity

Table 4.5 Different types of Eccentricity

The rotor eccentricity configurations considered in the ANSYS model are categorized based on the nature and direction of the rotor displacement relative to the original shaft axis. Static Axis Eccentricity is obtained by shifting the rotor in the X/Y direction while keeping its rotational axis fixed. In this case, the rotor centre is offset from the original axis, but the offset does not change its orientation during rotation. Dynamic Axis Eccentricity is represented by shifting the rotor in X/Y and allowing it to rotate about the original axis, causing the eccentricity direction to vary with rotor rotation. Axial Eccentricity is produced by shifting the rotor along the Z-axis, resulting in an axial offset between the rotor and its reference position without introducing a transverse displacement.

CLOSED LOOP CONTROLLER RFPMSM

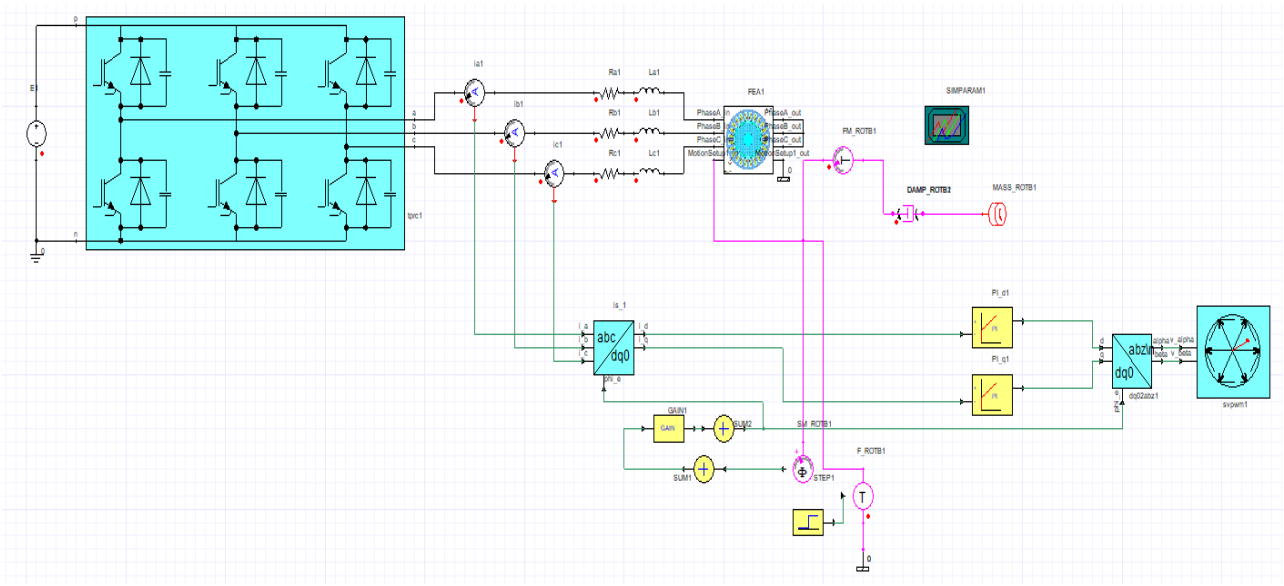


Fig- 4.55 Closed Loop Circuit Diagram

Field-Oriented Control (FOC), also called vector control, is a closed-loop control technique used for precise control of speed, torque, and flux in AC motors, particularly Permanent Magnet Synchronous Motors (PMSMs). FOC transforms the three-phase stator currents into a rotating d-q reference frame aligned with the rotor magnetic field. This makes it possible to control the motor flux and torque independently.

In the ANSYS Twin Builder implementation, the three-phase PMSM is supplied through a three-phase inverter connected to a DC voltage source. The motor phase currents Ia , Ib and Ic are measured and fed to the abc-to-dq transformation block. The rotor electrical position is also obtained from the PMSM and supplied to the transformation block. The transformed currents Id and Iq are compared with their reference values. The normal operation of a surface-mounted PMSM, Id is generally set to zero, while Iq is generated by the speed-control loop according to the required torque. The current errors are processed by PI controllers to generate the required d and q-axis voltage references.

The voltage references are converted back to the stationary reference frame using the inverse Park transformation and then supplied to the SVPWM block. The SVPWM block generates the switching pulses for the six switches of the three-phase inverter. The inverter applies the controlled three-phase voltage to the PMSM. Motor speed and position are fed back to the control system, forming a closed-loop FOC system.

4.7 COMPARATIVE DISCUSSION

The RFPMSM shows relatively small changes in stator current and average torque with increasing eccentricity. The three-phase currents remain largely balanced, with peaks around ±8 A, while the average torque remains close to the healthy operating value. However, the air-gap flux distribution becomes increasingly asymmetric as eccentricity increases, particularly at higher severity. Therefore, current and average torque alone provide limited separation of eccentricity severity in the RFPMSM, making harmonic and frequency-domain analysis important for identifying the fault.

The AFPMSM shows clearer changes in its electromagnetic characteristics when eccentricity is introduced. The healthy motor has an average RMS current of approximately 3.014 A and an average torque of 3.6715 N·m. Under the 1 mm rotor displacement, the average RMS current decreases to 2.4502 A, representing an 18.70% reduction, while the average torque decreases to 3.1868 N·m, corresponding to a 13.20% reduction. The faulty condition also produces increased current asymmetry and torque oscillations, indicating a stronger observable change in the time-domain characteristics.

4.8 CONCLUSION

The RFPMSM analysis shows that eccentricity primarily produces noticeable changes in the air-gap magnetic-field distribution, while the time-domain current and average torque remain comparatively less sensitive. Hence, additional features such as air-gap flux harmonics, FFT components, and other signal-processing characteristics are required for reliable eccentricity classification and severity estimation.

The AFPMSM results demonstrate that eccentricity causes measurable reductions in stator current and electromagnetic torque, along with increased torque fluctuations. These parameters therefore provide useful fault-sensitive indicators for detecting abnormal operation. Combining these changes with FFT, harmonic analysis, and AI/ML techniques can further improve eccentricity fault classification and severity estimation.

Chapter 5

RADIAL FLUX PMSM DIAGNOSIS FRAMEWORK

5.1 INTRODUCTION

This chapter presents the final diagnosis framework developed for the radial flux PMSM (RFPMSM). The framework works at the level of a complete recording. Every recording produces one feature vector, the training and the test recordings are kept strictly apart, and the models are checked by leaving a whole severity level out. The framework estimates the eccentricity severity (in percent) and classifies the fault as static eccentricity (SE) or dynamic eccentricity (DE) from the three-phase stator currents alone, without any additional sensor.

5.2 OVERALL PROCESS FLOW

The input to the framework is the set of three-phase stator currents (Ia, Ib, Ic) and the outputs are the fault type (SE or DE) and the fault severity in percent. The process, shown in Fig. 5.1, has seven stages: data acquisition from the ANSYS Maxwell simulation of the healthy, SE and DE machines; pre-processing (resampling, filtering and removal of transients); signal processing (FFT harmonics, current asymmetry and discrete wavelet transform); feature extraction (one vector per recording, scaled with training data only); validation (leave-one-severity-level-out cross-validation followed by one test on new recordings); the machine-learning models; and the final output.

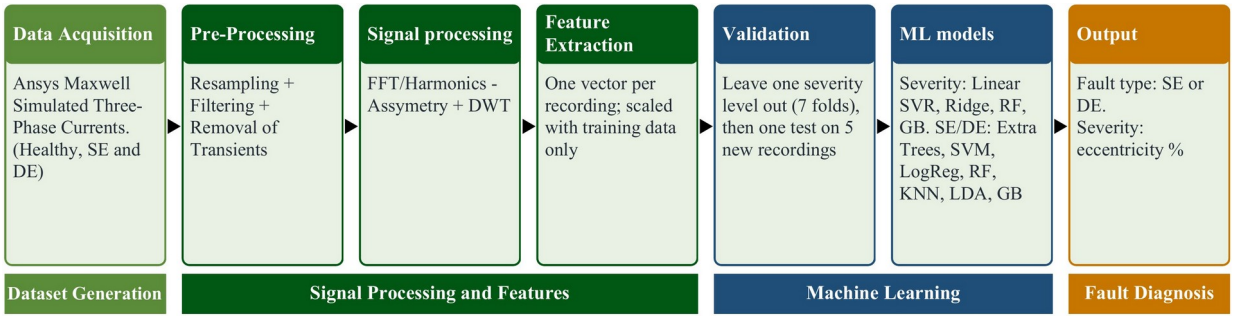


Fig. 5.1. Process flow of the radial flux PMSM diagnosis framework.

5.3 DATASET

The dataset consists of 20 CSV files, each holding one simulated recording of the three phase currents. A recording has between 23,600 and 24,500 raw samples and a duration of 0.3 s at a supply frequency of 100 Hz. The file name carries the label (healthy, SE or DE) and the eccentricity severity. The dataset was divided into 15 training recordings (75 %) and 5 test recordings (25 %), and the test recordings were never used for fitting, scaling or model selection. Fig. 5.2 and Table 5.1 show the inventory.

- Training set: 1 healthy recording, 7 SE recordings and 7 DE recordings, at 10, 12.5, 20, 22.5, 35, 40 and 60 % severity.
- Test set: 2 SE recordings (15 and 25 %) and 3 DE recordings (15, 25 and 50 %).
- Every test severity is a level that does not occur in training, so the test measures interpolation to unseen severities, which is the practical situation for an unknown machine.

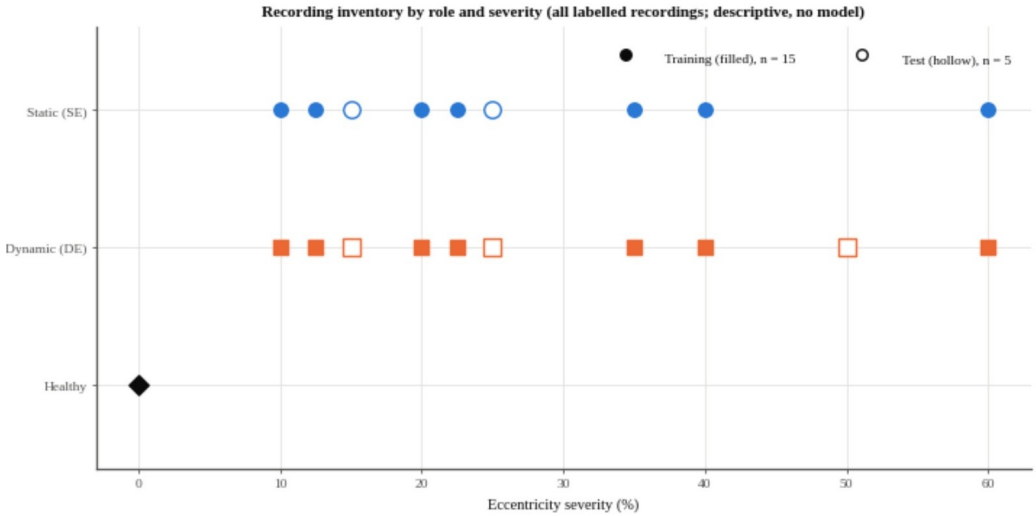


Fig. 5.2. Recording inventory by role and severity (filled markers: training, hollow markers: test).

Set	Healthy	SE	DE	Total
Training	1	7	7	15
Test	0	2	3	5
Total	1	9	10	20

Table 5.1 Dataset split of the radial flux PMSM recordings

5.4 SIGNAL ACQUISITION AND PRE-PROCESSING

The three-phase stator currents Ia, Ib and Ic were obtained from the ANSYS Maxwell transient simulation (stop time 0.3 s, supply frequency 100 Hz). The simulator writes samples at irregular time instants, which cannot be analysed directly by an FFT, so four pre-processing steps are applied to every recording (Fig. 5.3).

- Uniform resampling: the irregular simulator samples are interpolated onto a uniform 400 kHz grid.
- Filtering and down-sampling: an anti-alias filter is applied and the sampling rate is reduced to 20 kHz (6,000 samples for 0.3 s). Filtering before down-sampling prevents high-frequency switching content from folding into the band of interest.
- Removal of transients: the first 20 ms (start-up transient) and the final 1 ms are discarded so that only the steady-state current is analysed.
- Whole-cycle selection: an integer number of fundamental cycles (26 to 27 cycles of 100 Hz) is analysed, which keeps the harmonic lines of the Fourier analysis exactly on frequency bins and avoids leakage.

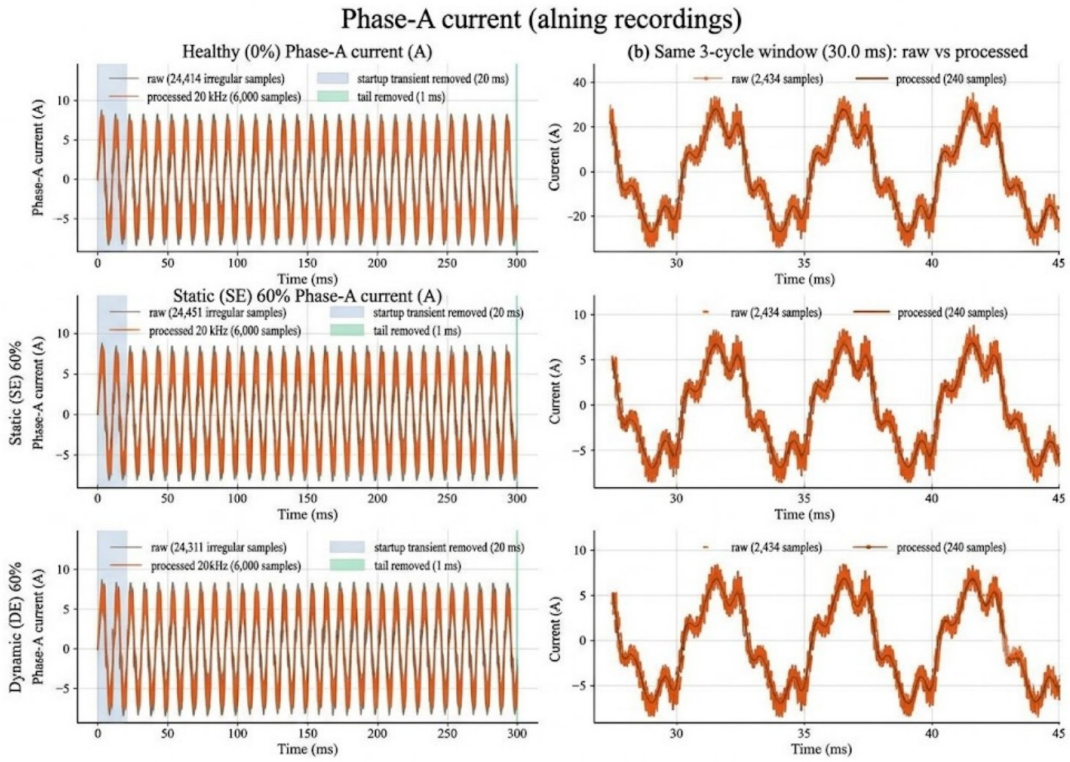


Fig. 5.3. Phase-A current of the healthy, SE 60 % and DE 60 % recordings: raw and processed (20 kHz) signals, and a 30 ms window.

5.5 FEATURE EXTRACTION

5.5.1 Space-vector representation and fundamental frequency

The three phase currents are combined into a complex space vector (Clarke/Park representation). The supply fundamental, about 100 Hz, is located from the vector, and the Fourier coefficients are then evaluated over complete steady-state cycles. Because the machine is fed from a converter and operated at a fixed supply frequency, the spectrum has energy only at multiples of 100 Hz (Fig. 5.4a), so the harmonic amplitudes can be read at known frequencies.

5.5.2 FFT-based harmonic analysis

The fast Fourier transform converts the current from the time domain to the frequency domain. The fundamental and the 5th, 7th, 11th and 13th harmonics are extracted and expressed as a percentage of the fundamental. Eccentricity distorts the air-gap field, which changes the harmonic content of the stator current, so these amplitudes are fault-sensitive. Fig. 5.5 compares the spectrum before and after pre-processing and shows that the analysed harmonic amplitudes are largely preserved by the anti-alias filtering and down-sampling, while the high-frequency content above the new Nyquist frequency is removed.

5.5.3 Current asymmetry and harmonic phase

An eccentric rotor makes the air gap uneven, which disturbs the symmetry of the three phase currents. Five asymmetry features measure the three-phase imbalance. Eight harmonic-phase features (sine and cosine components of the phase of the 5th, 7th, 11th and 13th harmonics) are also computed because phase relations may help to distinguish SE from DE; they are used only in the SE/DE classifier.

5.5.4 Discrete wavelet transform (DWT)

The discrete wavelet transform decomposes the current into frequency bands. A Daubechies-4 (db4) wavelet with four levels is applied at 20 kHz [41], giving the approximation A4 (0 to 625 Hz, containing the fundamental and about 97 % of the energy) and the details D4 (625 to 1250 Hz), D3 (1.25 to 2.5 kHz), D2 (2.5 to 5 kHz) and D1 (5 to 10 kHz). The percentage of energy in each band forms five DWT features (Fig. 5.6). They describe how the energy is distributed over frequency and complement the single-frequency harmonic amplitudes.

Feature group	Count	Features	Used for
Amplitude	5	Fundamental and 5th, 7th, 11th, 13th harmonics (% of fundamental)	Severity and SE/DE
Asymmetry	5	Three-phase current imbalance	Severity and SE/DE
Harmonic phase	8	Sine and cosine of the 5th, 7th, 11th and 13th harmonic phases	SE/DE only
DWT energy	5	Energy percentage in five bands (db4, 4 levels)	Severity and SE/DE

Table 5.2 Feature groups used in the radial flux framework

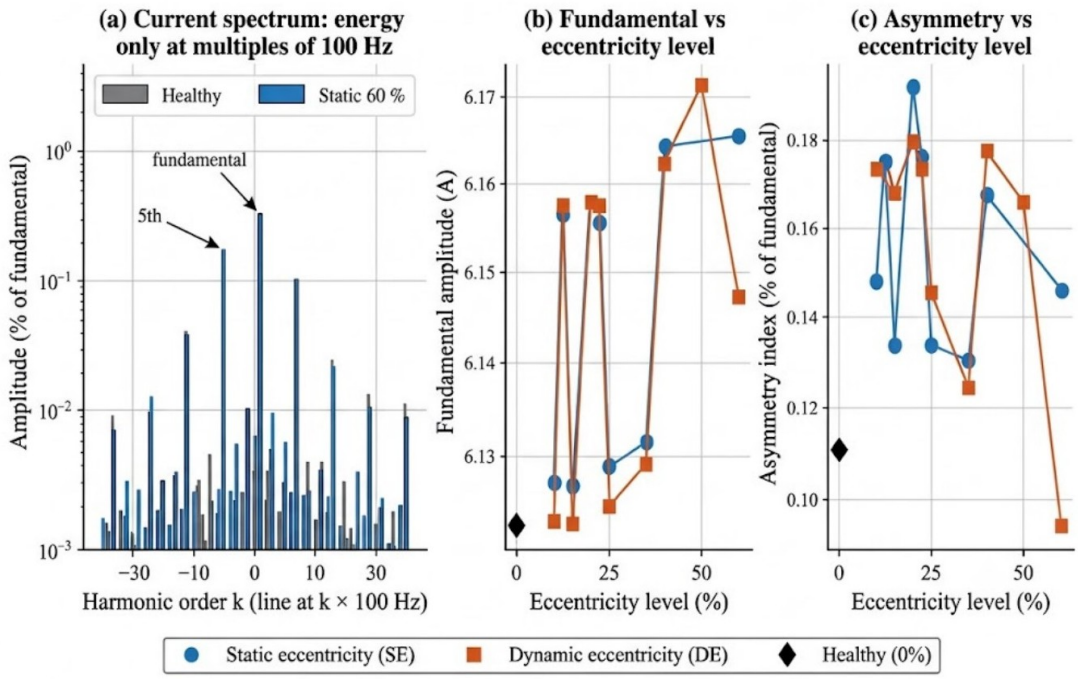


Fig. 5.4. (a) Current spectrum of the healthy and SE 60 % recordings, (b) fundamental amplitude and (c) asymmetry index against eccentricity level.

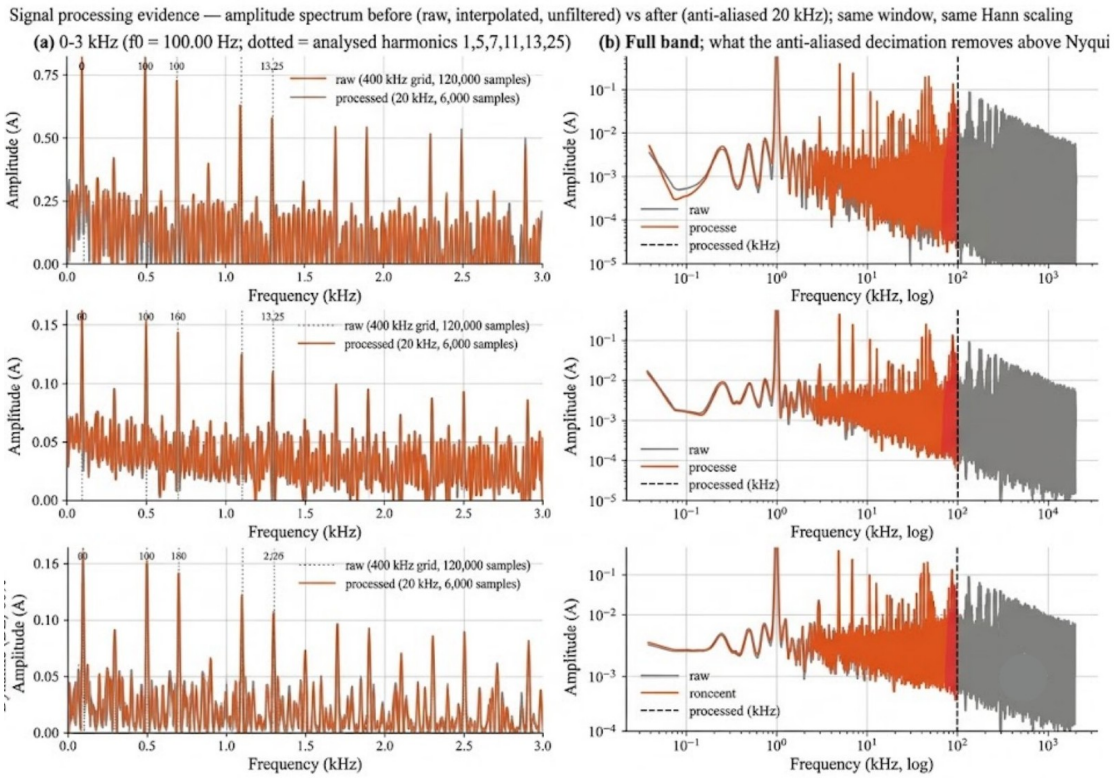


Fig. 5.5. Amplitude spectrum before (raw, 400 kHz grid) and after (anti-aliased, 20 kHz) pre-processing for the healthy, SE 60 % and DE 60 % recordings.

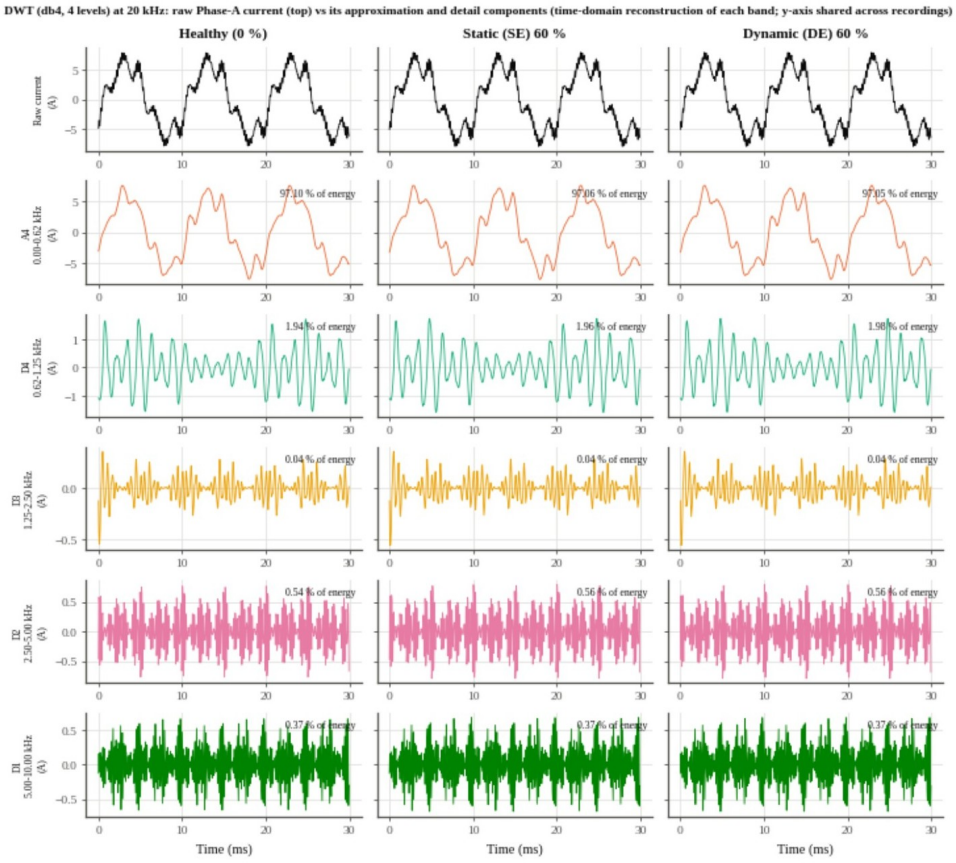


Fig. 5.6. DWT (db4, four levels) of the phase-A current: original signal and the reconstructed approximation and detail bands, with the percentage of energy in each band.

5.6 MACHINE-LEARNING MODELS

Two tasks are solved with separate models: a regression model estimates the severity in percent, and a classifier decides between SE and DE. Because only 15 training recordings are available, the models were chosen to be robust on small tables and the primary model of each task was fixed in advance, before the results were seen. All models were implemented with scikit-learn [34]. Tables 5.3 and 5.4 list the models.

Model	How it decides	Role
Extra Trees [35]	Many randomised decision trees vote; robust on small tables and hard to over-fit	Selected final model
RBF-SVM	Smooth curved boundary between the classes	Comparison
Logistic regression	Linear probability model	Comparison
Random forest [36]	Bagged decision trees	Comparison
KNN (k = 3)	Majority of the three nearest recordings	Comparison
LDA	Linear discriminant analysis	Comparison
Gradient boosting	Sequentially built trees	Comparison
Majority-class baseline	Always answers the most common class	Baseline

Table 5.3 Classification models (SE versus DE)

Model	How it predicts	Role
Linear SVR	Straight-line rule from features to severity; ignores errors smaller than a tolerance; regularisation tuned inside the training folds	Selected final model
Ridge regression	Straight-line rule with a penalty on large weights	Comparison
Random forest	Average of many decision trees	Comparison
Mean-predictor baseline	Always predicts the average training severity	Baseline

Table 5.4 Regression models (severity estimation)

Linear SVR was selected for severity because the features change smoothly with the severity level, and a model with few parameters suits 14 faulty recordings. Extra Trees was selected for SE/DE classification because an ensemble of decision trees can capture non-linear relations between the features without over-fitting small data. The two baselines show what a model without any skill would score.

5.7 VALIDATION STRATEGY

- Training-only cross-validation: leave-one-severity-level-out with 7 folds on the 15 training recordings (14 faulty recordings). In each fold both recordings of one severity level are held out, so the model is always tested on a level it has not seen.
- Scaling and the regularisation constant of the linear SVR are fitted inside the training folds only, so no information from the held-out recording enters the model.
- Independent test: the models fitted on the 15 training recordings are applied once to the 5 unseen test recordings.
- Metrics: mean absolute error (MAE), root-mean-square error (RMSE) and R^2 with 95 % confidence intervals for severity; accuracy, balanced accuracy, macro-F1, Matthews correlation coefficient (MCC) [43] and area under the ROC curve (AUC) for classification, always next to the baseline.

5.8 RESULTS

5.8.1 Severity estimation

Table 5.5 gives the cross-validation result on held-out levels and Table 5.6 the result on the independent test recordings. In cross-validation the selected Linear SVR with DWT features reduces the MAE from 15.93 points (mean predictor) to 8.26 points with $R^2 = 0.50$, and it is better than ridge regression and random forest. The random forest performs no better than the baseline (16.25 against 15.93 points). On the independent test the Linear SVR reaches an MAE of 1.98 points and $R^2 = 0.97$, against 10.00 points for the baseline.

Model	MAE (pts) [95 % CI]	RMSE	R^2
Mean-predictor baseline	15.93 [9.5, 24.0]	19.00	−0.35
Linear SVR + DWT (final)	8.26 [4.7, 11.8]	11.61	0.50
Ridge regression	8.97 [4.5, 13.2]	12.66	0.40
Random forest	16.25 [8.6, 23.2]	19.01	−0.35

Table 5.5 Severity regression metrics, training-only cross-validation (14 faulty recordings, held-out levels)

Model	MAE (pts) [95 % CI]	RMSE	R^2
Mean-predictor baseline	10.00 [3.7, 16.7]	12.82	−0.00
Linear SVR + DWT (final)	1.98 [1.0, 3.0]	2.31	0.97
Ridge regression	1.82 [0.9, 2.9]	2.17	0.97
Random forest	5.69 [3.1, 8.7]	6.56	0.74

Table 5.6 Severity regression metrics, independent test (5 recordings)

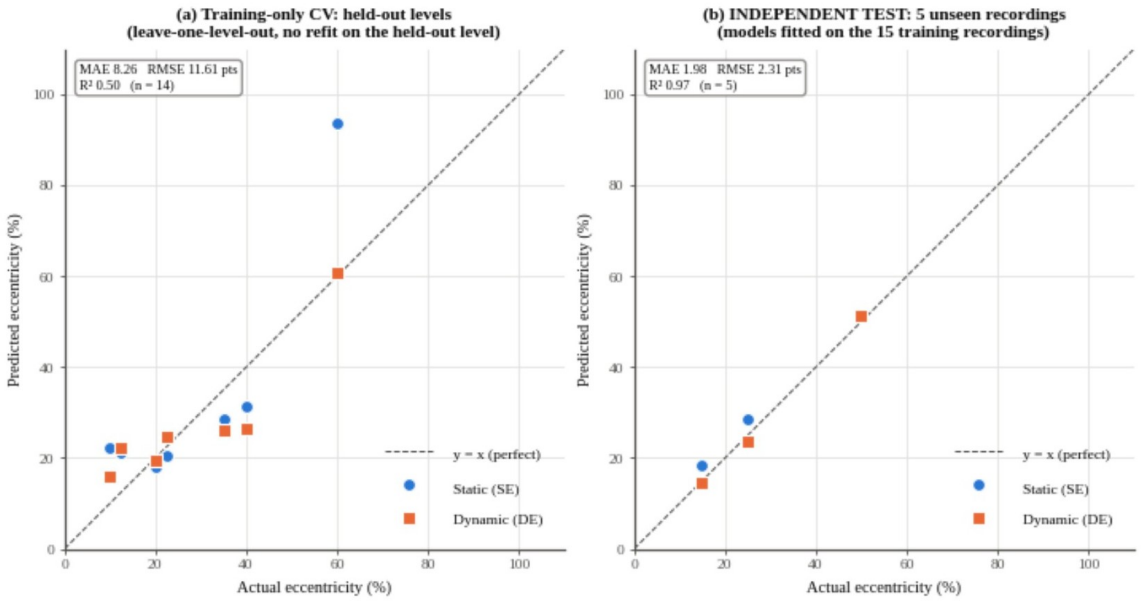


Fig. 5.7. Predicted against actual eccentricity: (a) cross-validation on held-out levels, (b) independent test on five unseen recordings.

Recording	True (%)	Estimated (%)	Error (pts)
DE 15 %	15	14.5	−0.5
DE 25 %	25	23.8	−1.2
DE 50 %	50	51.4	+1.4
SE 15 %	15	18.3	+3.3
SE 25 %	25	28.5	+3.5

Table 5.7 Estimated severity of every independent test recording

All five recordings are estimated within 3.5 points. The model slightly over-estimates both SE recordings. Only five recordings were available for the test, so the result is indicative and should not be read as a precise error bound.

5.8.2 SE/DE classification, cross-validation

Table 5.8 and Fig. 5.8 show the leave-one-level-out classification of the 14 faulty recordings. The majority baseline scores 0.50 because the two classes are balanced. The selected Extra Trees model reaches the same 0.50 (3 of 7 SE and 4 of 7 DE recordings correct), and its 95 % confidence interval [0.21, 0.79] contains the chance level. The RBF-SVM scores 0.64, but it answers DE for most recordings (6 of 7 DE correct against 3 of 7 SE), so its higher score reflects a bias towards one class and not a clear separation. No model separates SE from DE reliably when a whole severity level is held out.

Model	Accuracy	95 % CI	Bal. acc.	Macro-F1	MCC	AUC
Majority-class baseline	0.50	[0.50, 0.50]	0.50	0.33	0.00	0.50
Extra Trees + DWT (final)	0.50	[0.21, 0.79]	0.50	0.50	0.00	0.67
RBF-SVM	0.64	[0.36, 0.93]	0.64	0.63	0.32	0.57
Logistic regression	0.57	[0.36, 0.79]	0.57	0.57	0.14	0.57
Random forest	0.57	[0.36, 0.79]	0.57	0.57	0.14	0.69
KNN (k = 3)	0.43	[0.21, 0.64]	0.43	0.38	−0.17	0.43
LDA	0.50	[0.21, 0.79]	0.50	0.50	0.00	0.57
Gradient boosting	0.36	[0.14, 0.50]	0.36	0.35	−0.29	0.39

Table 5.8 SE/DE classification metrics, training-only cross-validation (14 faulty recordings)

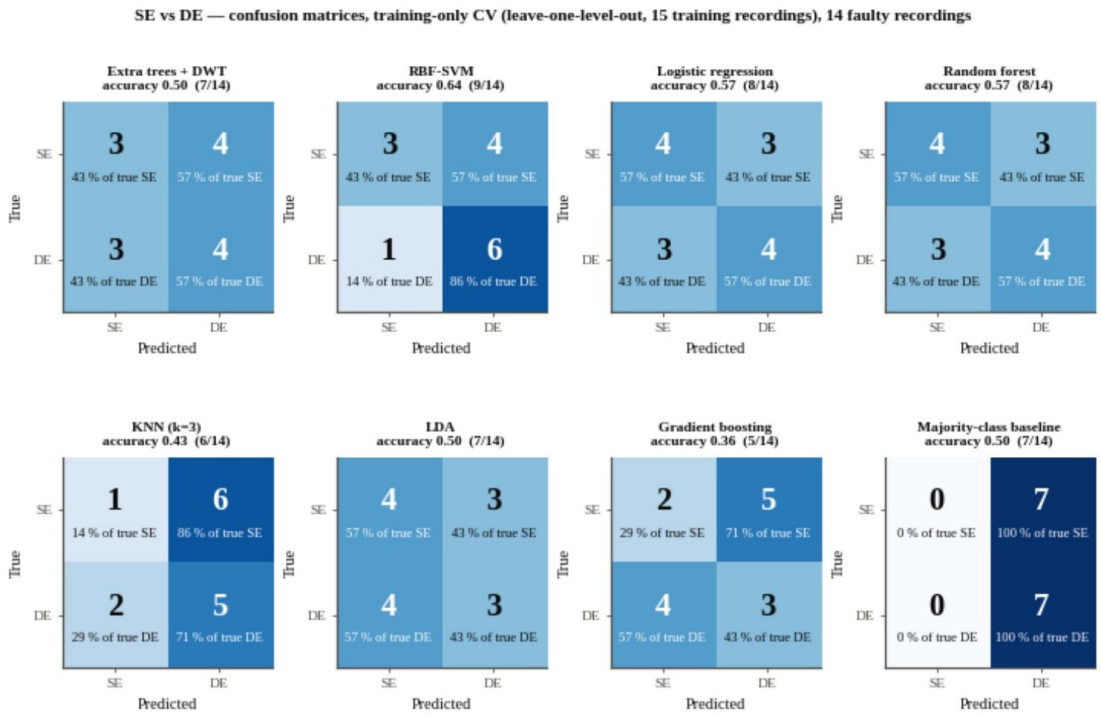


Fig. 5.8. Confusion matrices of the SE/DE classifiers in leave-one-level-out cross-validation (14 faulty recordings).

5.8.3 SE/DE classification, independent test

On the five independent test recordings the selected Extra Trees model, the RBF-SVM, the random forest and KNN classify all five recordings correctly (Table 5.9, Fig. 5.9), and logistic regression, LDA and gradient boosting make one error each. The majority baseline already scores 0.60 because three of the five recordings are DE.

Model	Acc.	Bal. acc.	Macro-F1	MCC	AUC
Majority baseline	0.60	0.50	0.38	0.00	0.50
Extra Trees + DWT (final)	1.00	1.00	1.00	1.00	1.00
RBF-SVM	1.00	1.00	1.00	1.00	1.00
Logistic regression	0.80	0.75	0.76	0.61	1.00
Random forest	1.00	1.00	1.00	1.00	1.00
KNN (k = 3)	1.00	1.00	1.00	1.00	1.00
LDA	0.80	0.75	0.76	0.61	0.83
Gradient boosting	0.80	0.83	0.80	0.67	1.00

Table 5.9 SE/DE classification metrics, independent test (2 SE and 3 DE recordings)

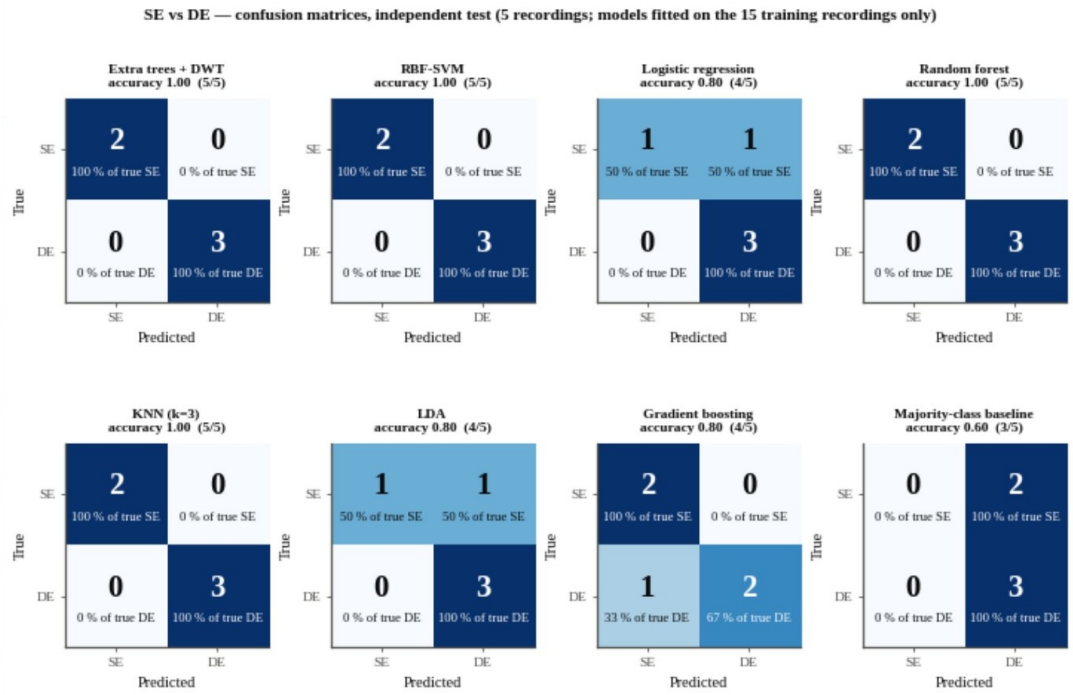


Fig. 5.9. Confusion matrices of the SE/DE classifiers on the five independent test recordings.

5.9 DISCUSSION: SEVERITY IS ACCURATE, FAULT TYPE IS UNCERTAIN

The results show two different levels of confidence. The severity of a recording can be estimated well, with an MAE of 1.98 points on unseen test recordings and a clear improvement over the baseline in cross-validation. The fault type (SE or DE) is reliable only on the small independent test; in the stricter cross-validation the selected classifier is at chance level. The framework therefore reports the severity as the dependable output and the fault type as a tentative one. Four reasons explain why severity can be learned while the type cannot be separated reliably:

- The current patterns of the two fault types are similar. Both SE and DE produce harmonics at multiples of the supply frequency (Fig. 5.4a), and the fundamental amplitude and the asymmetry index of the two types overlap over the whole severity range (Fig. 5.4b, c). The DWT energy shares of the SE 60 % and DE 60 % recordings differ only in the second decimal place (A4 97.06 % against 97.05 %, D4 1.96 % against 1.98 %, Fig. 5.6).
- The model learns severity-related patterns more easily than fault-specific patterns. A rise in the harmonic amplitudes and in the asymmetry accompanies a higher severity for both fault types, so one smooth rule covers both. Telling the types apart needs small differences that the extracted features hardly contain.
- The training data are limited. There are only seven recordings per fault type, and in each cross-validation fold one severity level (two recordings) is removed, so the classifier is trained on twelve faulty recordings. With so few examples a small number of recordings decides the outcome, as the wide confidence intervals in Table 5.8 show.
- Classification and regression optimise different objectives. The regression model minimises the error in percent, which the smooth feature trend supports, while the classifier must find a decision boundary between two classes whose feature clouds overlap; a model can reduce the first error without improving the second.

The independent test result (5 of 5 recordings correct) must be read with the same caution. With five recordings one error changes the accuracy by 20 points, the majority baseline already reaches 0.60, and the cross-validation on 14 recordings does not confirm the result. The fault type is therefore reported as tentative. More recordings per severity level and more informative features are needed before the type can be stated with confidence (see the future scope in Chapter 7).

5.10 SUMMARY OF THE RADIAL FLUX FRAMEWORK

- FFT and harmonic analysis is used because eccentricity alters the air-gap field and so changes the harmonic amplitudes of the stator current.
- The DWT is used to capture the energy distribution over frequency bands, which adds information that individual FFT harmonics may not capture.
- Current asymmetry is used because eccentricity disturbs the symmetry of the three-phase currents.
- Harmonic phase features are used because phase relations may help to distinguish SE from DE.
- Linear SVR with DWT features estimates the severity, because it predicts a continuous percentage and its regularisation controls the model complexity with limited training data.
- Extra Trees with DWT features classifies SE and DE, because an ensemble of decision trees captures non-linear relations between the extracted features.

Chapter 6

AXIAL FLUX PMSM DIAGNOSIS

6.1 INTRODUCTION

In the axial flux PMSM (AFPMSM) the rotor and the stator are discs facing each other across an axial air gap, so the eccentricity takes different forms from those of the radial machine. Two fault types are studied: rotor tilt, in which the rotor axis is inclined against the stator axis, and x-displacement, in which the rotor axis is shifted sideways (Fig. 6.1). Both change the air gap seen by the stator and therefore the stator currents.

The axial diagnosis uses the same three-phase stator currents as the radial machine, but the recordings are fewer and are sampled at a much higher rate. This chapter describes the dataset, the pre-processing, the 224 features, the five classical machine-learning models, the five deep-learning models, the ensemble decision, the results on held-out recordings and a comparison of eight signal-processing methods.

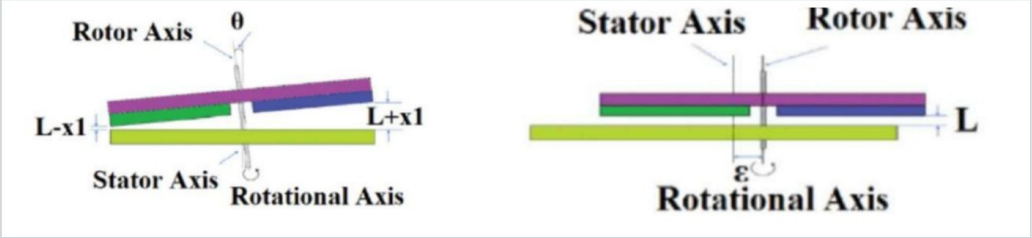


Fig. 6.1. Axial machine: rotor tilt (left) and rotor displacement (right).

6.2 DATASET

The axial dataset contains nine recordings (Table 6.1). Seven are used for training (one healthy, four rotor-tilt and two x-displacement recordings) and two are held out for testing. The class and the severity are read from the file names. Each recording was sampled at 526 kHz (time step 1.9 µs) and lasts about 22 ms (10,000 to 12,900 samples). Both test severities (x-displacement 10 % and tilt 17.5 %) do not occur in training, and the test set has no healthy recording. There is no separate validation file, so the training accuracy of the deep models and the test result are the only checks.

Split	Class	Severity (%)	Samples	Windows
Train	healthy	0	10,231	72
Train	tilt	7.5	11,843	85
Train	tilt	10	11,223	80
Train	tilt	20	11,987	86
Train	tilt	30	11,832	85
Train	x-displacement	4	12,073	87
Train	x-displacement	20	11,995	86
Test	x-displacement	10	12,864	93
Test	tilt	17.5	11,859	85

Table 6.1 Axial flux PMSM recordings

6.3 DIAGNOSIS PIPELINE

Two kinds of input are prepared from every window (Fig. 6.2). Hand-made features (224 numbers per window) feed five classical machine-learning models, and the raw waveform (three channels of 256 values) feeds four voting deep-learning models. A fifth deep model is trained and evaluated but does not vote. Every voting model gives a probability for healthy, tilt and x-displacement for each window, and the probabilities are averaged (soft voting). The window averages are averaged again over the file, the class with the highest value is the file verdict and its probability is the confidence. Separate regression models and deep-learning heads estimate the severity.

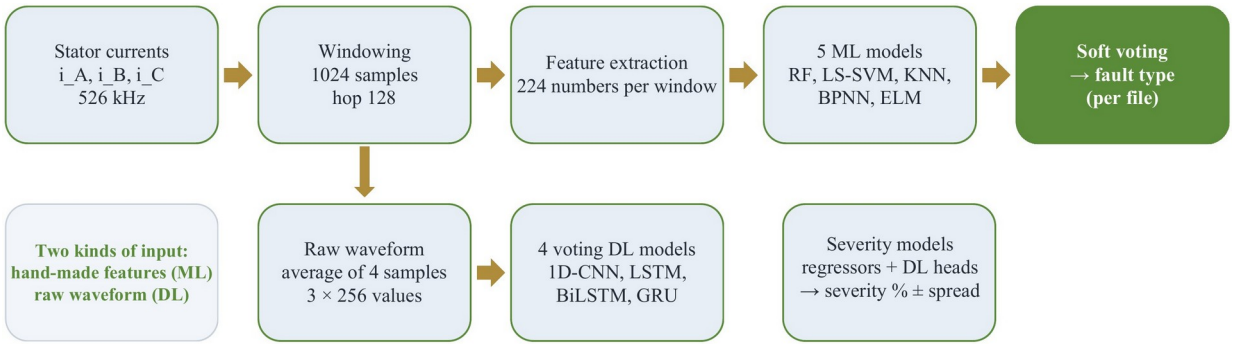


Fig. 6.2. Axial flux PMSM diagnosis pipeline.

6.4 SIGNAL PRE-PROCESSING

- Read and label: the time column gives the sampling rate; the next three columns are phases A, B and C; the file name gives the class and severity.
- Segmentation: windows of 1024 samples (1.95 ms at 526 kHz) with a hop of 128 samples (87.5 % overlap), at most 300 windows per file. Short windows give many examples from few recordings.
- Four channels: A, B, C and the sum A+B+C. In a balanced machine the sum is close to zero, so the sum channel exposes the unbalance that eccentricity causes.
- Normalisation: one global amplitude scale and then a z-score for every feature. Both are fitted on the training windows only, so that no test information leaks into the models.
- Input for deep learning: the raw window is averaged over every 4 samples (a light low-pass filter followed by decimation), giving 3 channels × 256 samples.

6.5 FEATURE EXTRACTION

Four channels (A, B, C and A+B+C) are described by 8 time-domain statistics and 48 frequency-domain values each, giving 4 × (8 + 48) = 224 features per window (Table 6.2). The time statistics describe the shape and size of the waveform, and the spectral band maxima describe the harmonic content, which is where eccentricity acts.

Domain	Per channel	Description
Time	8	Mean (DC offset), standard deviation, RMS, skewness (asymmetry of the waveform), kurtosis (peakiness), peak, peak-to-peak and crest factor (peak/RMS)
Frequency	48	Hann window [42] to limit spectral leakage; FFT magnitude; the range 0 to fs/8 (0 to 65.8 kHz, bin width 514 Hz) is split into 48 bands of about 1.4 kHz; the strongest line of each band is taken and log10 is applied to compress the dynamic range
Total	56 × 4 = 224	Channels A, B, C and A+B+C

Table 6.2 Composition of the 224 axial features

6.6 CLASSICAL MACHINE-LEARNING MODELS

Five classifiers are trained on the 224 standardised features of a window (Table 6.3). All five see exactly the same features and return a probability for each class.

Model	Configuration	Principle
Random forest	100 trees, maximum depth 12, √d features per split, balanced class weights	Combines many decision trees [36]
LS-SVM (RBF)	RBF kernel, σ^2 = median distance, γ = 60, one-versus-rest	Separates the classes with a non-linear boundary [40]
KNN	k = 7, distance-weighted	Classifies by the seven nearest training windows
BPNN (MLP)	1 hidden layer of 48 ReLU units, Adam (learning rate 0.002), 70 epochs	Learns by back-propagation
ELM	400 random sigmoid neurons, ridge output (C = 1000)	Random hidden layer, closed-form output weights for fast training [39]

Table 6.3 Classical machine-learning classifiers

6.7 DEEP-LEARNING MODELS

Five networks are trained on the raw waveform (Table 6.4), using PyTorch [37]. The 1D-CNN extracts local waveform patterns, and the LSTM [38], BiLSTM and GRU learn temporal relations along the 32 time steps of 24 values. A deep model votes only if its training accuracy reaches 70 % (voting gate). The CNN-LSTM reached 65.6 %, so it was evaluated but does not vote; the ensemble therefore has nine voters (five machine-learning and four deep-learning models).

Model	Architecture	Training acc.	Test acc. (window)	Votes
1D-CNN	Conv(8, k9) → pool 4 → Conv(16, k7) → pool 4 → FC 32	87.8 %	36.0 %	Yes
LSTM	32 steps × 24 values → LSTM(16)	98.8 %	74.2 %	Yes
BiLSTM	32 steps × 24 values → BiLSTM(12 + 12)	98.0 %	56.7 %	Yes
GRU	32 steps × 24 values → GRU(16)	97.8 %	52.8 %	Yes
CNN-LSTM	Conv(8, k9) → pool 4 → LSTM(16)	65.6 %	60.7 %	No

Table 6.4 Deep-learning models on the raw waveform

6.8 ENSEMBLE DECISION, SEVERITY AND HEALTH INDEX

The fault type is decided by soft voting over the nine voters. For the severity, separate regressors are trained for each fault type together with the healthy recording (0 %): a random-forest regressor,

a kernel ridge regressor, a KNN regressor, an ELM regressor and four deep-learning heads. The severity of a file is the median of the window estimates. The headline severity is taken from the best single estimator of each fault type (the random-forest regressor for x-displacement and the LSTM head for tilt), and the other models are reported beside it.

A health index between 0 and 100 summarises the condition, where 100 is a healthy machine:

$$H = 100 \cdot [\, P(\text{healthy}) + (1 - P(\text{healthy})) \cdot (1 - 0.9 \cdot s / s_{max}) \,], \quad s_{max} = 30\ \%$$

where s is the estimated severity and P(healthy) is the probability of the healthy class. For example, with P(healthy) = 0.1 and s = 15 %, $H = 100 \cdot [0.1 + 0.9 \cdot (1 - 0.9 \cdot 15/30)] = 59.5$.

6.9 RESULTS

6.9.1 Fault-type classification

Both held-out files were classified correctly (file accuracy 2/2). At window level 176 of 178 windows were classified correctly, an accuracy of 98.88 %, with a macro-F1 of 0.9887, a Matthews correlation coefficient of 0.9777 and a macro AUC of 0.9994 (Table 6.5). The confusion matrix (Fig. 6.3) shows that the only errors were two of the 85 tilt windows that were called x-displacement; all 93 x-displacement windows were correct. The ROC curves (Fig. 6.4) lie close to the top-left corner, with AUC = 1.0000 for tilt against the rest and 0.9989 for x-displacement against the rest.

Level	Accuracy	Macro precision	Macro recall	Macro-F1	Balanced acc.
File (2 files)	1.0000	1.0000	1.0000	1.0000	1.0000
Window (178)	0.9888	0.9895	0.9882	0.9887	0.9882

Table 6.5 Axial fault-type classification metrics on the held-out files (nine voting models)

Level	Class	Precision	Recall	F1-score	Specificity	Support
File	tilt	1.0000	1.0000	1.0000	1.0000	1
File	x-disp.	1.0000	1.0000	1.0000	1.0000	1
Window	tilt	1.0000	0.9765	0.9881	1.0000	85
Window	x-disp.	0.9789	1.0000	0.9894	0.9765	93

Table 6.6 Per-class results of the axial classification

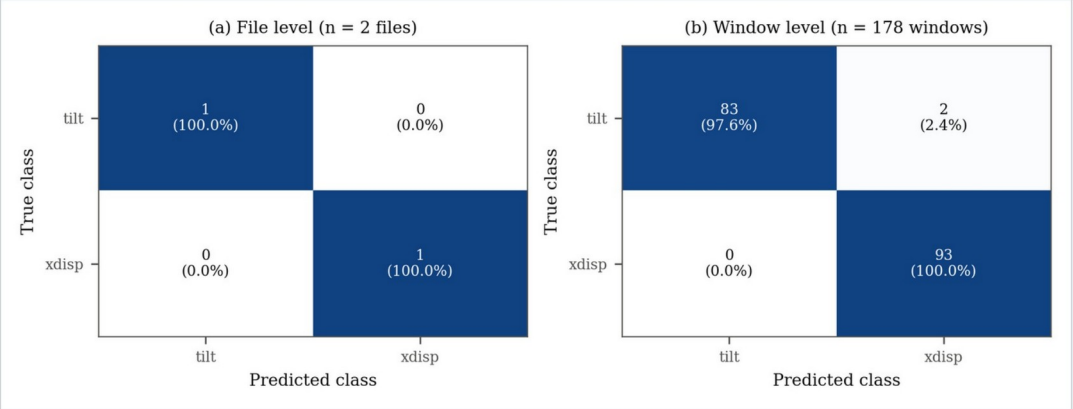


Fig. 6.3. Confusion matrices of the axial classification: (a) file level, (b) window level.

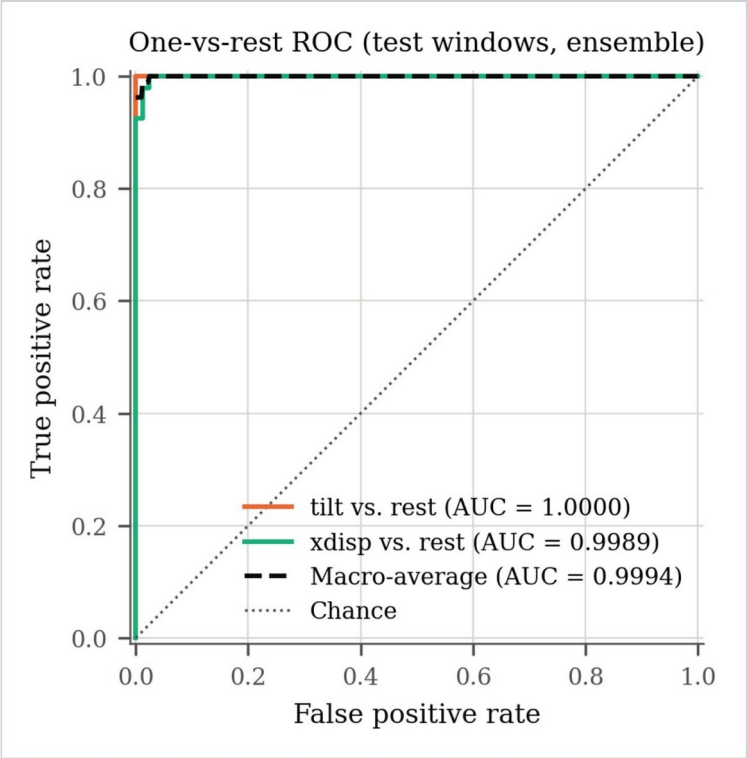


Fig. 6.4. One-versus-rest ROC curves of the ensemble on the test windows.

6.9.2 Ten models and the ensemble

Fig. 6.5 compares the ten models and the ensemble. The classical machine-learning models score between 0.89 and 0.99; the best single model is KNN with a window accuracy of 0.9944. The ensemble reaches 0.9888, close to the best single model and more robust than depending on one model. The deep-learning models score only between 0.36 and 0.74.

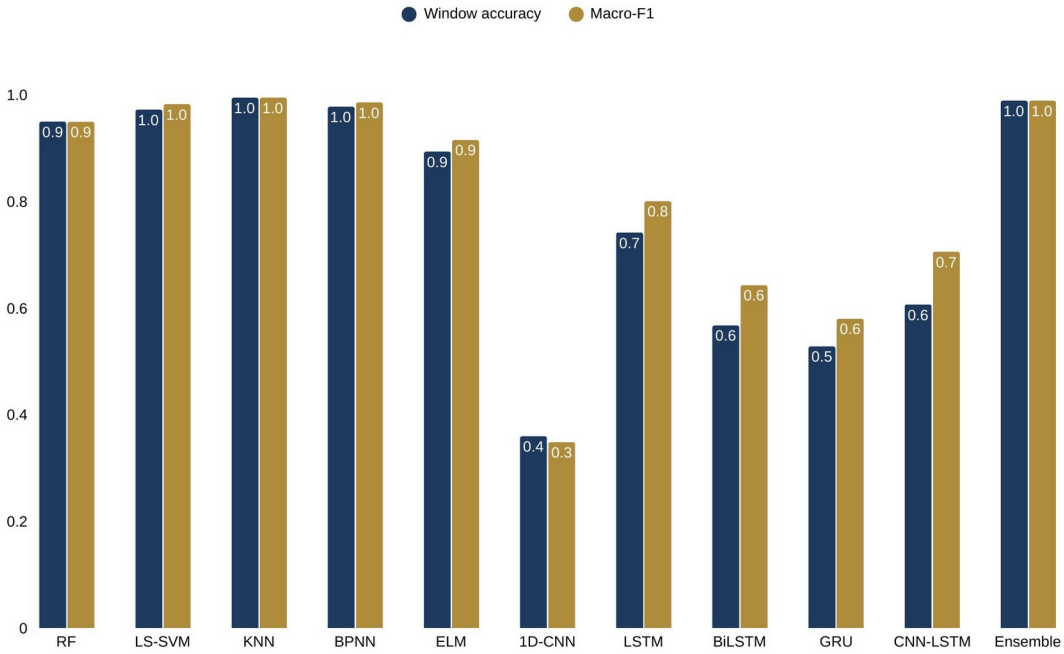


Fig. 6.5. Window accuracy and macro-F1 of the ten models and the ensemble on the test windows.

6.9.3 Deep-learning training behaviour

The training loss of the 1D-CNN, LSTM, BiLSTM and GRU falls steadily and their training accuracy rises above 87 %, while the CNN-LSTM stays below the 70 % voting threshold (Fig. 6.6). The test accuracy of the same networks is far lower (36.0 to 74.2 %, Table 6.4). This large gap shows that the networks over-fit the seven training recordings: with so few recordings, the windows of one file are almost identical, and a network can memorise the file instead of learning the fault. Deep models of this kind normally need far more recordings than were available here. The deep-learning models are therefore kept in the ensemble only where they pass the voting gate, and the classical models, which rely on the engineered features, carry the decision.

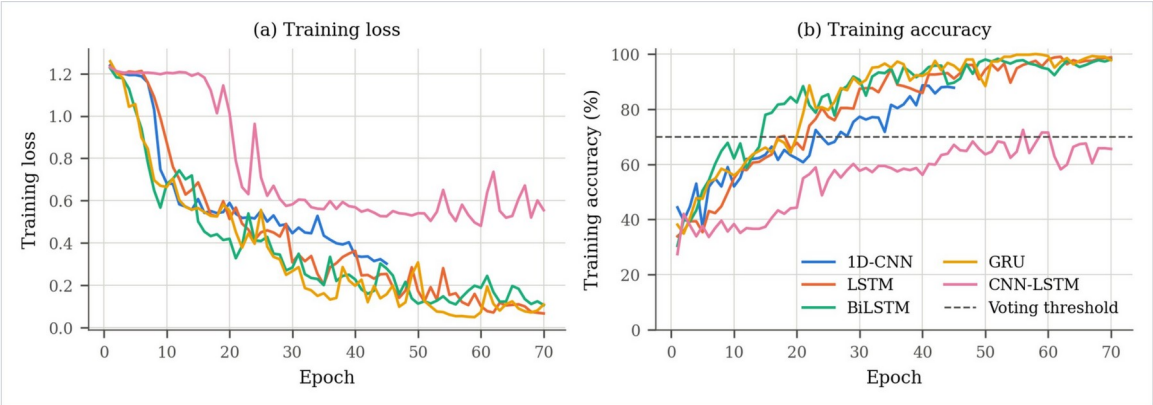


Fig. 6.6. Deep-learning training loss and training accuracy per epoch.

6.9.4 Severity estimation

The severity of both test files was estimated from severities that never occur in training (Table 6.7, Fig. 6.7), so the result is an interpolation. The x-displacement 10 % file was estimated at 6.54 % (error 3.46 points) and the tilt 17.5 % file at 15.78 % (error 1.72 points). The mean absolute error is 2.59 points, the root-mean-square error 2.73 points and the maximum absolute error 3.46 points.

Test file	True type / %	Predicted type / %	|Error| (pts)
x(disp)-10%	x-disp. / 10.00	x-disp. / 6.54	3.46
tilt-17.5%	tilt / 17.50	tilt / 15.78	1.72

Table 6.7 Axial severity estimation on the held-out files (equal-weight ensemble)

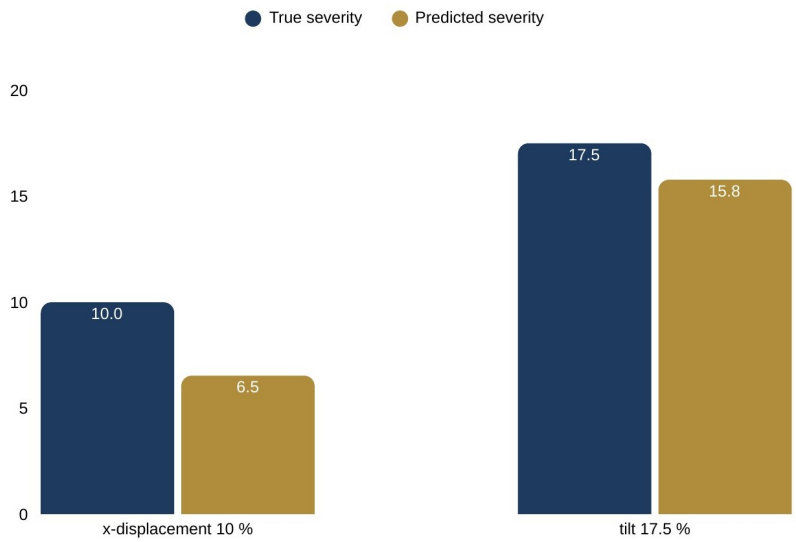


Fig. 6.7. True and predicted severity of the two held-out test files.

Fig. 6.8 compares the severity estimators. The equal-weight ensemble has the lowest error (MAE 2.59). The best single estimator is the random-forest regressor (3.48), followed by the LSTM head (3.68), and the least reliable is the ELM regressor (9.63, maximum error 16.6). The random-forest value for x-displacement and the LSTM value for tilt are therefore used as the headline severity, with every model reported beside it.

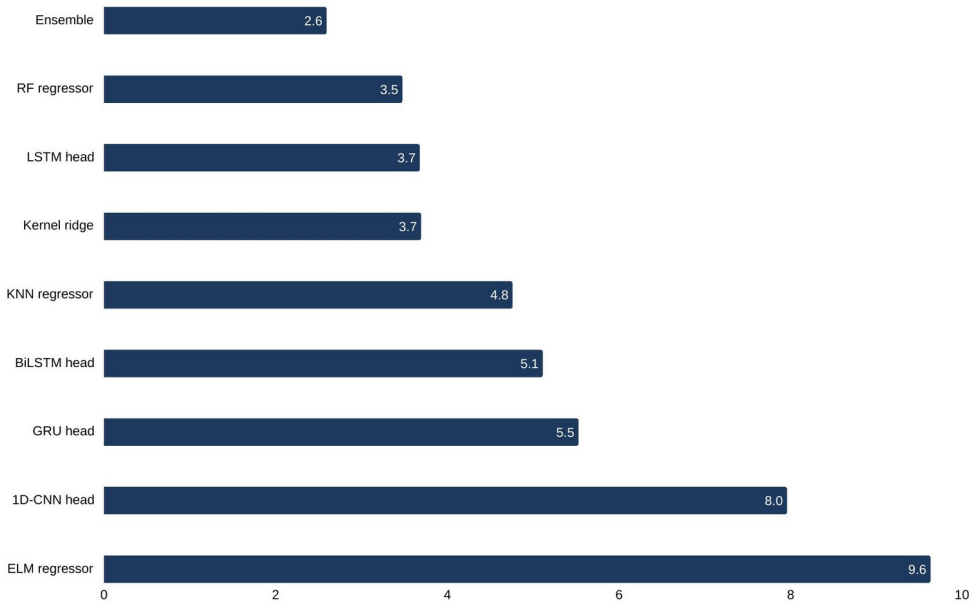


Fig. 6.8. Mean absolute severity error of each estimator and of the ensemble.

6.10 ANSYS MAXWELL SIMULATION RESULTS OF THE AXIAL FLUX PMSM

6.10.1 Eccentricity configurations of the axial flux machine

In the ANSYS model, angular eccentricity has two forms. Static angular (inclined) eccentricity is modelled by tilting the rotor while keeping the tilted axis fixed, so that the rotor axis stays inclined at a constant angle to the original shaft axis. Dynamic angular eccentricity occurs when the rotor is tilted and the tilt itself rotates during operation, which gives a continuously changing angular orientation. Mixed angular–axial eccentricity combines the rotor inclination with a displacement

along the Z-axis, and therefore represents the combined effect of angular misalignment and axial offset.

These implementations are a direct representation of the eccentricity conditions described in the literature. By varying the translational displacement, the axial displacement and the angular inclination independently, the model can separate the effects of static, dynamic, axial, angular and combined eccentricity, and gives a consistent framework for comparing the simulations with the terminology used in earlier studies.

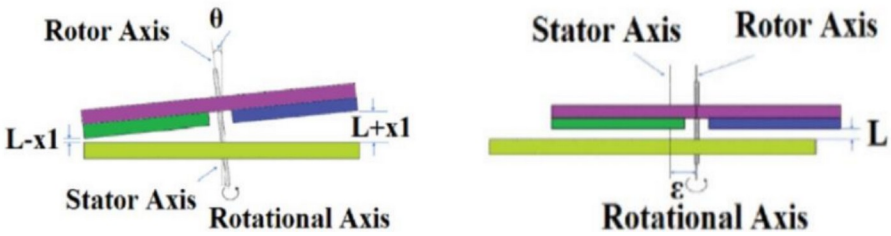


Fig. 6.9. Angular eccentricity (left) and static axis eccentricity (right) in the single-sided rotor topology of the axial flux PMSM.

Fig. 6.9 shows two important forms of rotor eccentricity, based on the relative position of the rotor and stator axes. In angular eccentricity (left) the rotor axis is inclined by an angle θ to the stator axis. The air gap then varies along the axial length of the rotor, with different clearances $L-x_1$ and $L+x_1$ at the two ends. If the inclination stays fixed during operation it is static angular eccentricity, and a rotating inclination represents dynamic angular eccentricity. In static axis eccentricity (right) the rotor axis stays parallel to the stator axis but is displaced from it by a distance ε. This gives a non-uniform air gap around the rotor circumference, with the smallest and largest gaps at different angular positions. The axes do not meet at an angle; they remain parallel with a fixed offset.

6.10.2 Tilt eccentricity (20 %)

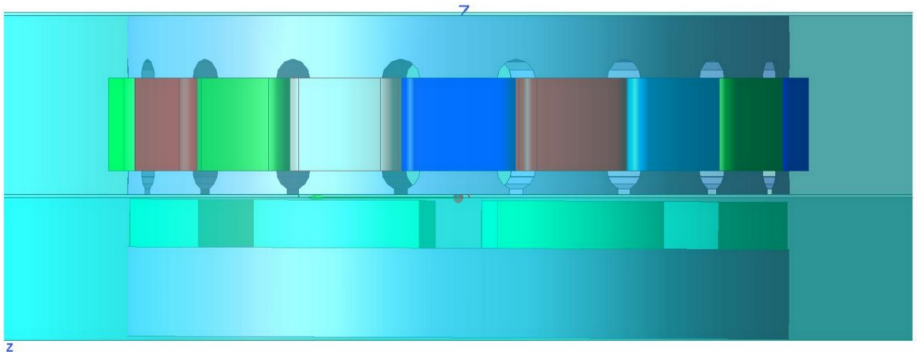


Fig. 6.10. Axial flux PMSM with 20 % tilt eccentricity (ANSYS Maxwell model).

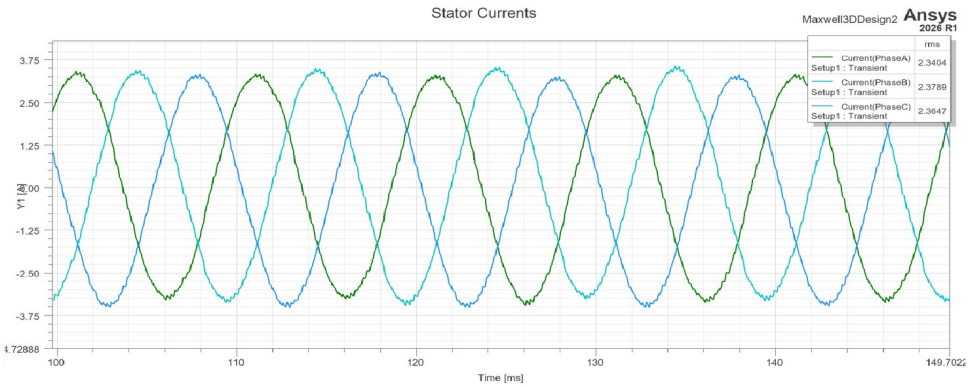


Fig. 6.11. Three-phase stator currents of the faulty axial flux PMSM with tilt eccentricity.

The stator currents of the AFPMSM were analysed for the healthy and the eccentric machine in ANSYS Maxwell. In healthy operation the three-phase currents are nearly sinusoidal and balanced, with RMS values of 3.0140 A, 3.0126 A and 3.0146 A for phases A, B and C. The very small difference between the phases indicates a symmetrical air-gap field and balanced operation. With eccentricity the air gap becomes non-uniform and the RMS currents change to 2.3404 A, 2.3789 A and 2.3647 A, a clear reduction in magnitude.

The average RMS current falls from about 3.014 A in the healthy motor to 2.361 A under eccentricity, a reduction of about 21.6 %. The phase-to-phase RMS difference increases from about 0.002 A to 0.039 A, which shows more current asymmetry caused by the non-uniform air gap. The faulty waveform remains approximately sinusoidal, but the lower amplitude and the larger phase imbalance show how eccentricity changes the electromagnetic behaviour of the machine. These changes are useful fault-sensitive indicators. Because eccentricity information may also be present in specific harmonic components, FFT and other signal-processing analyses are needed to find the characteristic fault signatures, and the extracted current features can then be used as inputs to the AI/ML models for detection, classification and severity estimation.

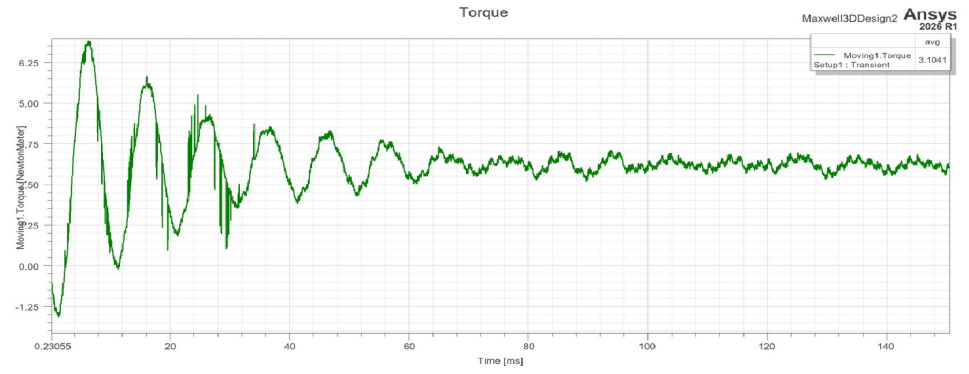


Fig. 6.12. Electromagnetic torque of the axial flux PMSM with tilt eccentricity.

The torque was analysed in the same way. In healthy operation the average electromagnetic torque is 3.6715 N·m. The torque shows transient oscillations during start-up, settles to a nearly steady value and then shows only small periodic variations, which indicates stable torque production. Under the eccentric condition the average torque falls to 3.1041 N·m, a reduction of 15.46 %. The faulty torque also shows larger oscillations and irregular variations, in the transient and in the steady-state region. This is due to the non-uniform air gap, which alters the magnetic-field distribution and the electromagnetic force. The lower average torque together with the higher

torque pulsation indicates degraded performance and can increase vibration and mechanical stress. Average torque reduction and torque ripple are therefore important fault-sensitive features for eccentricity diagnosis.

6.10.3 Dynamic axis eccentricity (1 mm rotor displacement)

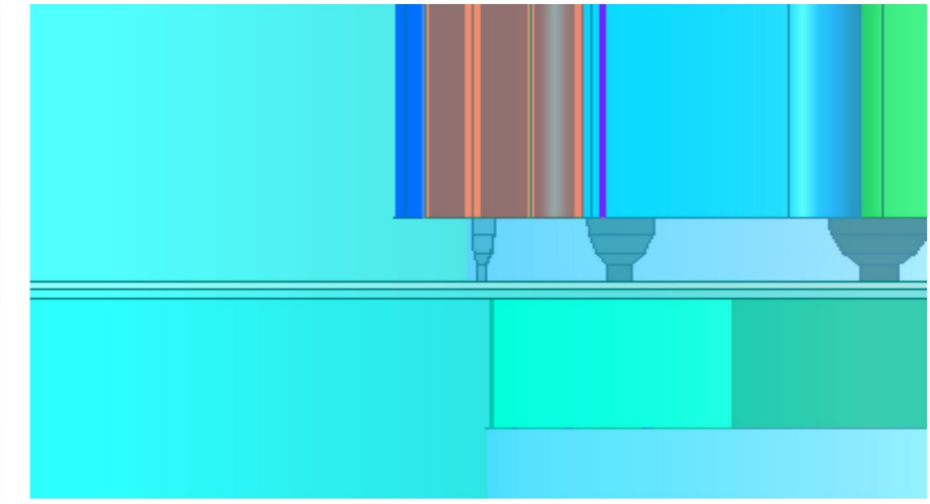


Fig. 6.13. Axial flux PMSM with dynamic axis eccentricity (ANSYS Maxwell model).

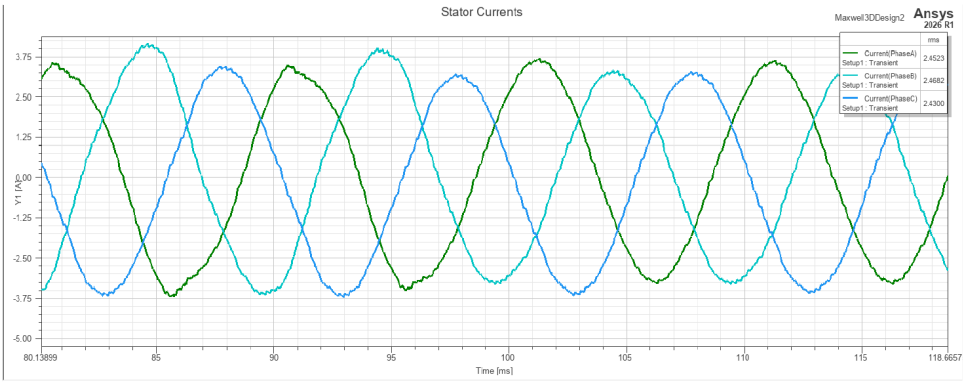


Fig. 6.14. Three-phase stator currents of the faulty axial flux PMSM with dynamic axis eccentricity.

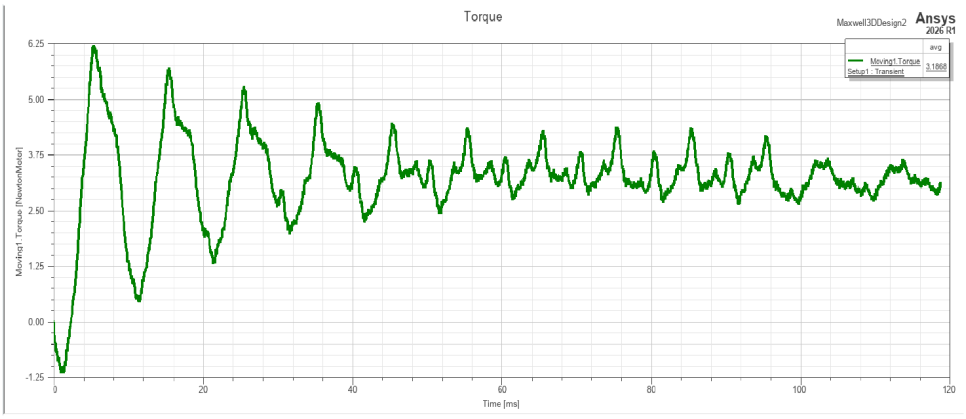


Fig. 6.15. Electromagnetic torque of the axial flux PMSM with dynamic axis eccentricity.

A displacement of 1 mm of the rotor components along the x-axis was introduced in the AFPMSM model. The stator current stays three-phase sinusoidal but its amplitude differs clearly from the healthy case. The RMS currents of phases A, B and C are 2.4523 A, 2.4682 A and 2.4300 A, an

average of about 2.4502 A. Against the healthy average of about 3.0137 A this is a reduction of about 18.70 %. The phase-to-phase RMS difference grows from about 0.002 A to 0.0382 A, which shows increased current asymmetry caused by the non-uniform air gap.

The torque confirms the effect. The average torque under the eccentric condition is 3.1868 N·m against 3.6715 N·m for the healthy motor, a reduction of about 13.20 %. The torque shows pronounced oscillations in the initial transient and noticeable periodic fluctuations later in the simulation. These variations come from the asymmetric air-gap distribution produced by the rotor displacement, which modifies the magnetic-field distribution and the electromagnetic force. The lower average torque, the higher torque pulsation and the current imbalance indicate degraded electromagnetic performance, and these changes can be used as fault-sensitive features for the signal-processing and AI/ML diagnosis described in this chapter.

Condition	Average RMS current (A)	Change (%)	Phase RMS difference (A)	Average torque (N·m)	Change (%)
Healthy	3.014	–	0.002	3.6715	–
Tilt eccentricity (20 %)	2.361	−21.6	0.039	3.1041	−15.46
Dynamic axis eccentricity (1 mm)	2.4502	−18.70	0.0382	3.1868	−13.20

Table 6.8 Simulated stator current and torque of the axial flux PMSM (healthy and eccentric)

Table 6.8 summarises these results. All three quantities respond to eccentricity in the axial machine, which is why the axial diagnosis in the earlier sections works directly on the stator-current waveform and its spectrum.

6.11 COMPARISON OF SIGNAL-PROCESSING METHODS

6.11.1 Motivation

The 224 features of Section 6.5 combine time statistics with FFT band maxima. To check that this choice is justified, seven other signal-processing methods that are common in motor-current diagnosis were implemented and compared with it under identical conditions. The aim is to identify which pre-processing performs best and to validate the answer with results.

6.11.2 Methods compared

Method	What is extracted	Features
Time-domain statistics	Eight statistics per channel (mean, standard deviation, RMS, skewness, kurtosis, peak, crest factor, peak-to-peak)	32
FFT band-max (adopted)	Time statistics plus 48 log10 band maxima of the Hann-windowed FFT up to fs/8	224
Welch PSD band power	Time statistics plus 32 log band powers of the Welch power spectral density (segment length 256, up to fs/4)	160
Wavelet DWT db4	Six-level db4 decomposition: log energy, relative energy and log standard deviation of each of the seven bands, plus time statistics	116
STFT band energy	Short-time Fourier transform (segment 128, overlap 64): mean and standard deviation over time of 16 log band magnitudes, plus time statistics	160
Hilbert envelope	Statistics of the Hilbert envelope, 32 log bands of the envelope spectrum, plus time statistics	192
Park vector (α-β)	Three phases converted to the complex Park vector: four statistics of its magnitude, 48 log bands of its complex spectrum and 24 bands of the magnitude spectrum	76
Fusion FFT+DWT+Park	Concatenation of the FFT, DWT (without statistics) and Park features	384

Table 6.9 Signal-processing methods compared (per window, four channels A, B, C and A+B+C)

6.11.3 Protocol

All methods used the same windows (1024 samples, hop 128, at most 300 per file) and the same four classifiers (random forest with 200 trees, KNN with k = 5, an RBF-SVM with C = 10 and an MLP with 64 hidden units), with the features standardised on the training windows only. The comparison was run on the original axial recordings: five training recordings (healthy, tilt 10 %, tilt 30 %, x-displacement 4 % and x-displacement 20 %) and two test recordings (tilt 20 % and x-displacement 10 %). Three checks were made for every method:

- Test accuracy and macro-F1 on the two held-out recordings, averaged over the four classifiers.
- Leave-one-file-out accuracy: each faulty recording is held out in turn and the models are trained on the other six recordings. This is the stricter check, because the held-out severity is always new.
- Severity error: a random-forest regressor trained on the faulty recordings of the same fault type, with the file value taken as the median over its windows, evaluated on the two test files and by leave-one-file-out.

6.11.4 Results

Method	Features	Test acc.	Macro-F1	LOFO acc.	Sev. MAE test	Sev. MAE LOFO
FFT band-max (adopted)	224	97.8	0.895	81.4	2.34	8.20
Fusion FFT+DWT+Park	384	93.2	0.930	79.7	2.19	8.33
Park vector (alpha-beta)	76	97.8	0.977	79.6	2.29	8.24
Welch PSD band power	160	93.2	0.858	79.5	1.61	7.90
Hilbert envelope	192	96.5	0.887	77.5	4.36	8.93
Time-domain statistics	32	84.2	0.562	77.4	5.90	10.01
STFT band energy	160	91.8	0.700	72.1	5.05	9.94
Wavelet DWT db4	116	87.4	0.731	71.1	6.00	10.08

Table 6.10 Comparison of signal-processing methods (accuracy in %, severity error in percentage points)

Method	RF	KNN	SVM	MLP	RF LOFO	KNN LOFO	SVM LOFO	MLP LOFO
FFT band-max (adopted)	100.0	99.4	92.2	99.4	83.0	82.8	79.1	80.5
Fusion FFT+DWT+Park	100.0	90.5	90.5	91.6	82.0	79.1	78.1	79.3
Park vector (alpha-beta)	100.0	99.4	91.6	100.0	79.9	79.5	76.2	82.7
Welch PSD band power	94.4	89.4	91.1	97.8	79.8	79.5	80.7	78.1
Hilbert envelope	100.0	94.4	91.6	100.0	80.6	75.5	73.5	80.3
Time-domain statistics	92.7	58.7	89.9	95.5	83.0	61.2	86.1	79.2
STFT band energy	98.9	81.6	89.4	97.2	81.8	63.1	71.9	71.7
Wavelet DWT db4	98.9	76.5	88.3	86.0	86.2	60.9	69.1	68.1

Table 6.11 Window accuracy of each method with each classifier (%)

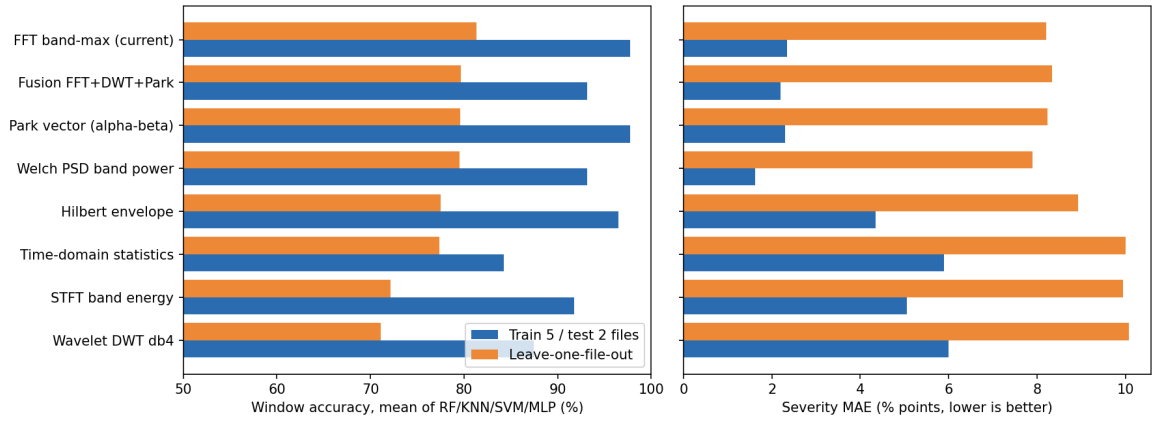


Fig. 6.16. Signal-processing methods compared: classification accuracy (train/test and leave-one-file-out) and severity error.

6.11.5 Discussion and validation

- Best methods. The FFT band-max features used in this project and the Park-vector features share the highest test accuracy (97.8 %). The FFT features have the highest leave-one-file-out accuracy (81.4 %), and the Park-vector features the highest macro-F1 (0.977) with three times fewer features (76 against 224). Both spectrum-based methods are clearly better than the others, so the choice of an FFT-based feature set is supported by the results.
- Severity. Welch power spectrum gave the lowest severity error (1.62 points on the test files and 7.90 in leave-one-file-out), and the FFT and Park features were close (about 2.3 and 8.2). The

methods built on time statistics, STFT or wavelets were clearly worse (4.4 to 6.0 points on the test files).

- Weak methods. The wavelet and STFT features give the lowest classification accuracy (71 to 72 % in leave-one-file-out). Time statistics alone fail on the minority class, which gives a macro-F1 of 0.56. The Hilbert envelope is in between.
- Fusion. Combining the FFT, DWT and Park features did not beat the best single method (leave-one-file-out 79.7 %); the extra weak features add noise when only a few recordings are available.
- Limits. In the leave-one-file-out check every method reaches only about 71 to 81 %, and the differences of one or two points between the best methods are not significant with seven recordings. The ranking was obtained on the original 5/2 split of the recordings, so it should be repeated on the 7/2 split of Table 6.1 with the supplied script (compare_sp.py) before it is quoted as final.

The comparison therefore validates the choice made for the axial pipeline: spectral band features of the stator current are the most informative among the tested processing methods, and the Park-vector features are a compact alternative that could be added as a second feature set.

6.12 SUMMARY OF THE AXIAL FLUX FRAMEWORK

- The axial pipeline cuts each recording into 1024-sample windows, extracts 224 time and spectral features from four current channels, and combines five machine-learning and four deep-learning models by soft voting.
- On the two held-out recordings the ensemble gave 2/2 correct files, a window accuracy of 98.88 %, a macro-F1 of 0.9887 and a macro AUC of 0.9994, and estimated the unseen severities with an MAE of 2.59 points.
- The deep-learning models over-fit the seven training recordings (training accuracy up to 98.8 %, test accuracy 36 to 74 %) and the classical models on engineered features are the more dependable.
- The signal-processing comparison confirmed that the FFT band-max features are the best choice among those tested, with the Park-vector features as a close and compact alternative.

Chapter 7
CONCLUSION AND FUTURE SCOPE

7.1 CONCLUSION

This project developed and compared current-based diagnosis of eccentricity faults in axial and radial flux permanent magnet synchronous motors. Healthy and faulty machines of equal rating (0.55 kW, 220 V, 1500 rpm, 8 poles) were modelled in ANSYS Maxwell and driven from a PWM converter, and the three-phase stator currents were used as the only input. The finite-element study (Chapter 4) showed that eccentricity changes the air-gap field, the torque and the stator current, and the signal-processing and machine-learning study (Chapters 5 and 6) turned these changes into a diagnosis of the fault type and the severity.

For the radial flux machine, 23 features from FFT harmonics, current asymmetry, harmonic phase and the DWT were used. Linear SVR estimated the severity of five unseen recordings with an MAE of 1.98 points ($R^2 = 0.97$, against an MAE of 10.00 for a mean predictor) and 8.26 points in leave-one-level-out cross-validation (against 15.93). The SE/DE type was classified correctly on the five test recordings by the Extra Trees model, but in cross-validation on 14 recordings it was at chance level, so the type is reported as tentative.

For the axial flux machine, 224 time and spectral features per window and an ensemble of five machine-learning and four deep-learning models classified both held-out recordings correctly (window accuracy 98.88 %, macro-F1 0.9887, macro AUC 0.9994) and estimated the severity with an MAE of 2.59 points. The deep-learning models over-fitted the seven training recordings, and the classical models on the engineered features carried the decision. A comparison of eight signal-processing methods confirmed that spectral band features of the current are the most informative of those tested.

7.2 COMPARISON OF THE TWO MACHINES

Item	Radial flux PMSM	Axial flux PMSM
Fault types	Static (SE) and dynamic (DE) eccentricity	Rotor tilt and x-displacement
Recordings	15 training, 5 test (20)	7 training, 2 test (9)
Sampling	400 kHz grid, 20 kHz after resampling	526 kHz
Unit of analysis	One feature vector per recording (whole cycles)	Windows of 1024 samples (about 580 training windows)
Features	23 (amplitude, asymmetry, phase, DWT)	224 (time statistics and FFT bands, four channels)
Models	Extra Trees (type), Linear SVR (severity)	5 ML + 4 DL voting models, regressors and DL heads
Severity error, unseen levels	MAE 1.98 pts (test); 8.26 pts (cross-validation)	MAE 2.59 pts (test, ensemble)
Fault-type result	5/5 on test; 0.50 in cross-validation (tentative)	2/2 files; 98.88 % windows; AUC 0.9994
Main limitation	Few recordings; SE and DE patterns overlap	Only 2 test files; deep models over-fit

Table 7.1 Summary of the radial and axial flux diagnosis results

The two results are not directly comparable, because the machines, fault types, sampling and evaluation units differ. Both topologies could be diagnosed from the stator current: severity is predicted within about 2 to 3 percentage points for unseen levels, and the fault type is dependable in the axial machine on the held-out files but not yet in the radial machine.

7.3 LIMITATIONS

- All data are from finite-element simulation. Measured currents contain noise, supply distortion, load changes and manufacturing tolerances that the simulations do not fully represent.
- The number of recordings is small (20 radial and 9 axial). Windows from one recording are strongly correlated, so a window-level score overstates the performance, and results from two or five test recordings are indicative only.
- Only single eccentricity faults at steady state and one operating point were studied. Mixed eccentricity, load changes and other faults that produce similar current harmonics were not included.
- The SE/DE classification of the radial machine is not yet reliable when a whole severity level is held out.
- The deep-learning models over-fit the small training set.

7.4 FUTURE SCOPE

- Generate more recordings, with more severity levels, several load and speed conditions and mixed eccentricity, so that the type classification can be validated with many more files per class and the deep-learning models can be trained properly.
- Add features that separate SE from DE, for example from the Park vector (which was a compact and accurate alternative in the axial study), the sideband frequencies around the fundamental and the slot harmonics, and test them with the same leave-one-level-out protocol.
- Validate the framework on the hardware test bench of the radial flux machine and on measured currents, and study how well a model trained on simulation transfers to measurement (domain adaptation).
- Use the closed-loop converter model, in which the controller reacts to the fault, to generate data and study how closed-loop control affects the current signature.
- Extend the framework to read data directly from a current sensor for continuous monitoring and predictive maintenance, with an alarm when the estimated severity crosses a limit.

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